

## Practical: 4

**Aim: Deploy and Manage SDN infrastructure using SCVMM 2019.**

**Writeup:**

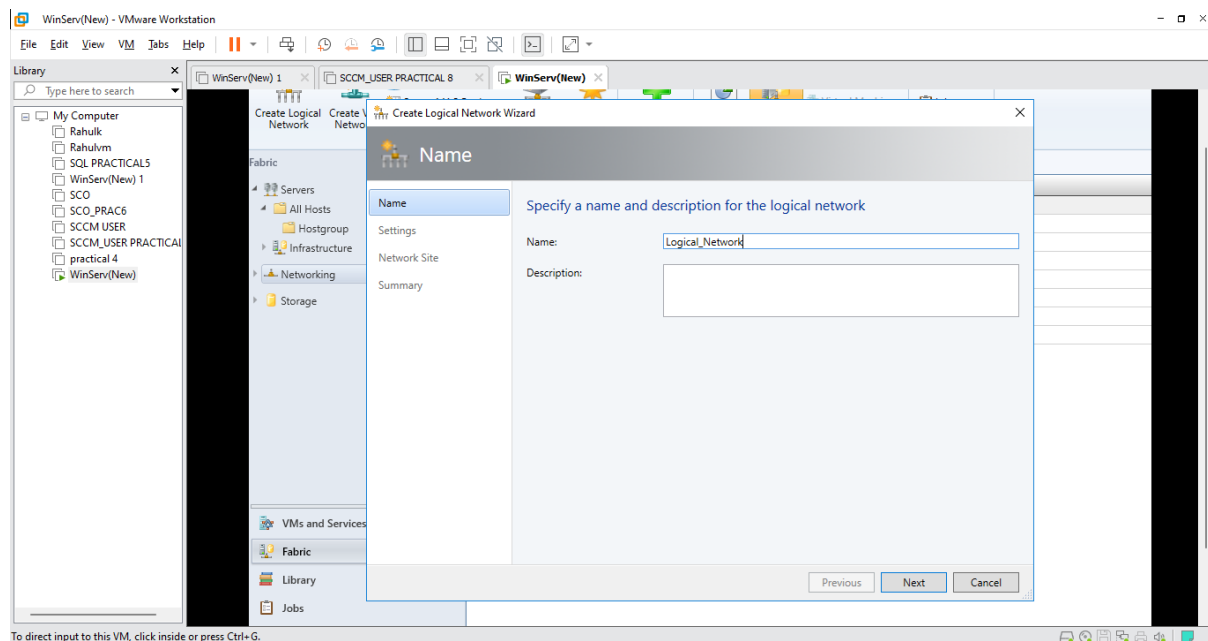
This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal black lines running across the width of the page. The lines are thin and consistent in thickness. There are no margins, text, or other markings on the paper.

**Pre-requisites:**

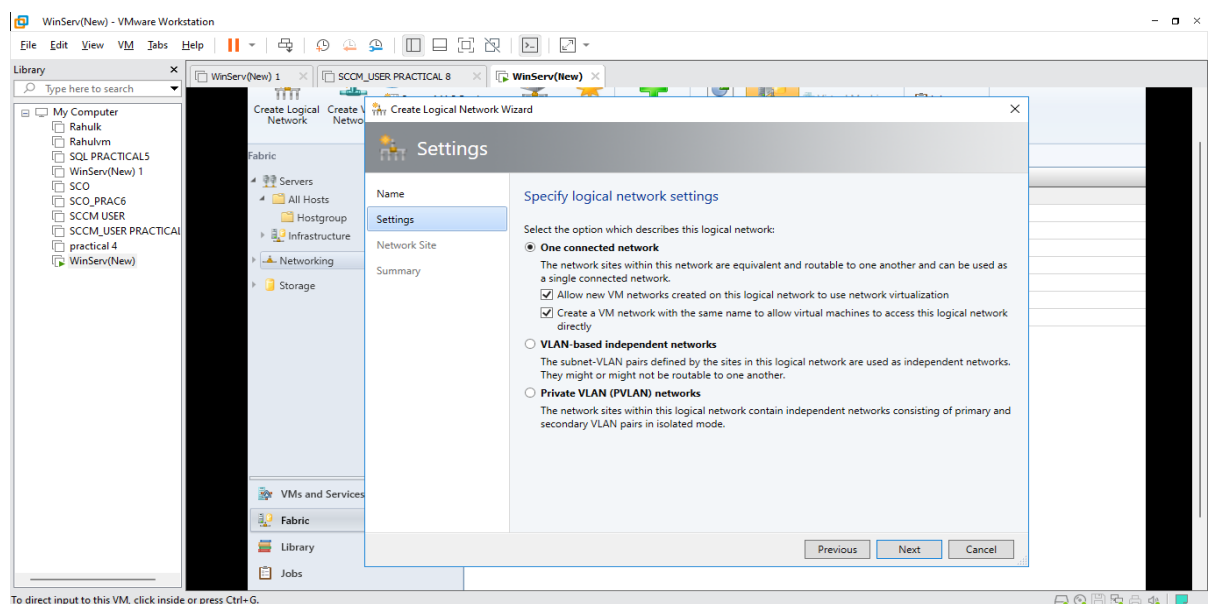
- SCVMM from practical 1

**4-A: Deploying Network Controller****Step 1: Create a Logical Network**

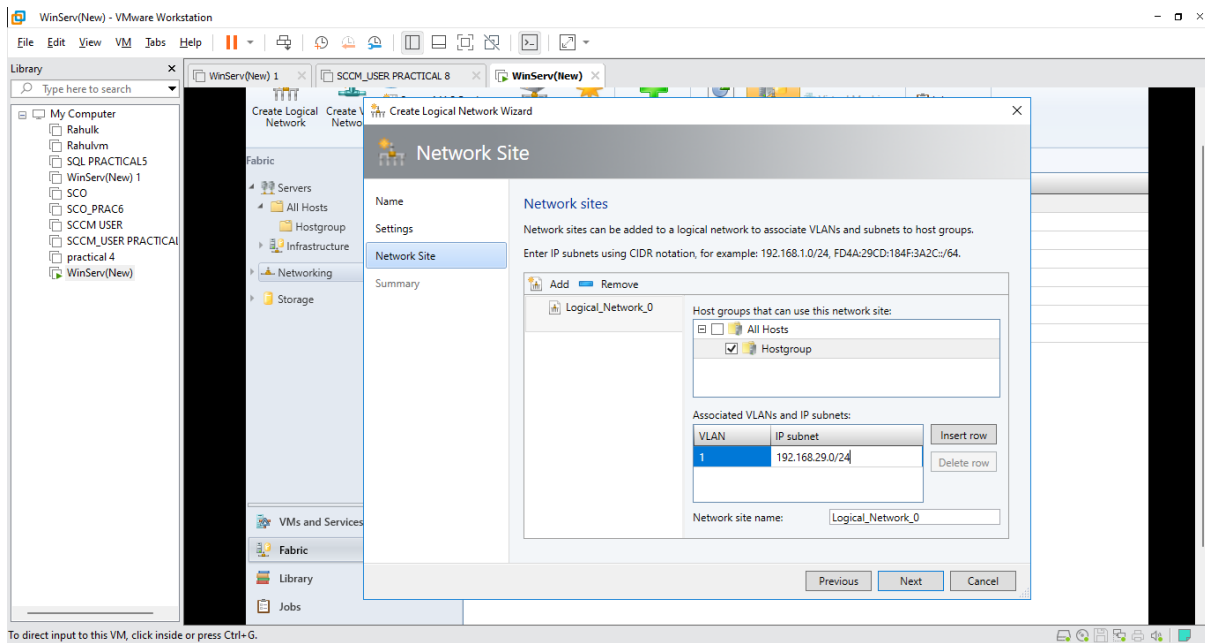
- Open SCVMM and **Click on the Fabric Tab** and Select **Networking**, within **Networking Right-Click on Logical Network** and Select **Create Logical Network** And Specify a name for the logical Network (Here it is: **Logical\_Network**) and **Click on Next**



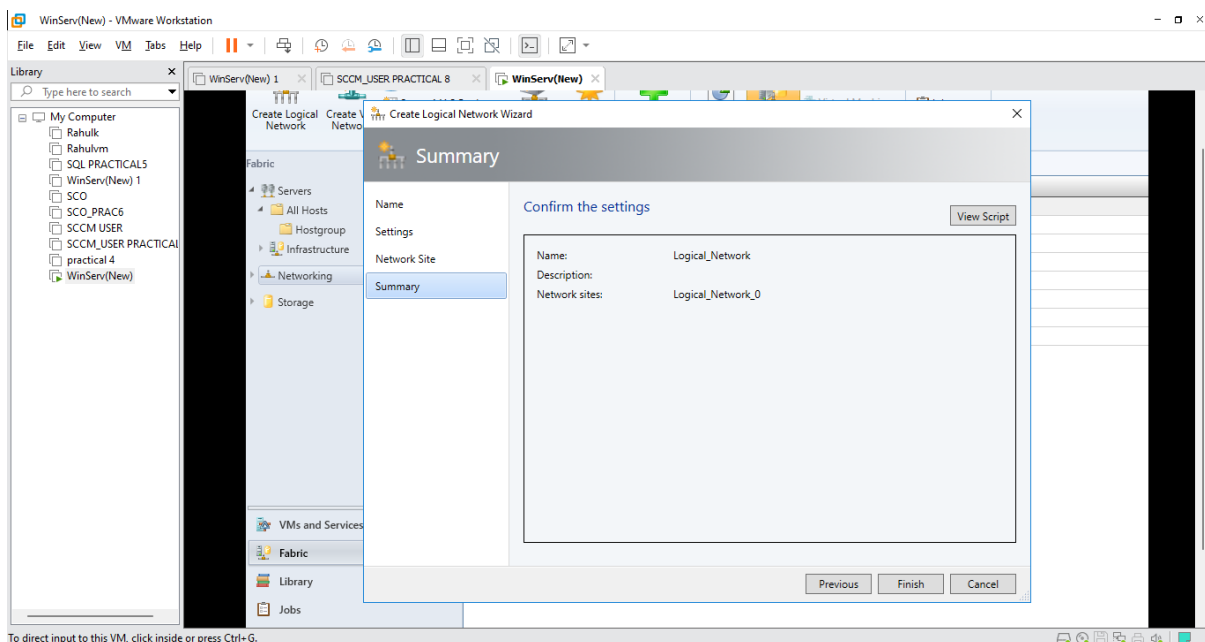
- Select **One-connected network** and **Tick both Allow new VM networks created on the logical network to use network virtualisation** and **Click on Next**



- Click on **Add** and Select **Hostgroup** and Click on **Insert Now** and type in the **VLAN number** and an **IP Address** and Click on **Next**

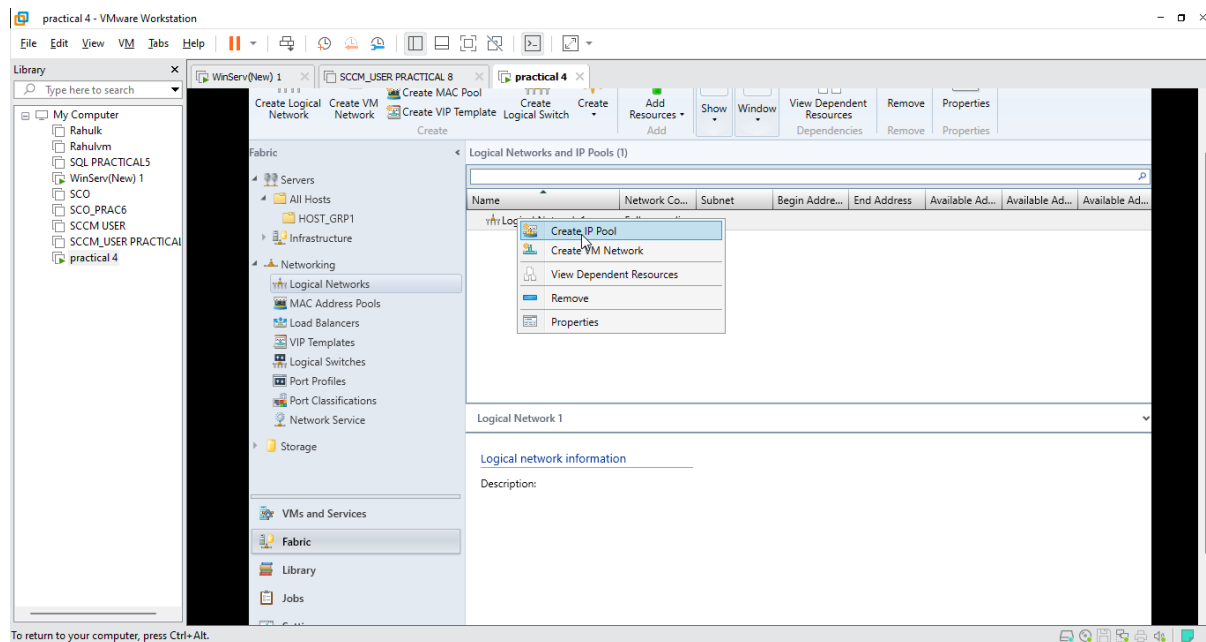


- Click on **Finish**

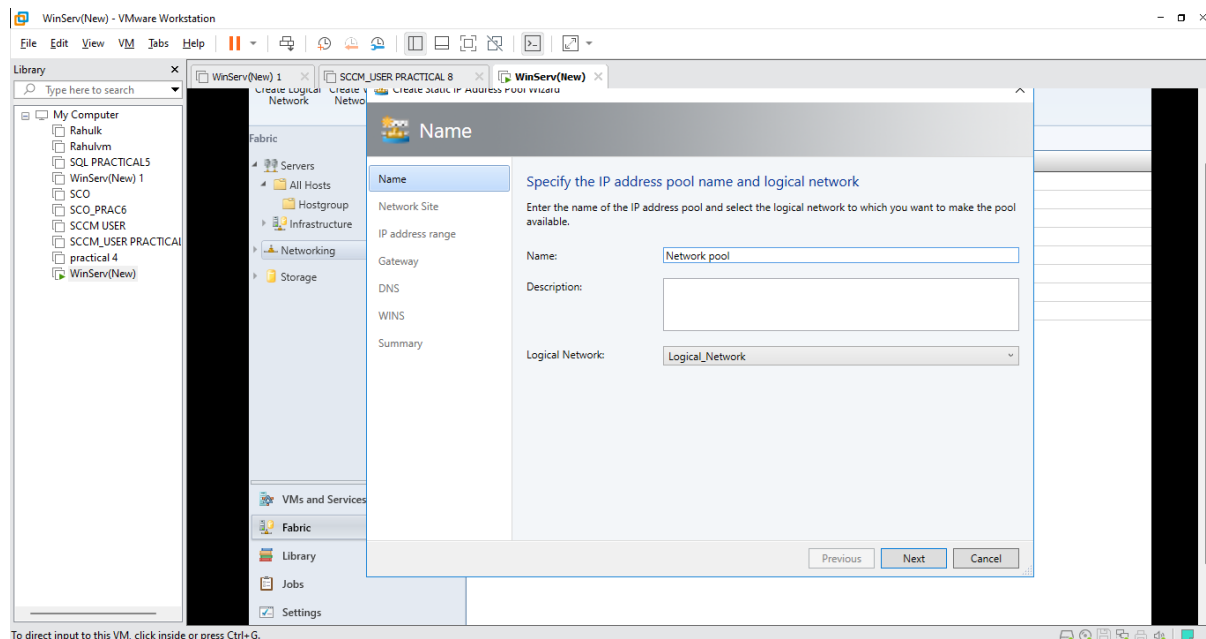


## Step 2: Creating IP Pools

- Right-Click on Logical Network and Select Create IP Pool

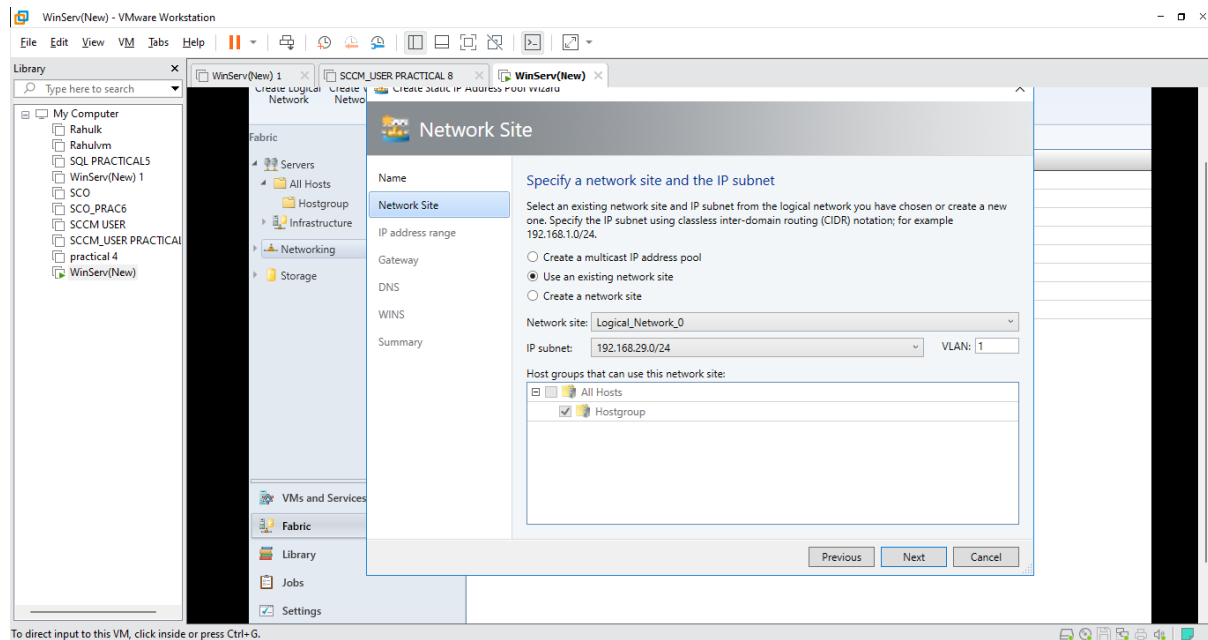


- Name it Pool and Make sure the Logical Network you just created has been selected and Click on Next

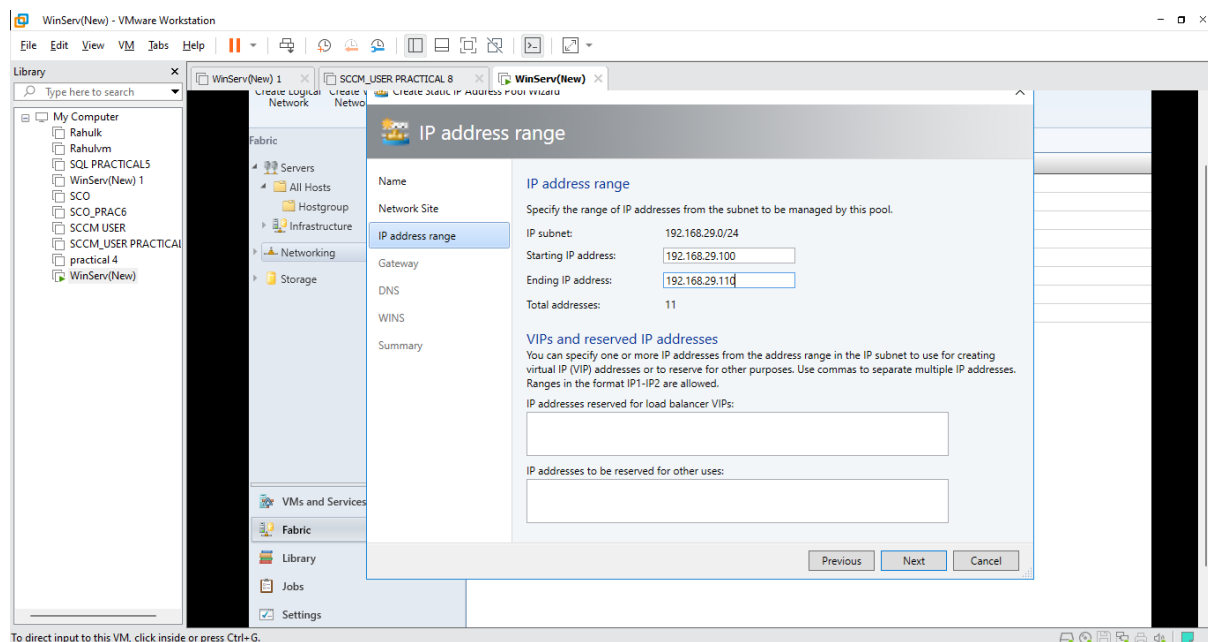




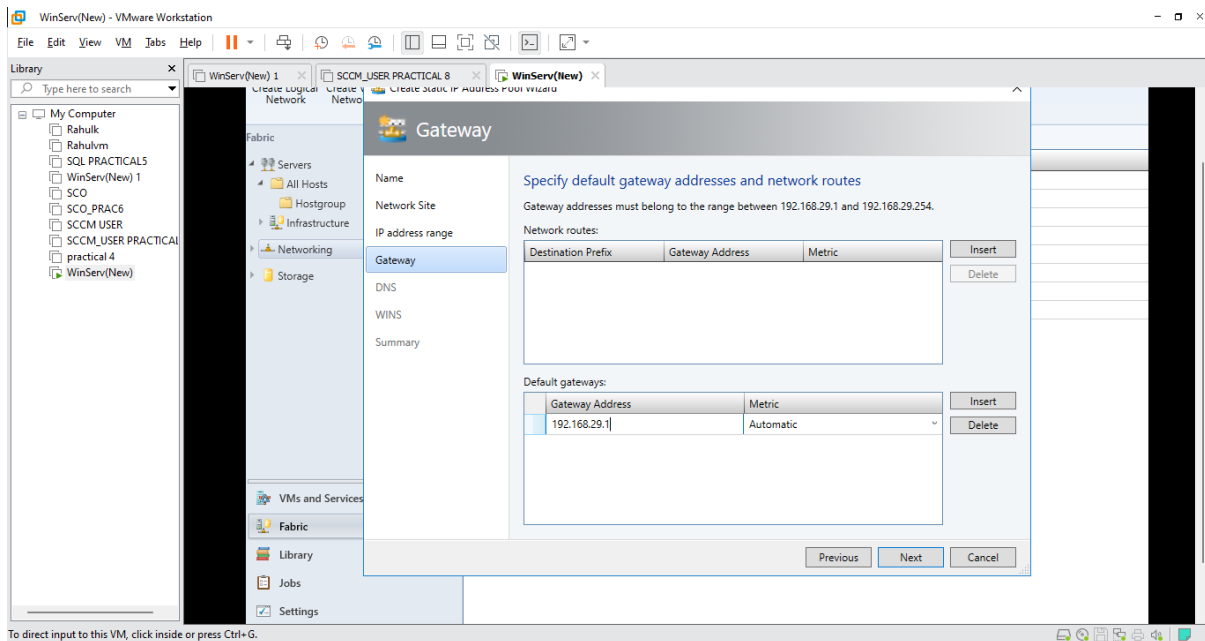
- Keep the default value and **Click on Next**



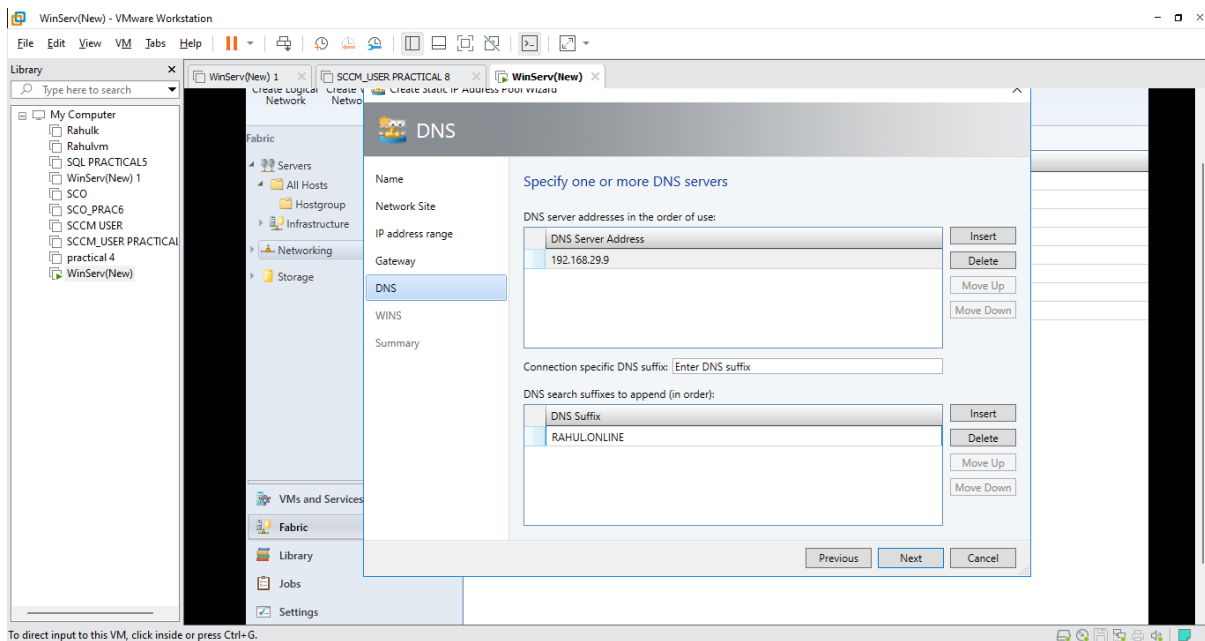
- In IP Address range Change the Starting IP Address to **192.168.29.100** and Ending IP Address Range to **192.168.29.110** and **Click On Next**



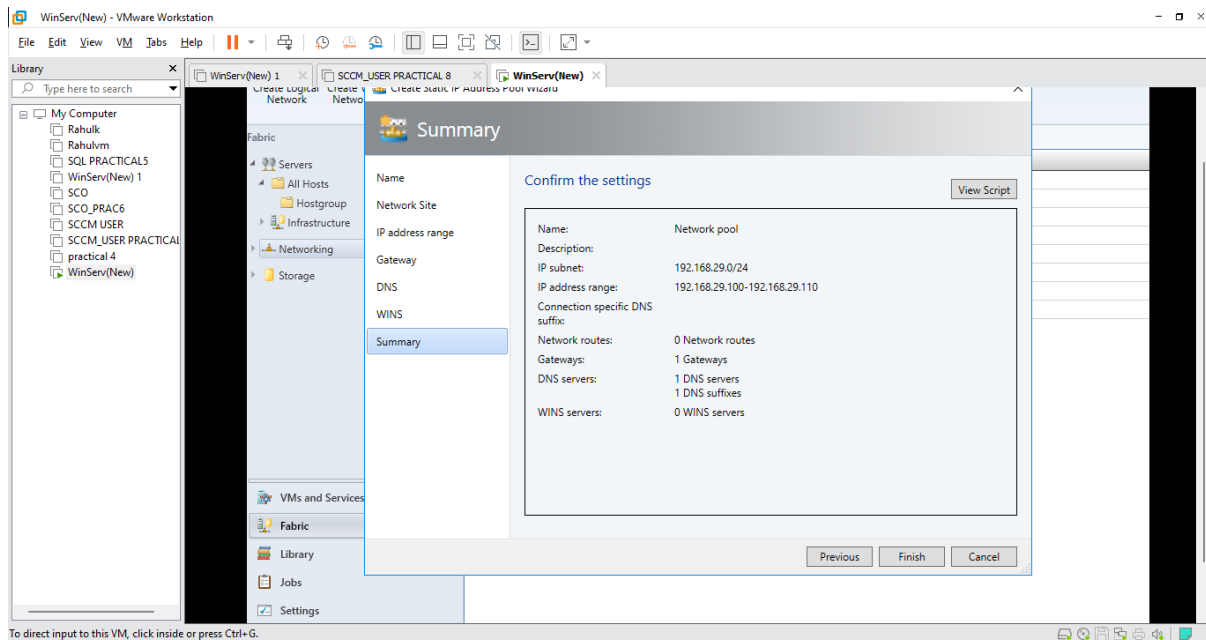
- Within Default Gateway **Click on Insert** and Enter the IP Address **192.168.29.1** and Keep the metric as Automatic and **Click on Next**



- In DNS, within DNS Server address in the order of use **Click on Insert** and Type in the IP Address 192.168.29.9 and **Click on Next**

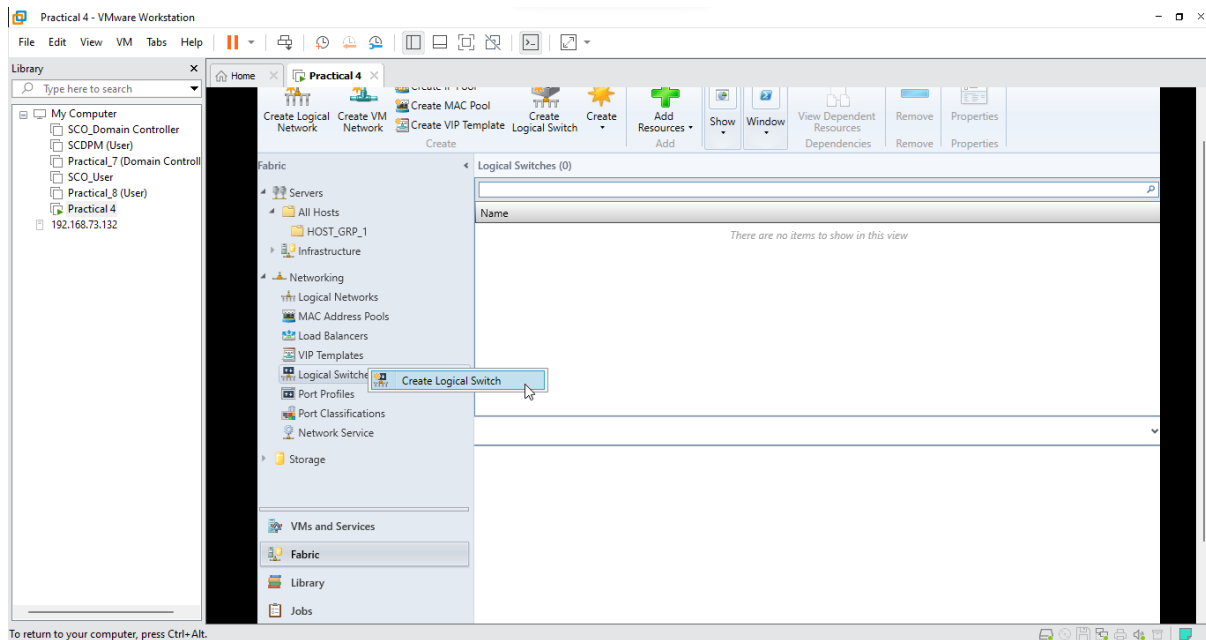


- **Click on Finish**

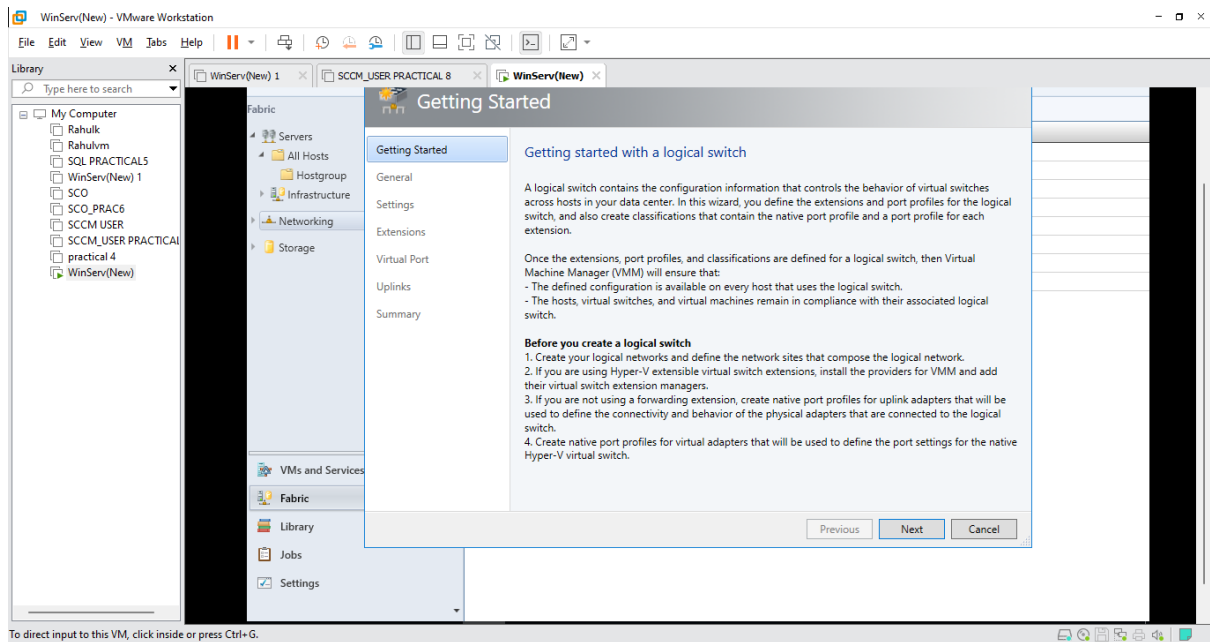


### Step 3: Creating Logical Switch

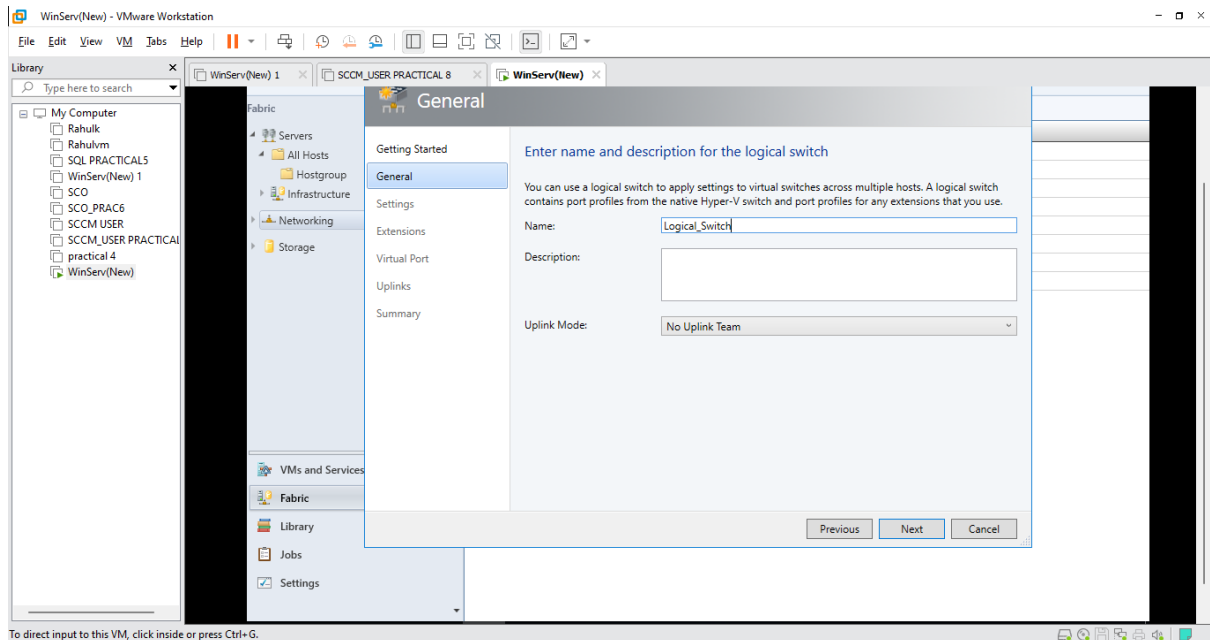
- **In Networking, Right-Click on Logical Switch and Select Create Logical Switch**



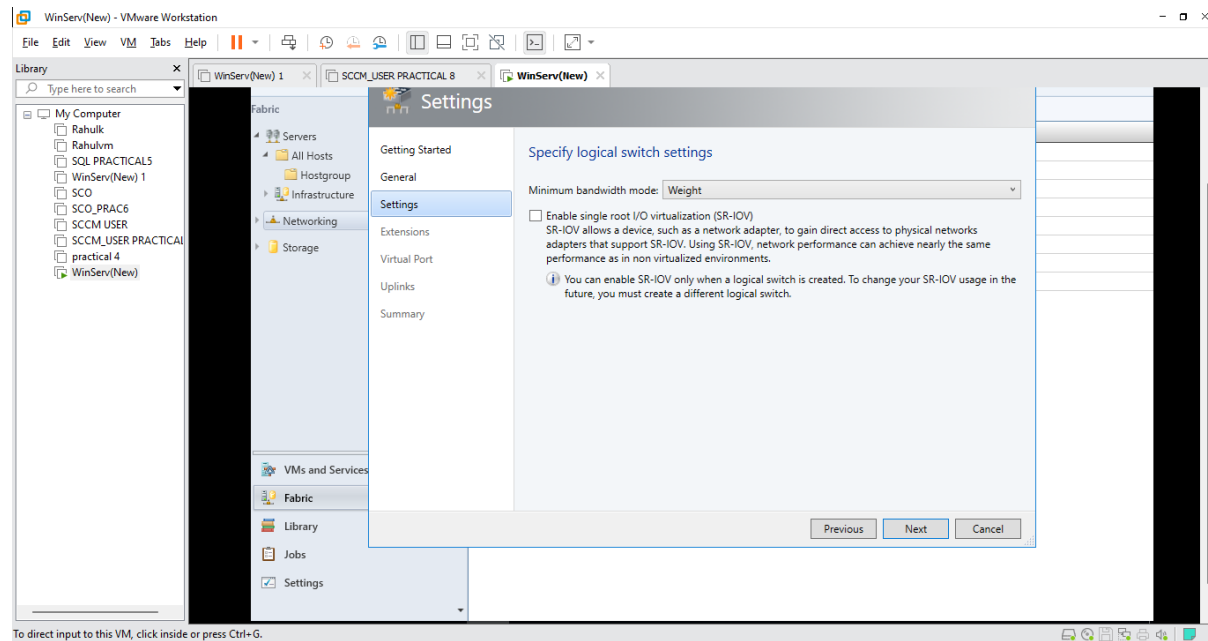
- **Click on Next**



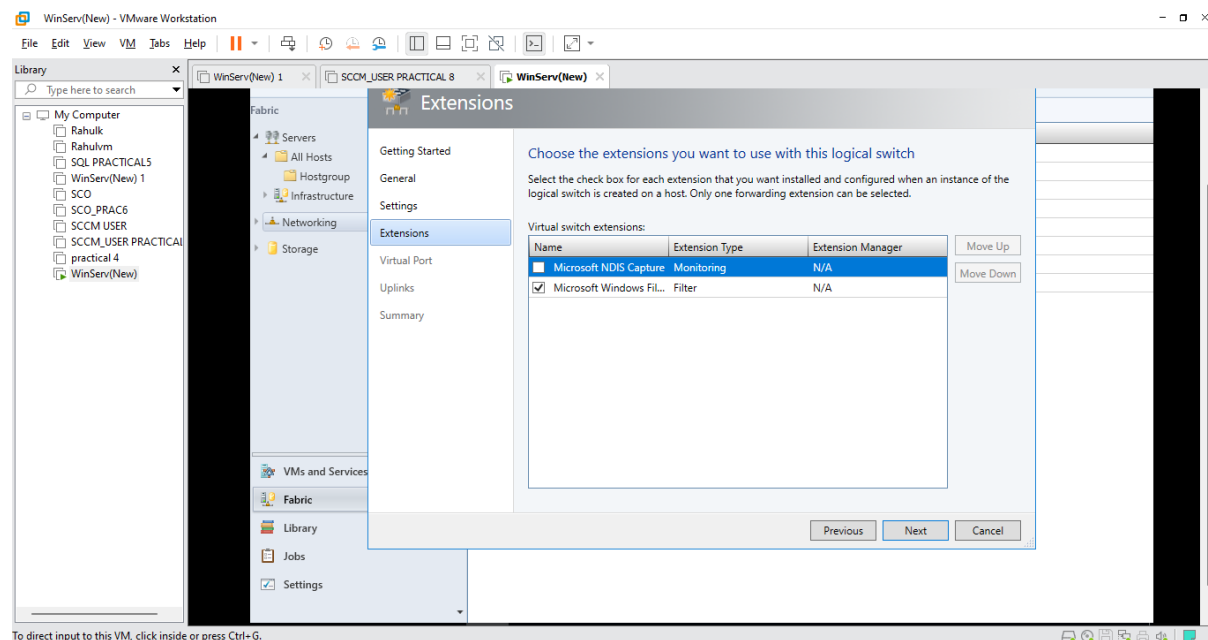
- **Name the Switch and Select No Uplink Team and Click on Next**



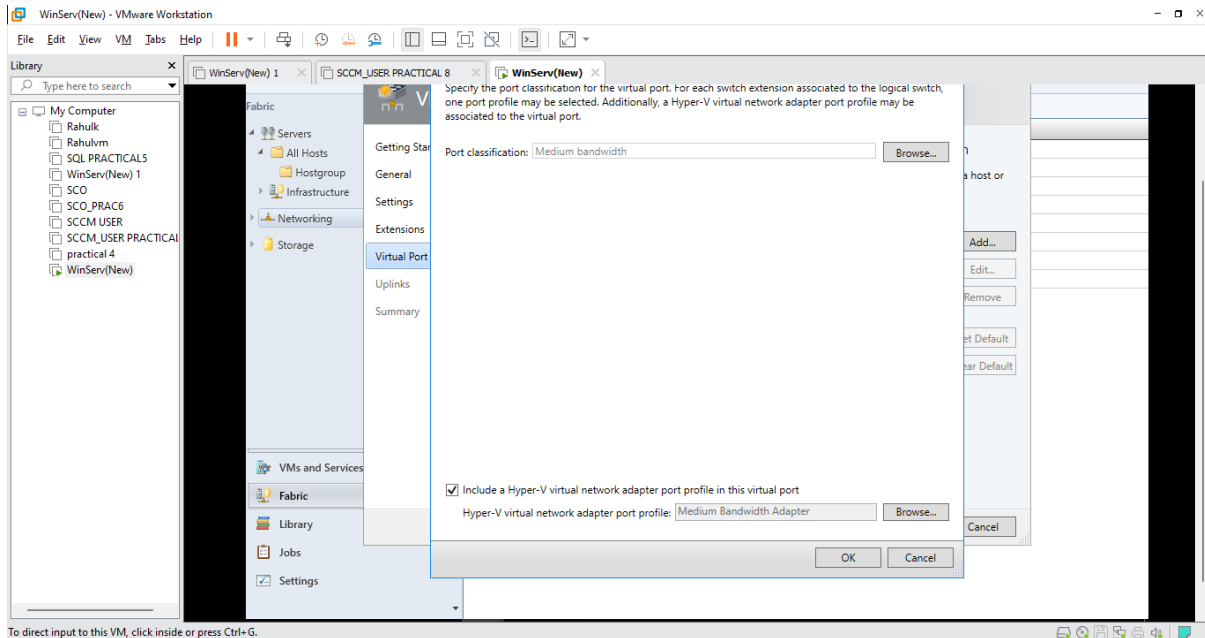
- **Keep the default Setting and Click on Next**



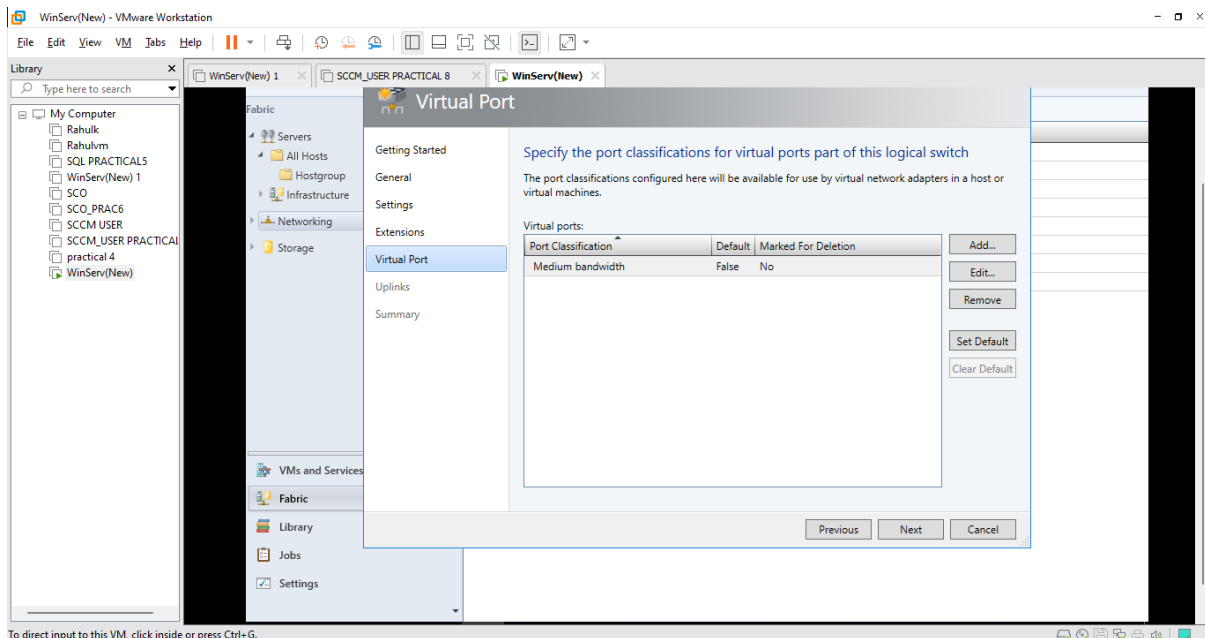
- **Keep the default Setting and Click on Next**



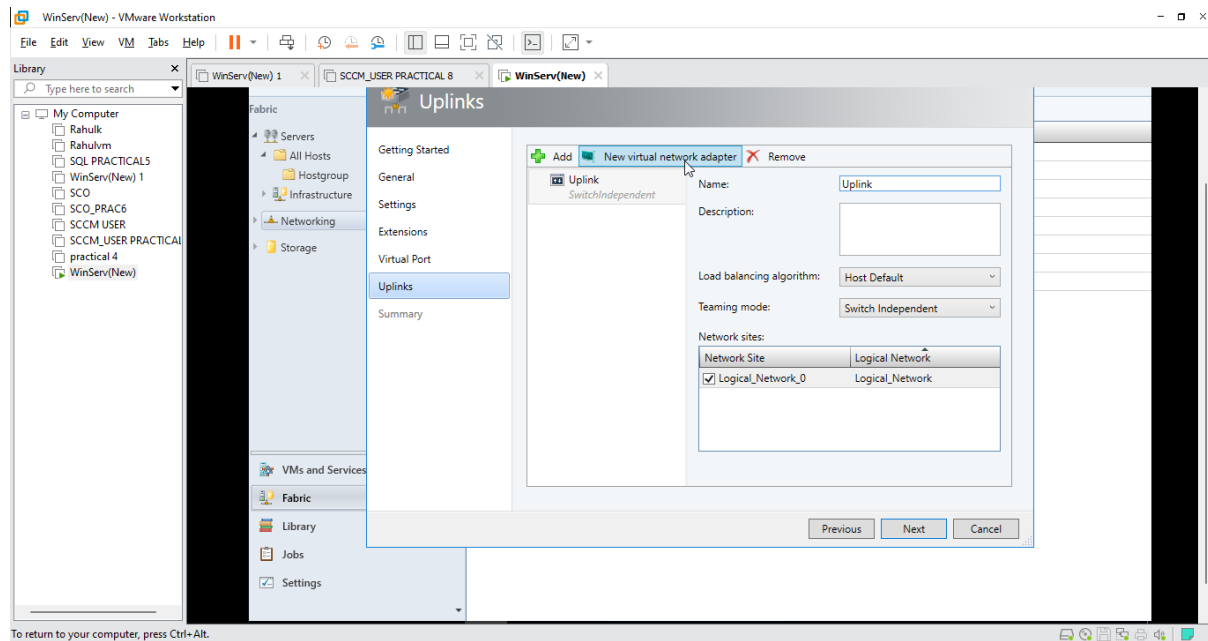
- Click on **Add**, in port classification Select **Medium bandwidth** and Select **Include a Hyper-V virtual network adapter port profile in this virtual port** and Select Medium bandwidth adapter and Click on **Ok**



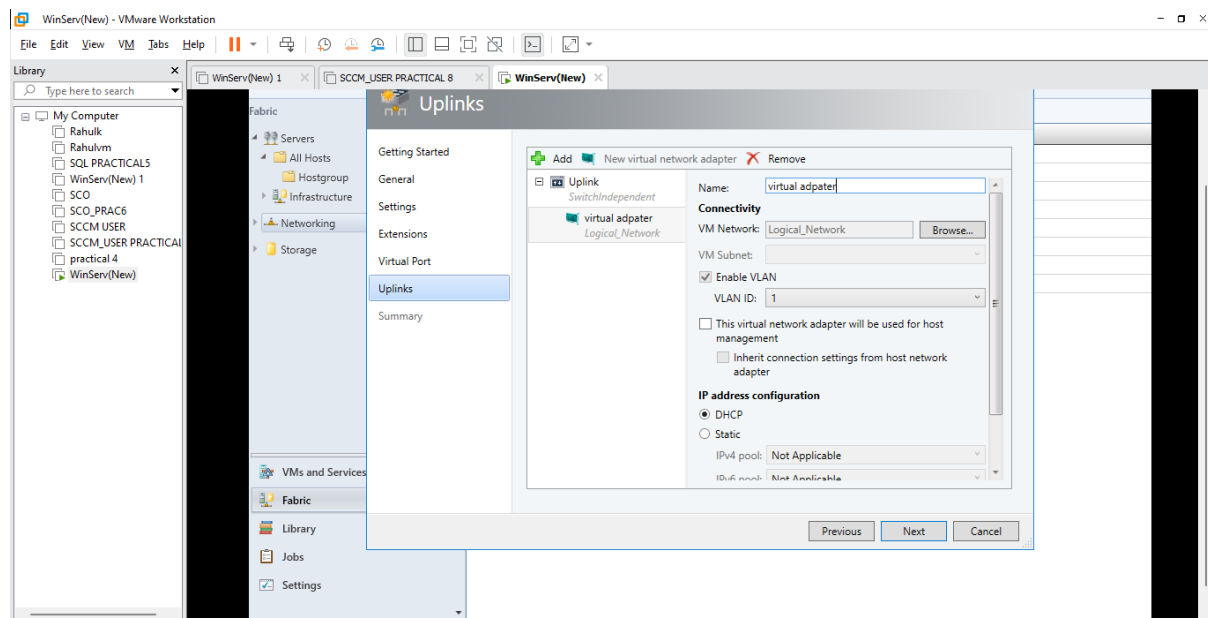
- Click on **Next**



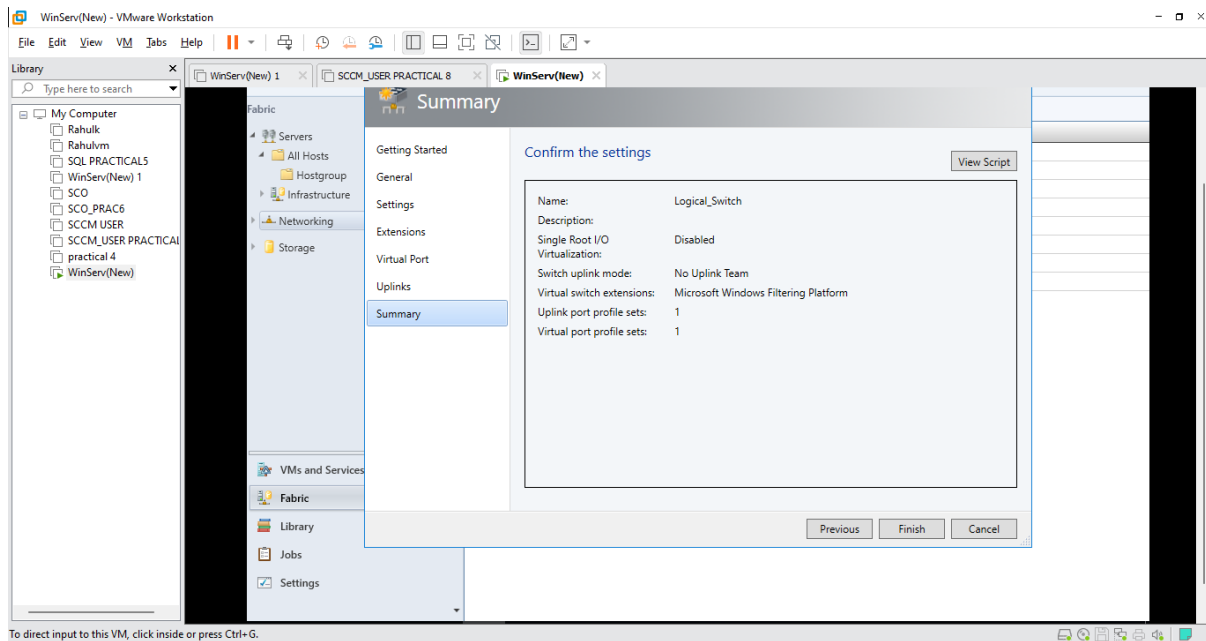
- Click on Add Give the Name (Here it is Uplink) and In Network sites: Check Logical\_Network\_0 and then Click on New virtual network adapter



- Name your virtual network adapter and Keep majority of the settings default and Select Medium bandwidth in Port Profile and Click Next

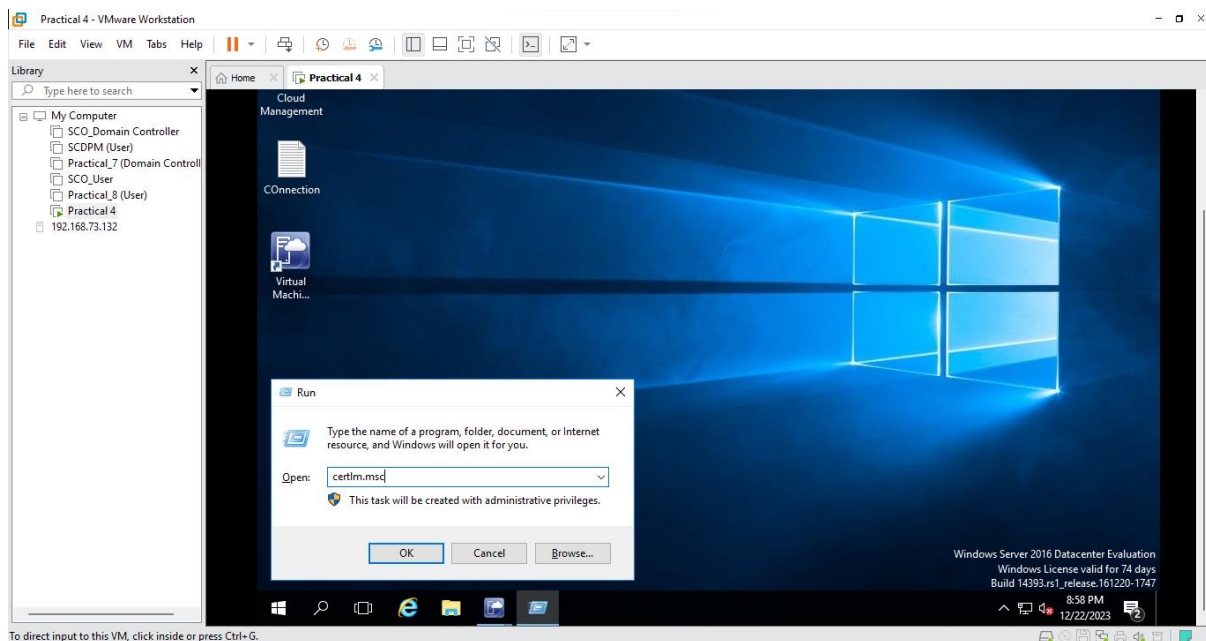


- **Click on Finish**



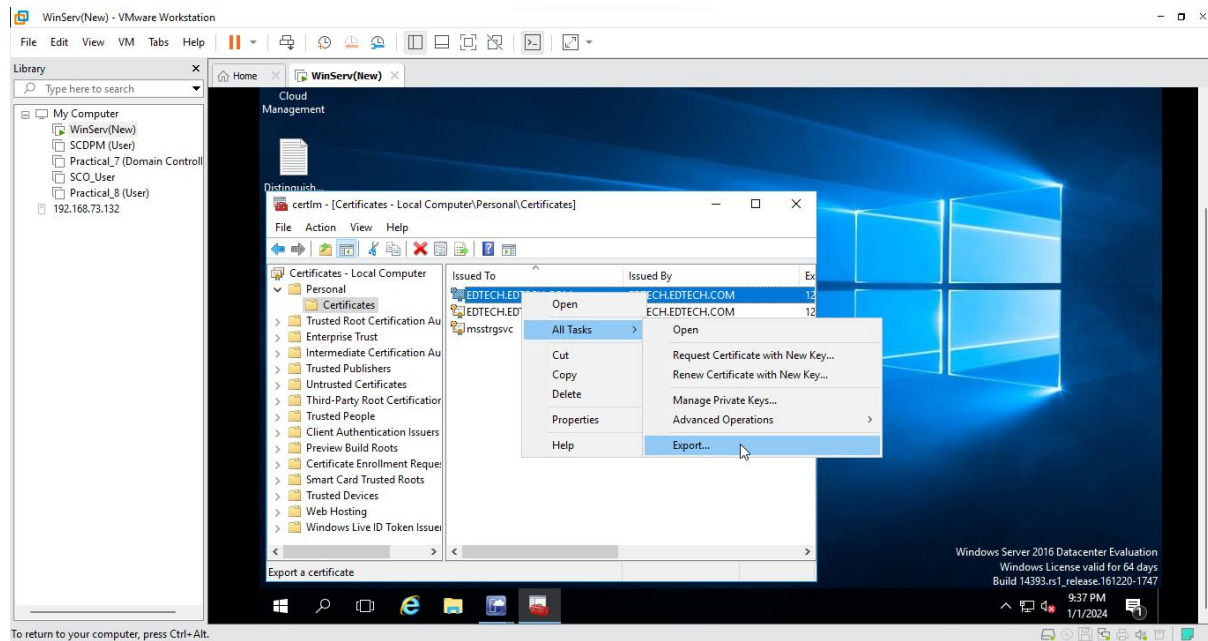
#### Step 4: Setting up the Security Certificate

- In your Windows Run type in **certlm.msc** and **Click On Ok**

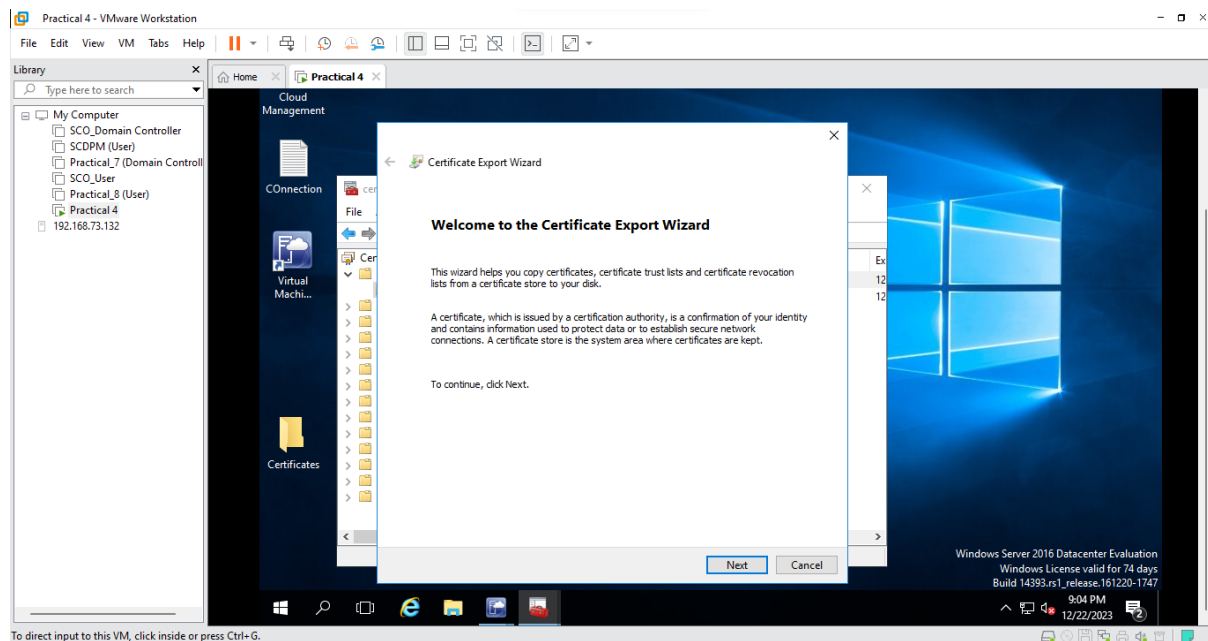




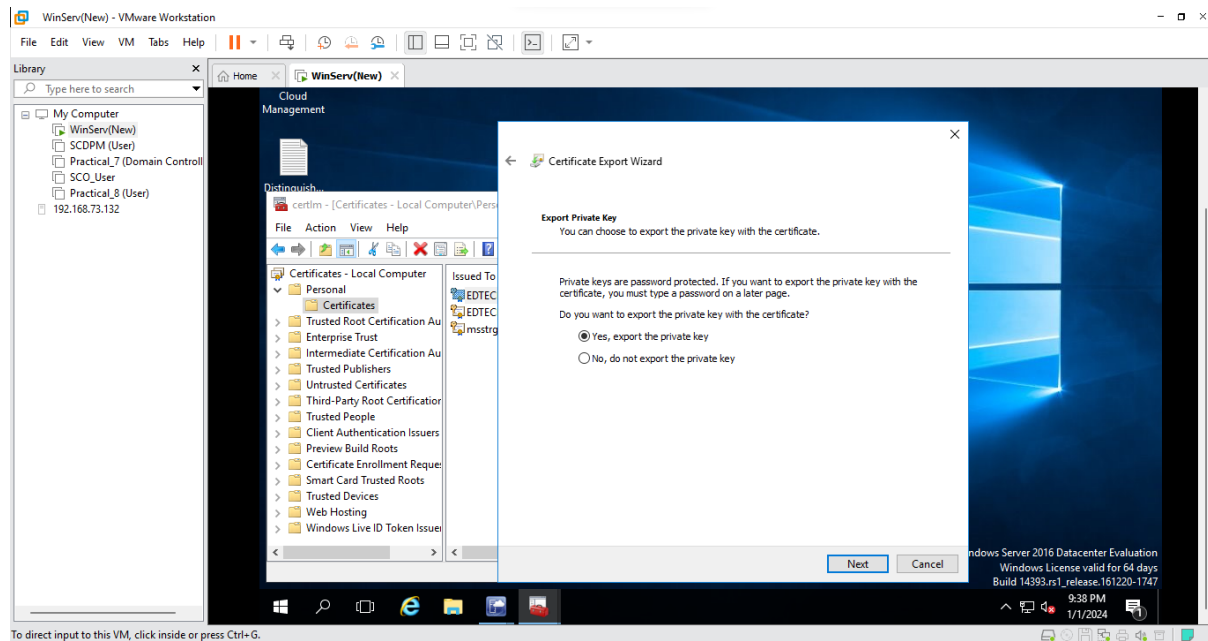
- Now Click on **Personal** and Select **Certificate** and within **Certificate**, Right-Click on **EDTECH.EDTECH.COM** and **msstrgsvc** and Click **All tasks** and Click **Export**



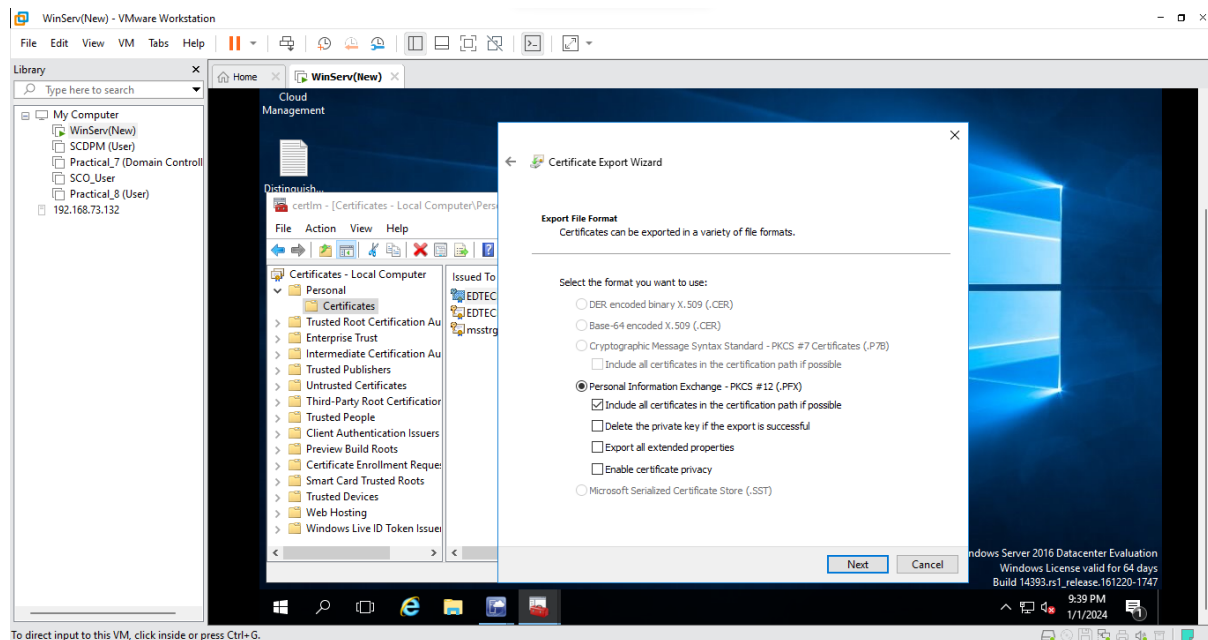
- Click Next



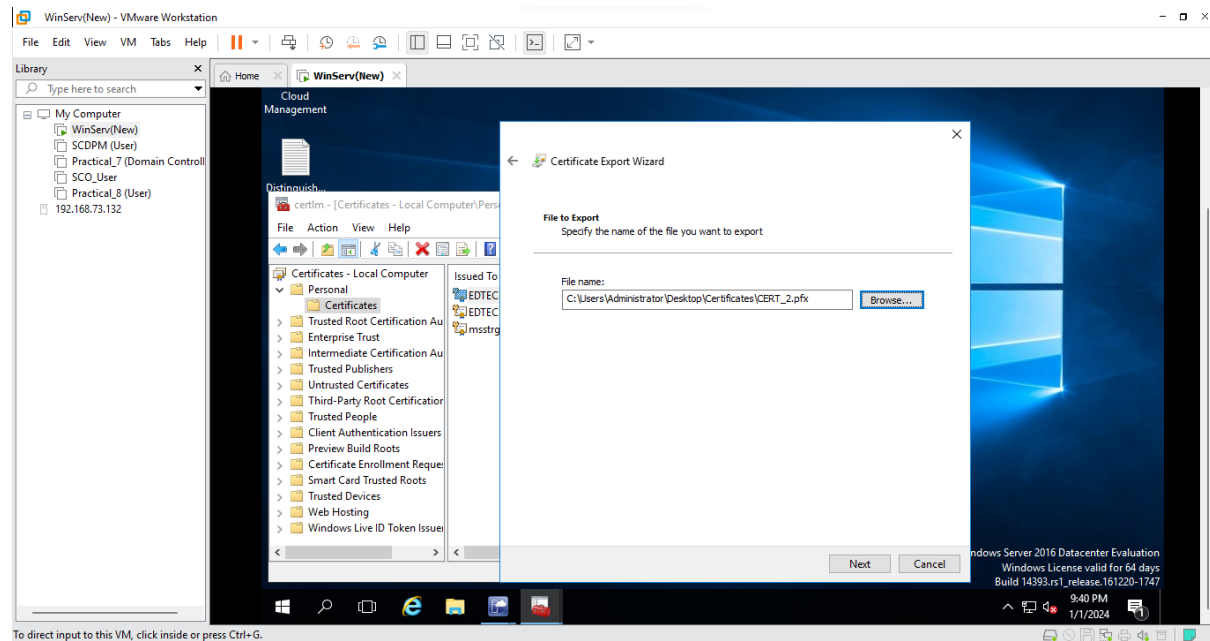
- **Select Yes, export the private key and Click on Next**



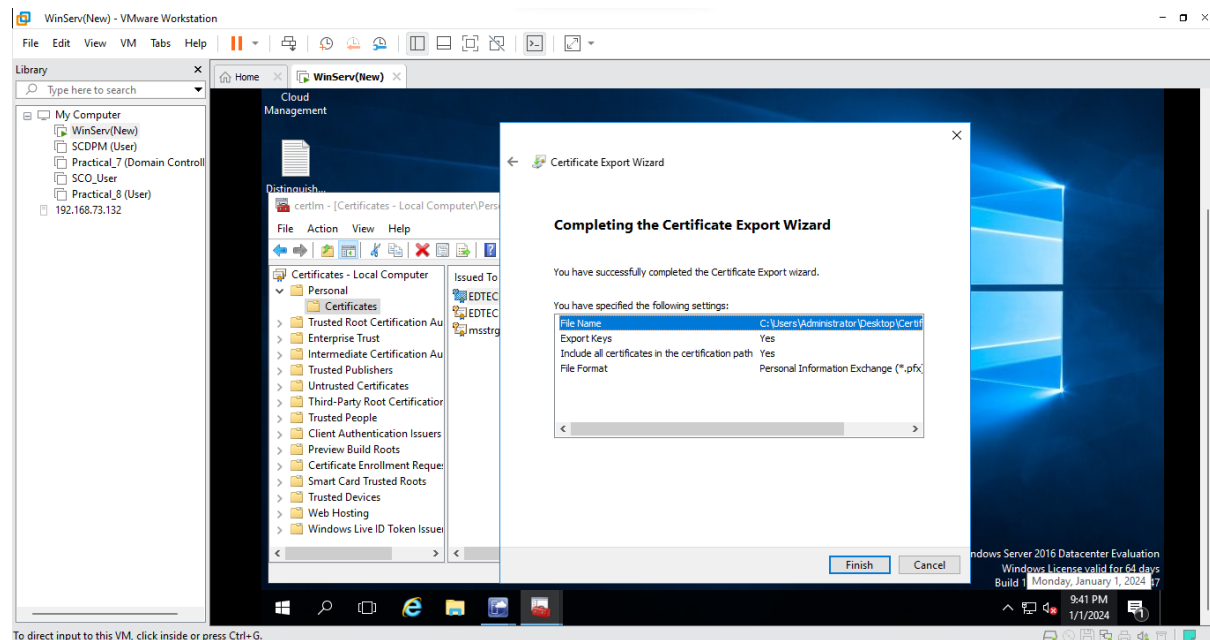
- **Keep default values and Click on Next**



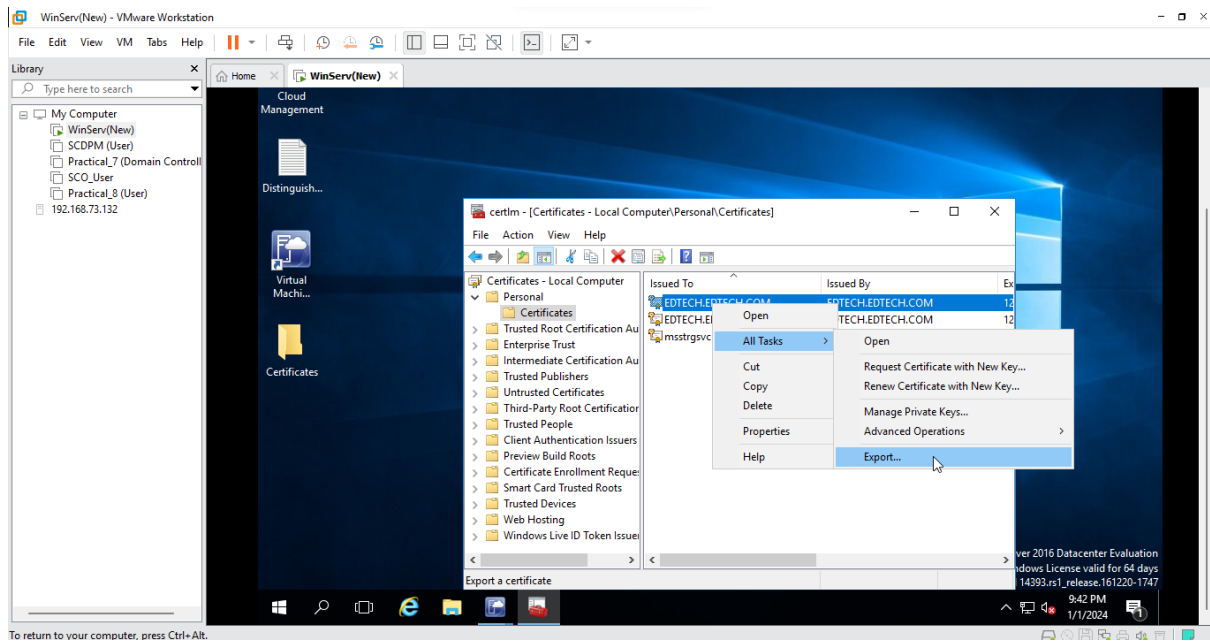
- Give it a file name and **Click on Next**



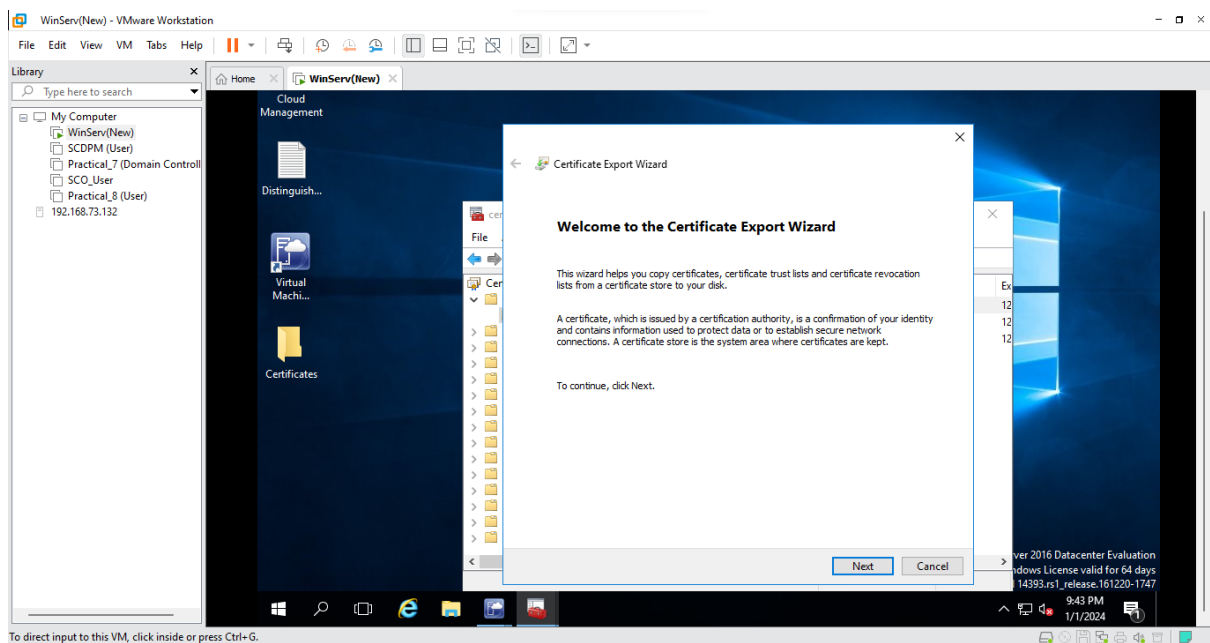
- **Click on Finish**



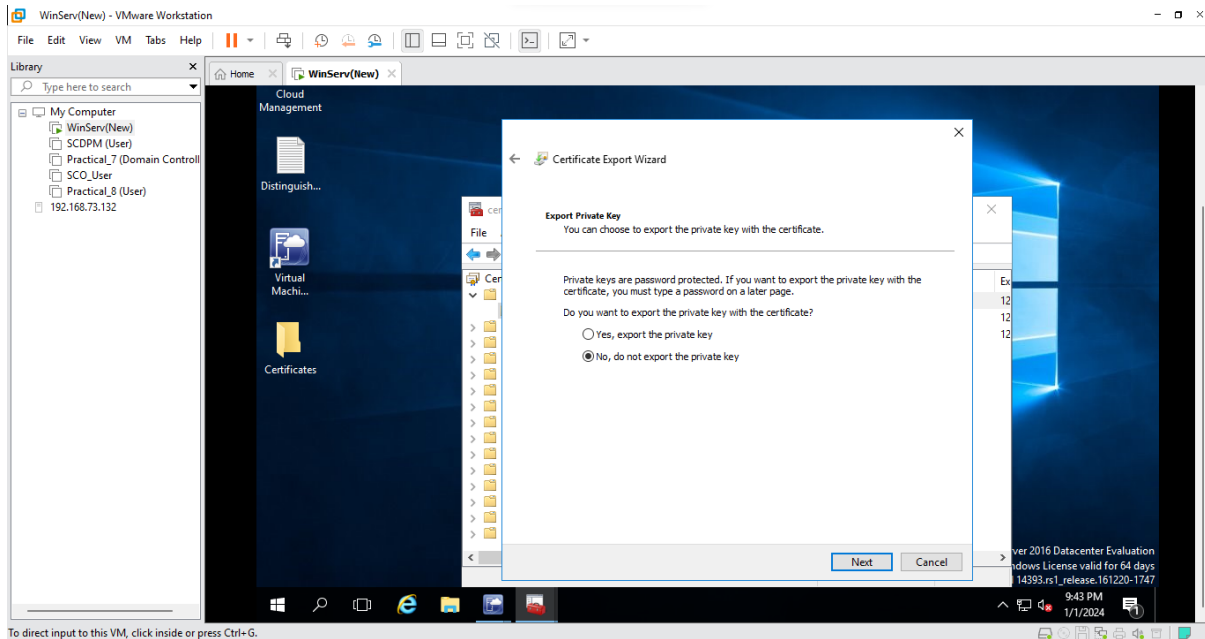
- Now Right-Click on the certificate again and Click on Export



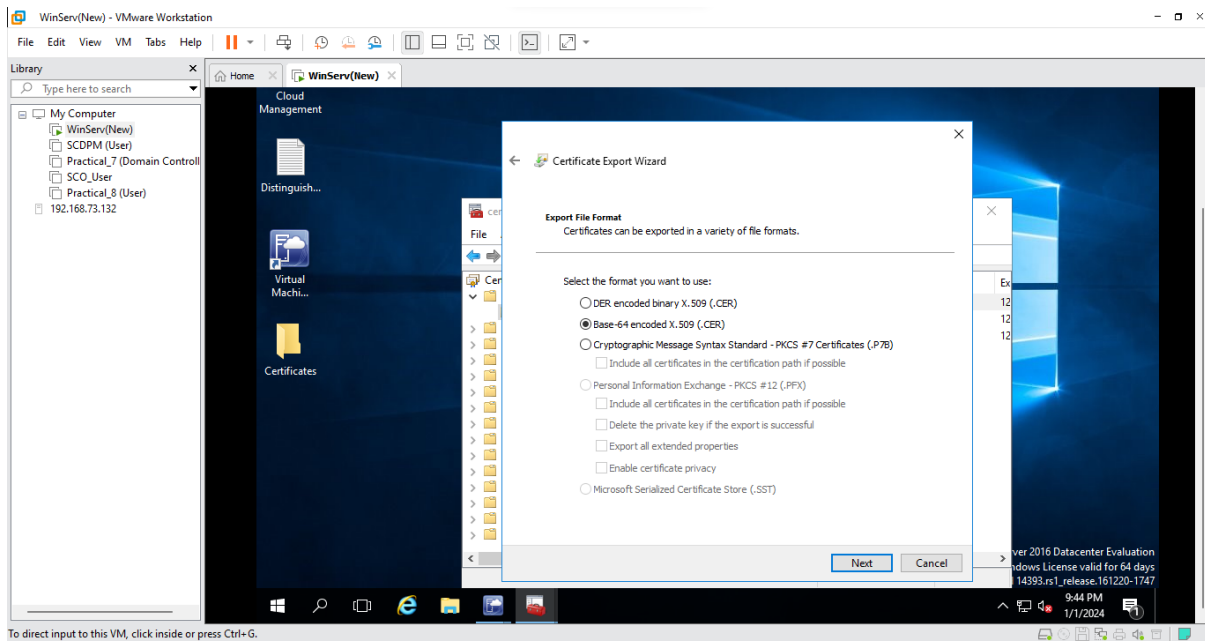
- Click On Next



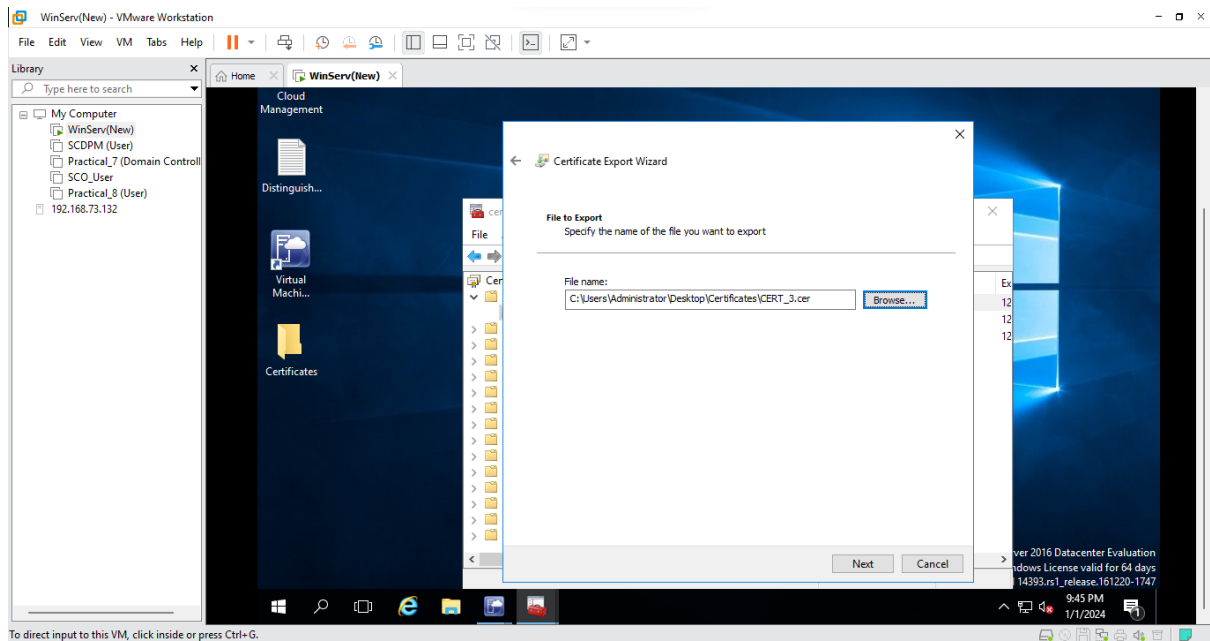
- **Select No, do not export the private key and Click On Next**



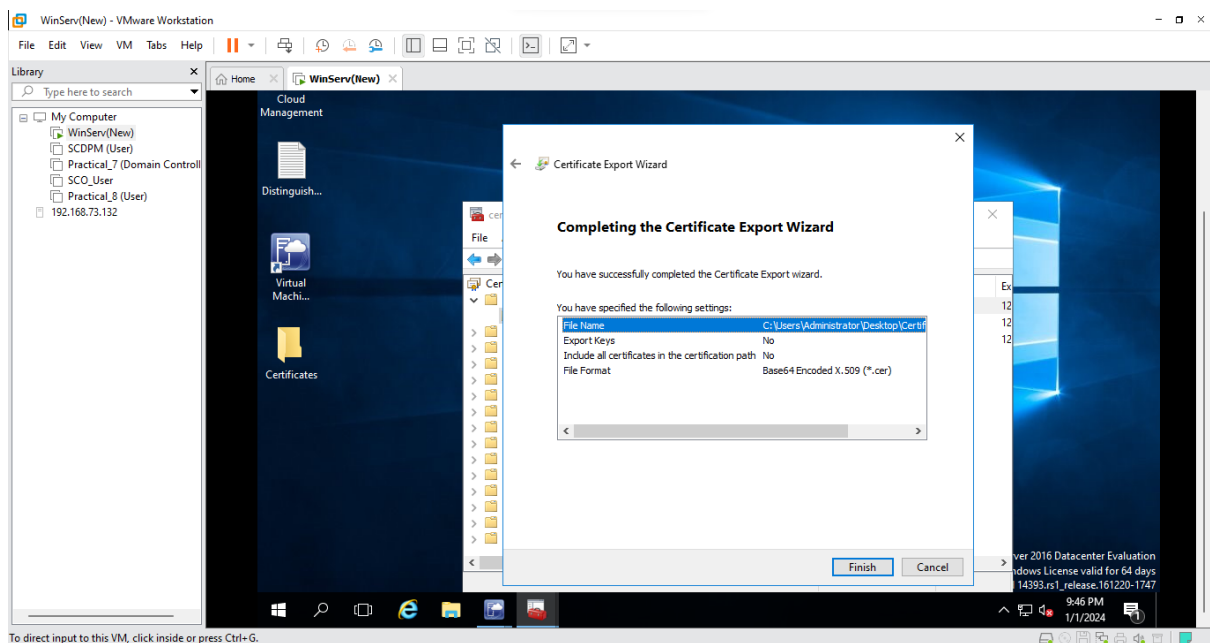
- **Select Base-64-encoded X.509 (.CER) and Click On Next**



- Give it a path and **Click On Next**

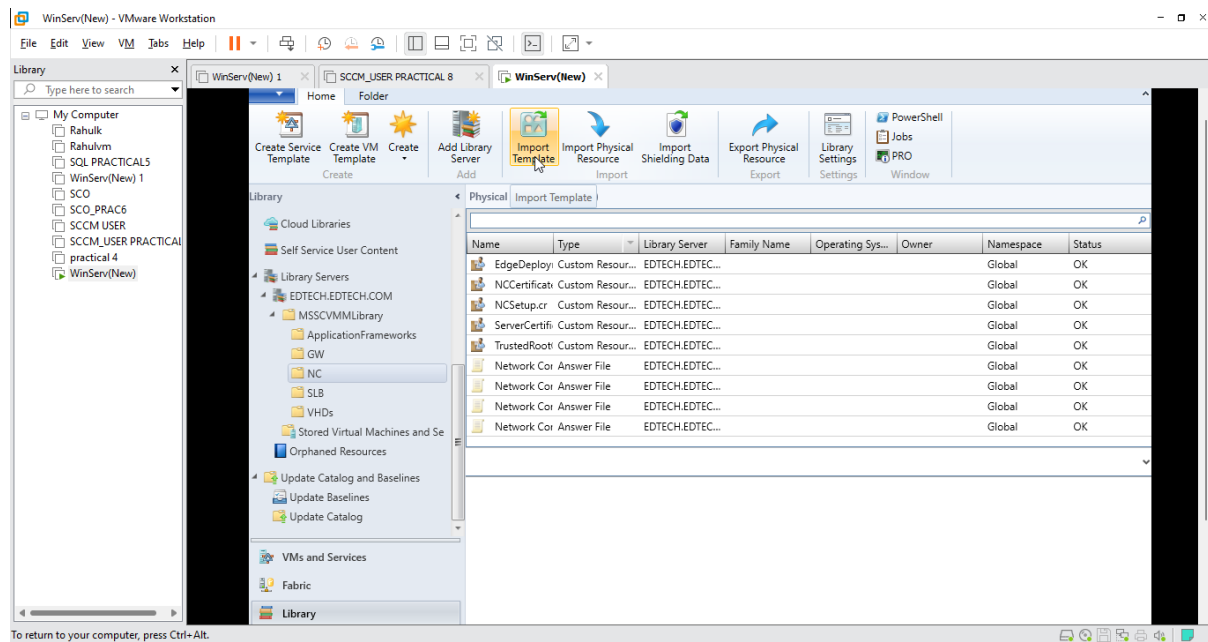


- **Click Finish**

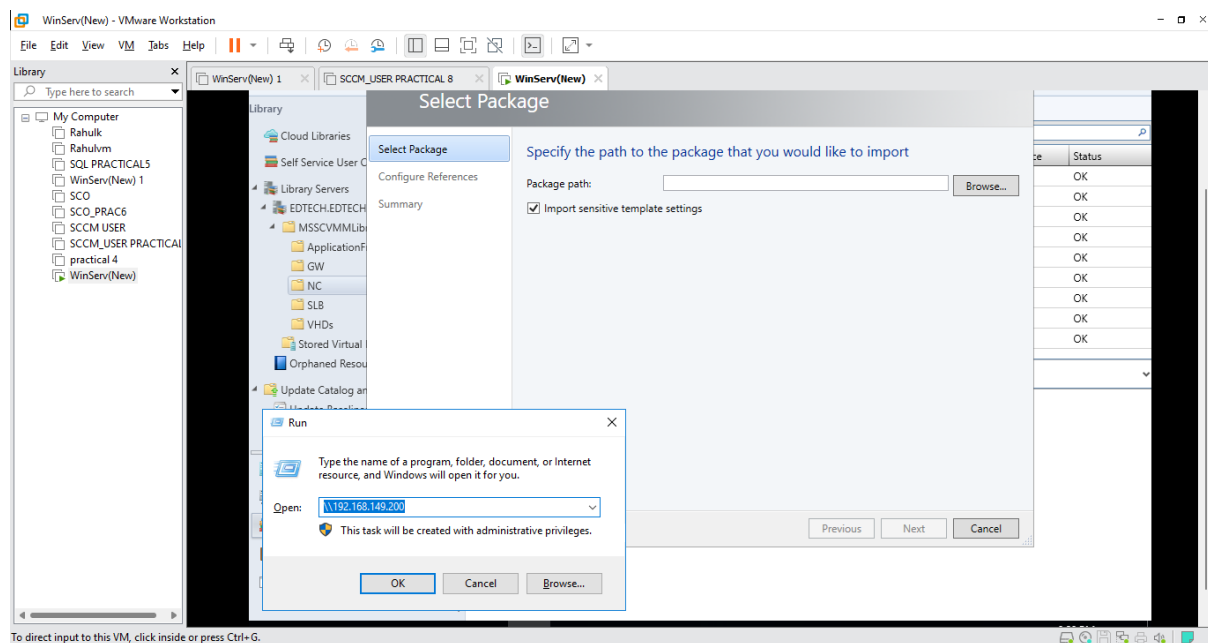


## Step 5: Set up Service Template

- In SCVMM Select the **Library Tab** and Click on **Import Template**

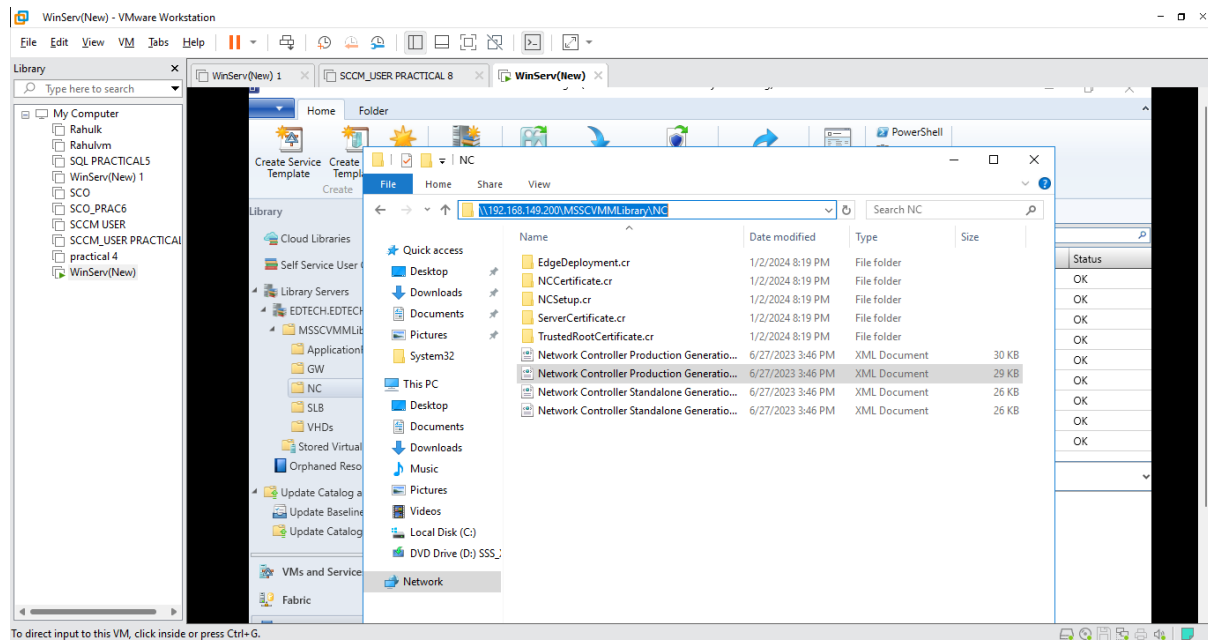


- In Select Package Open Run Window and then Type IP address of the WinServ(New) (Here it is \\192.168.149.200) Click on Ok

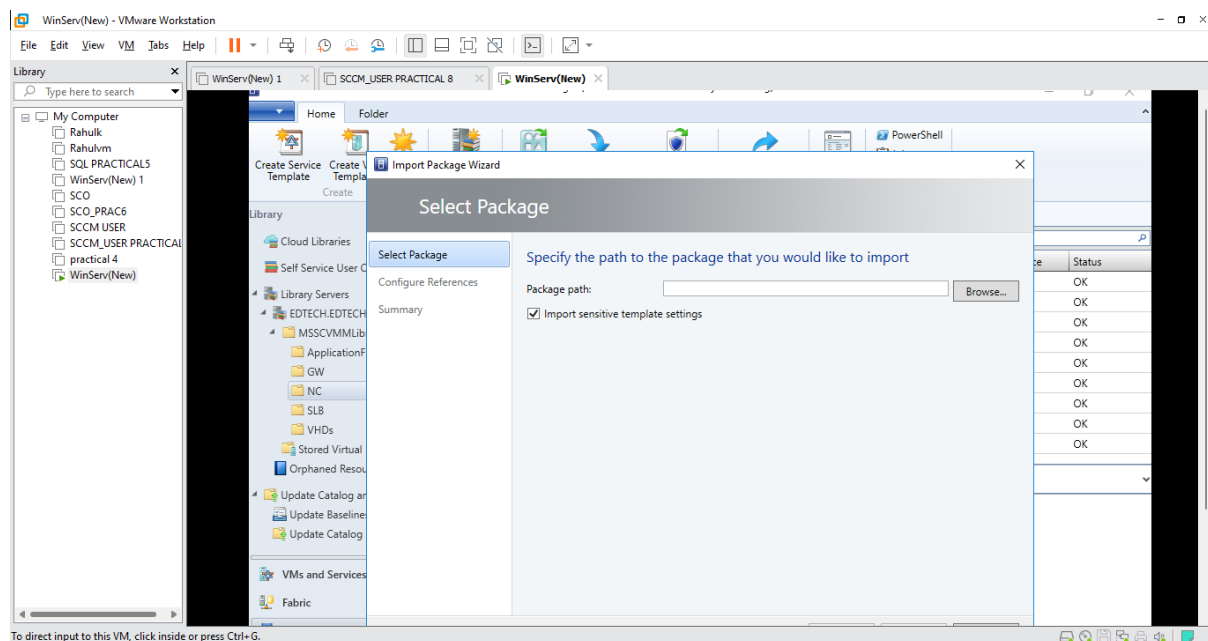




- Within MSSCMMLLibrary Select NC Folder and then Copy the Path

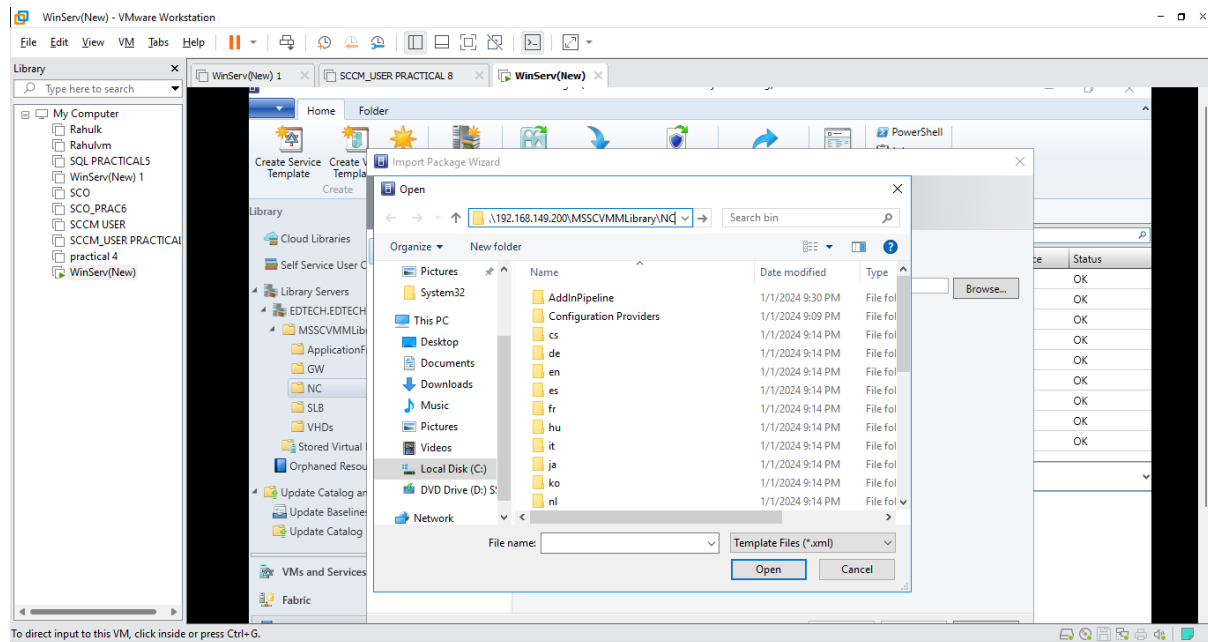


- Now again Go to the Select Package window Click on Browse

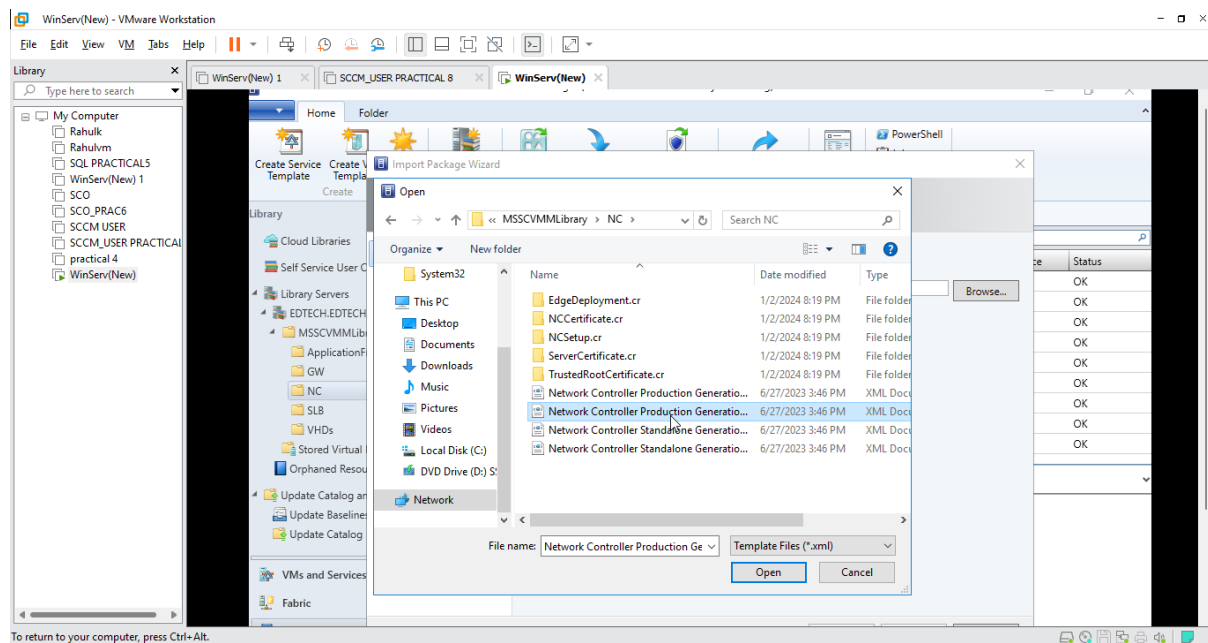




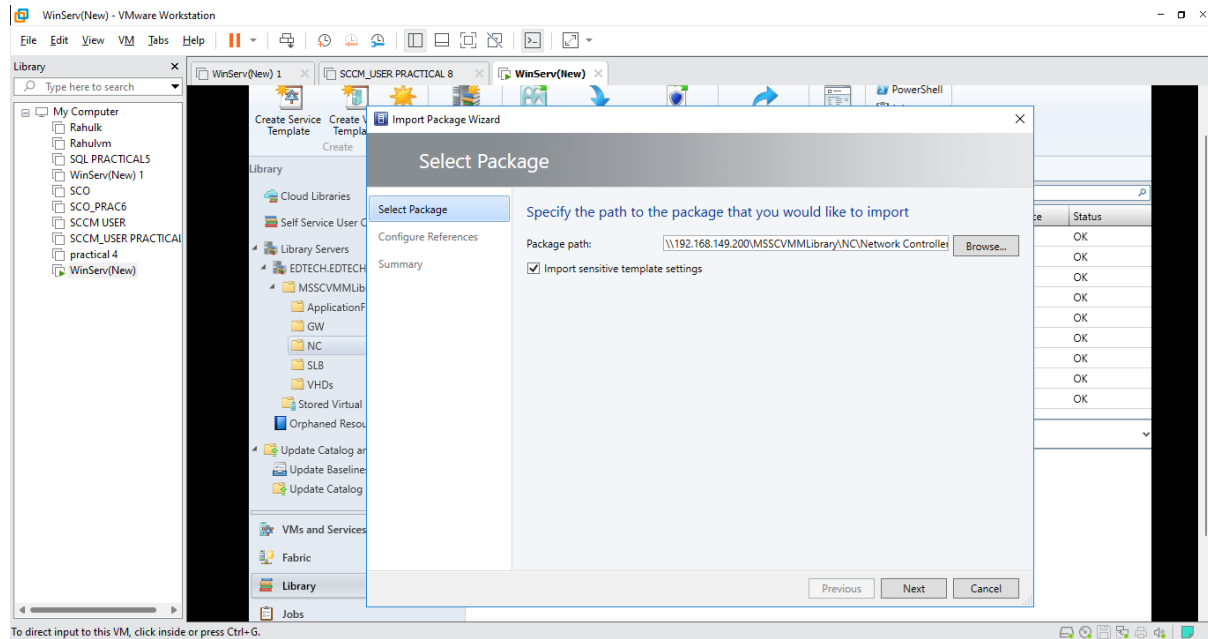
- Now paste the path for NC and then **Click on Enter**



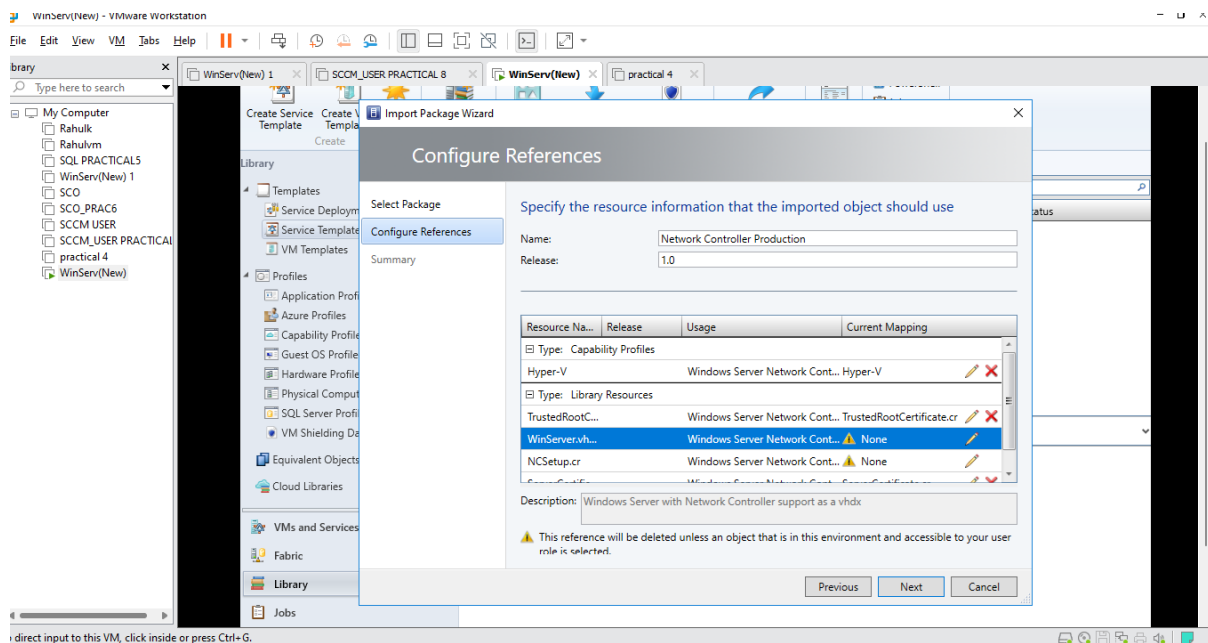
- Within NC Folder Select **Network Controller Production Generation 2** and then **Click on Ok**



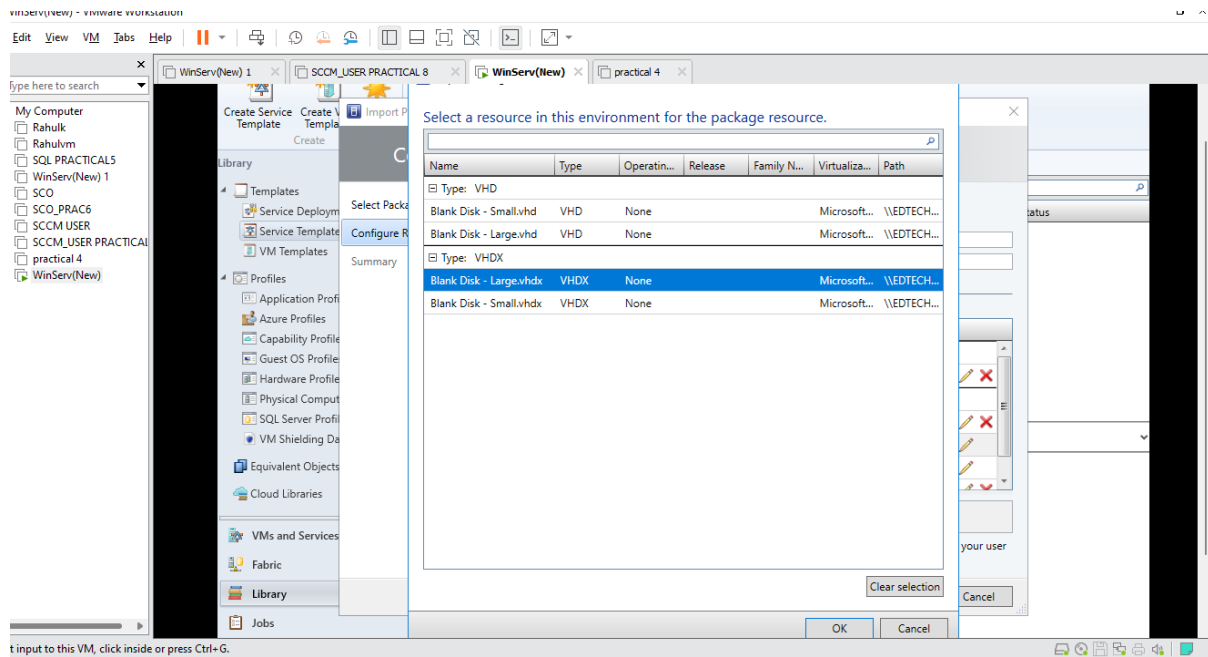
- **Click on Next**



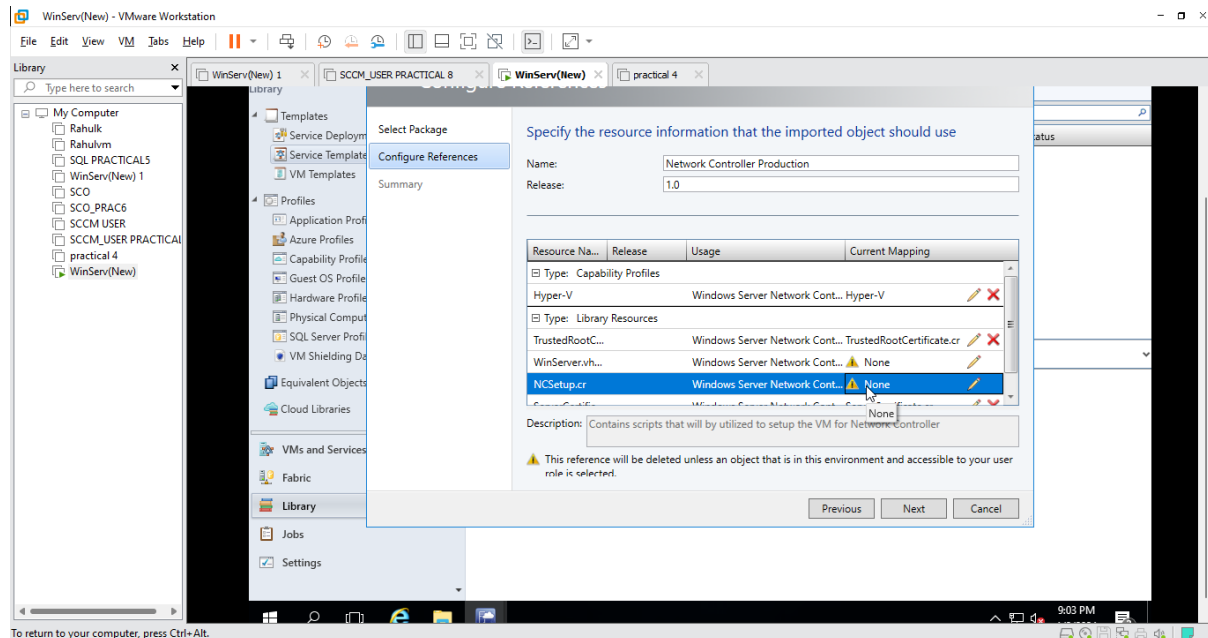
- **Within Configuration References Click on WinServer.vh and then Click on Edit**



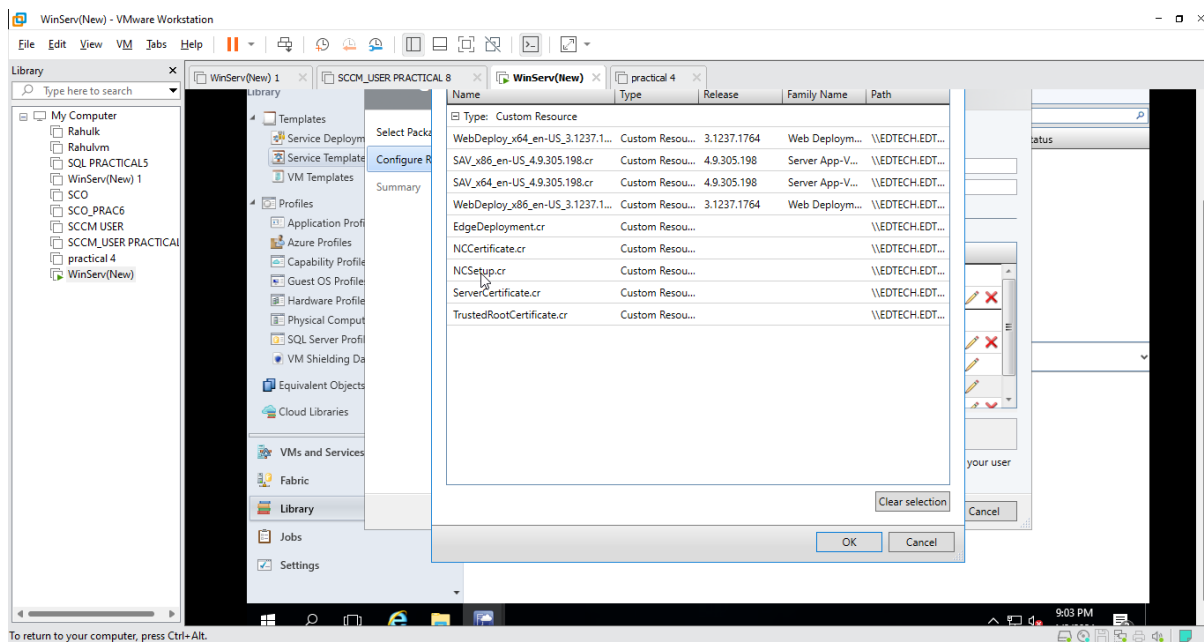
- **Within Type: VHDX Select Blank Disk – Large.vhdx and Click on OK**



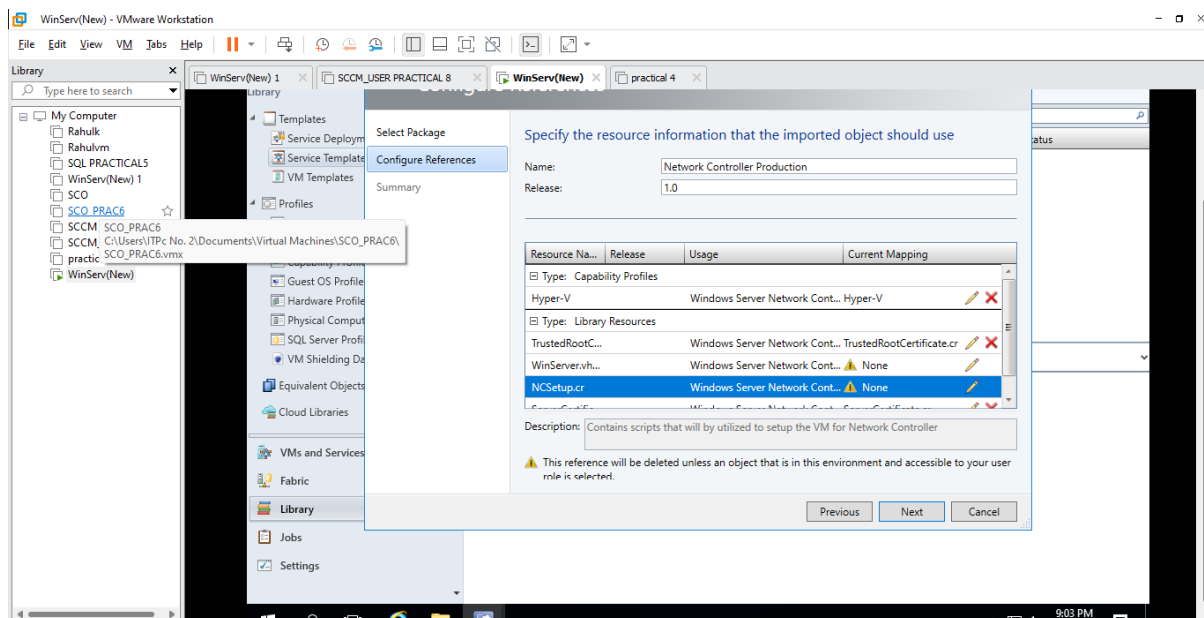
- **Now again within Configure Reference Click on NCSetup.cr and then Click on Edit**



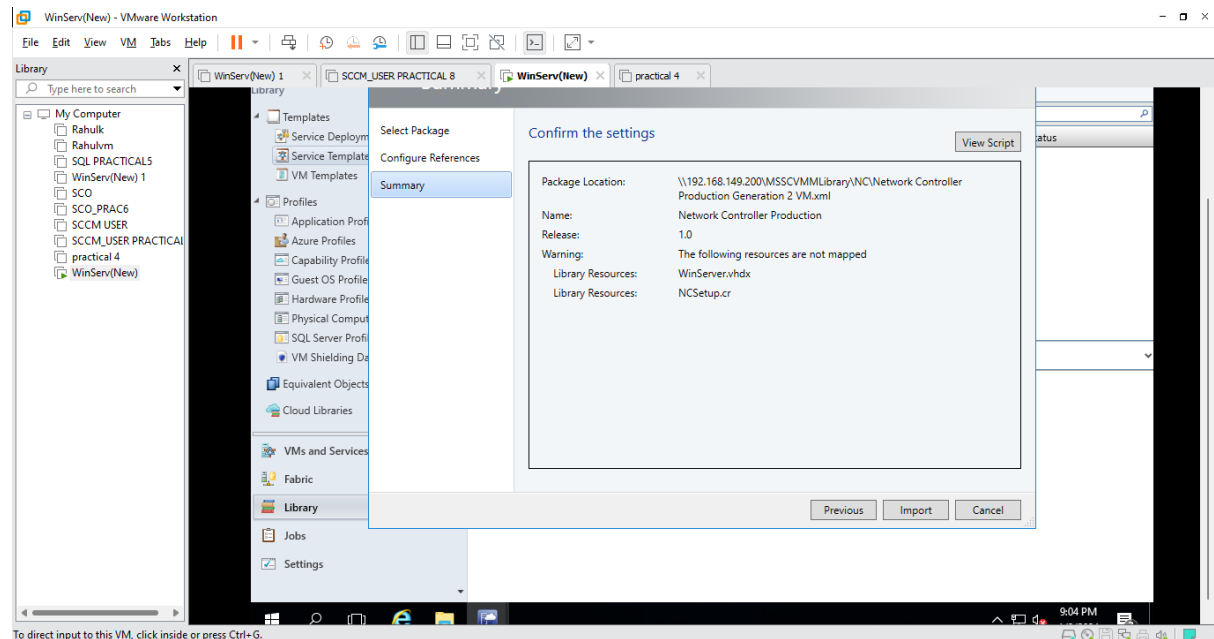
- Select NCSetup.cr and Click on Ok



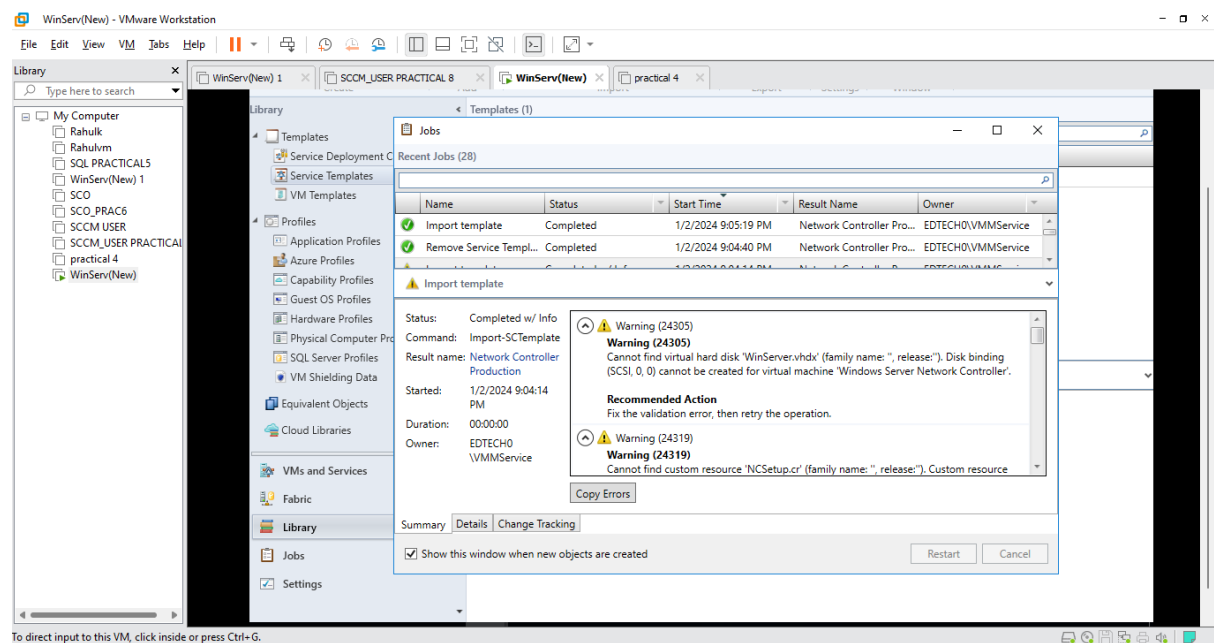
- Click on Next



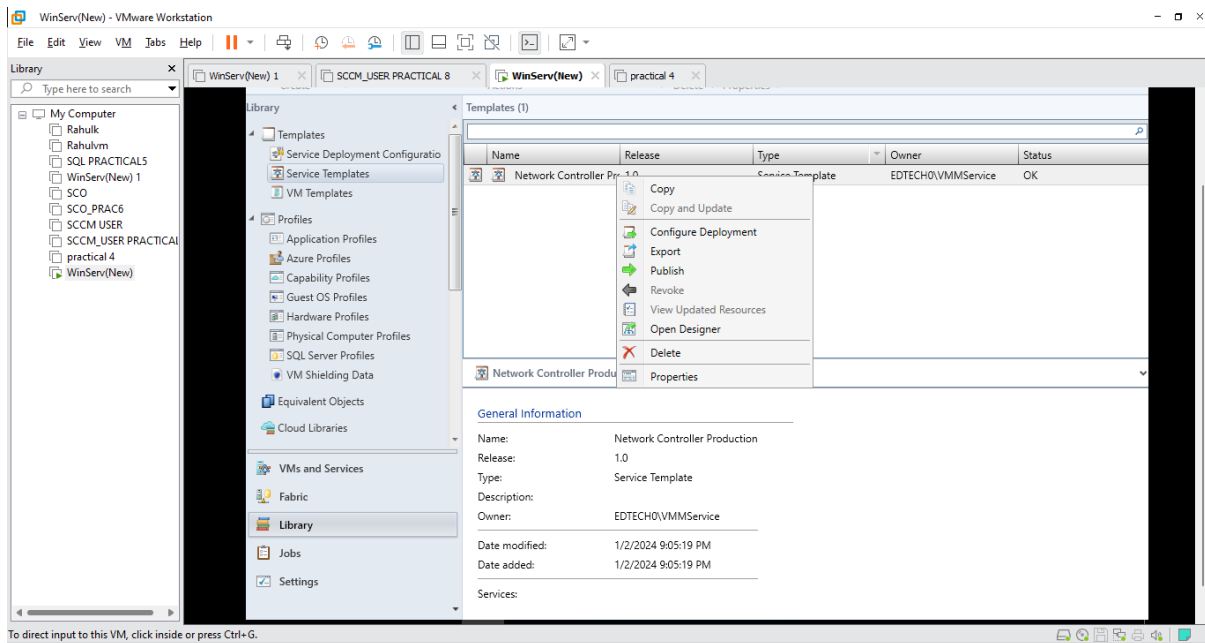
- Click on import



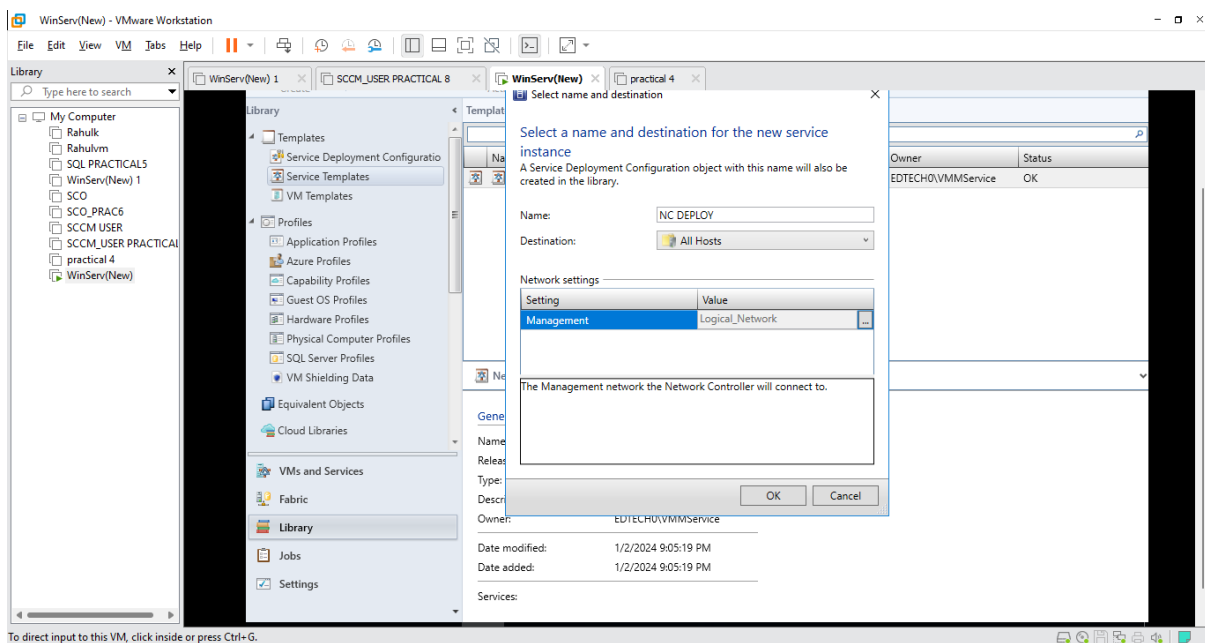
- Here we successfully import template



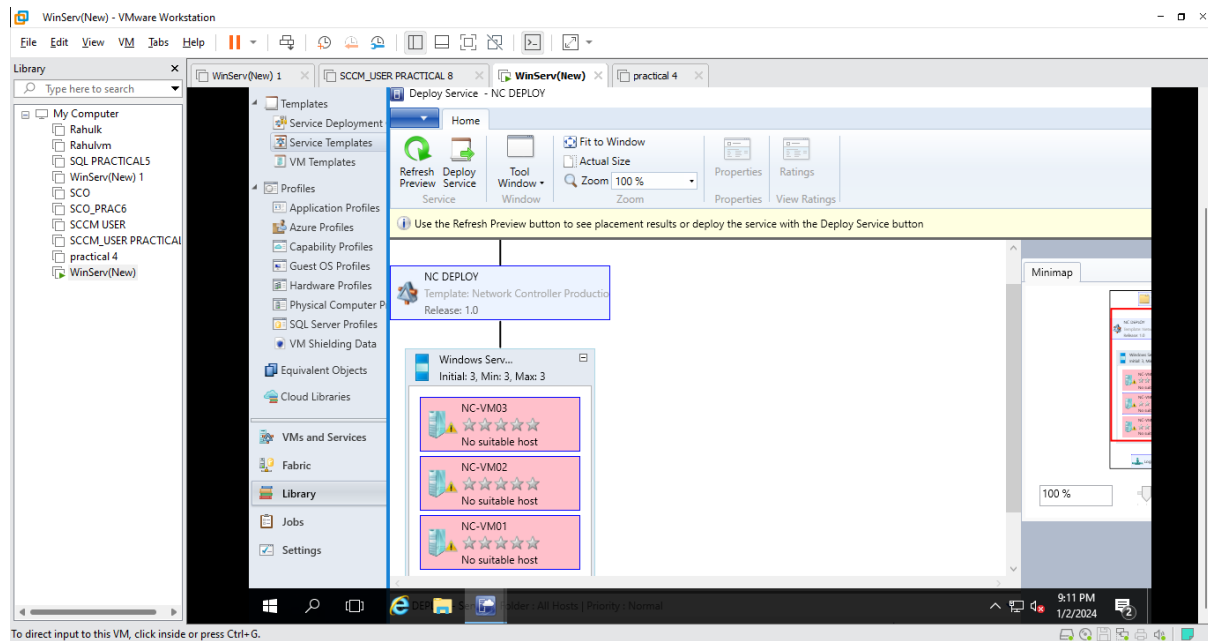
- Within Service Templates **Right-Click on Network Controller Production 1.0** and then **Click on Configure Deployment**



- Give it a name and Select you management as your **Logical Network** and Click Ok.



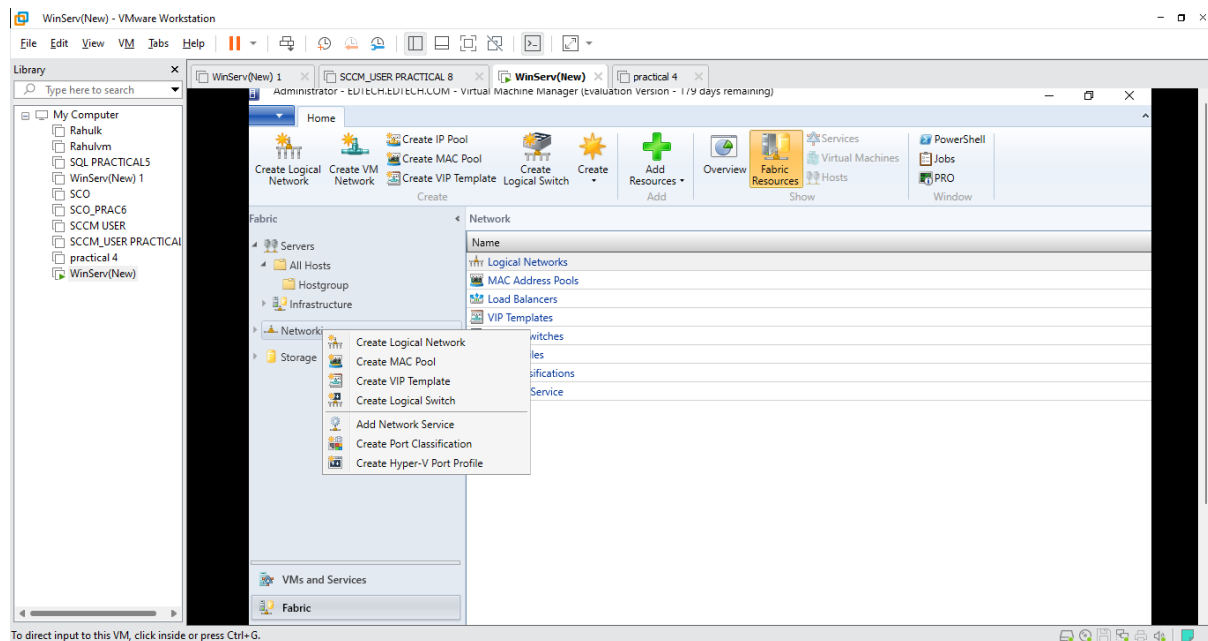
- This is the deployment of the Network Controller Service Template



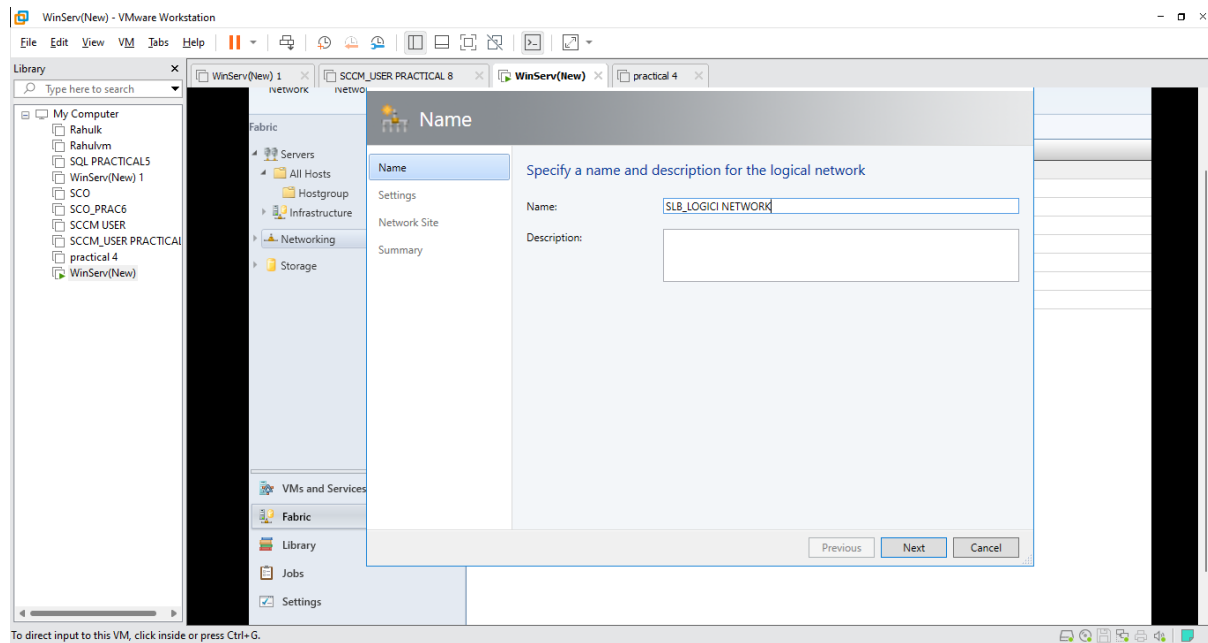
## 4-B: Deploying Software Load Balancer

### Step 1: Create a Logical Network

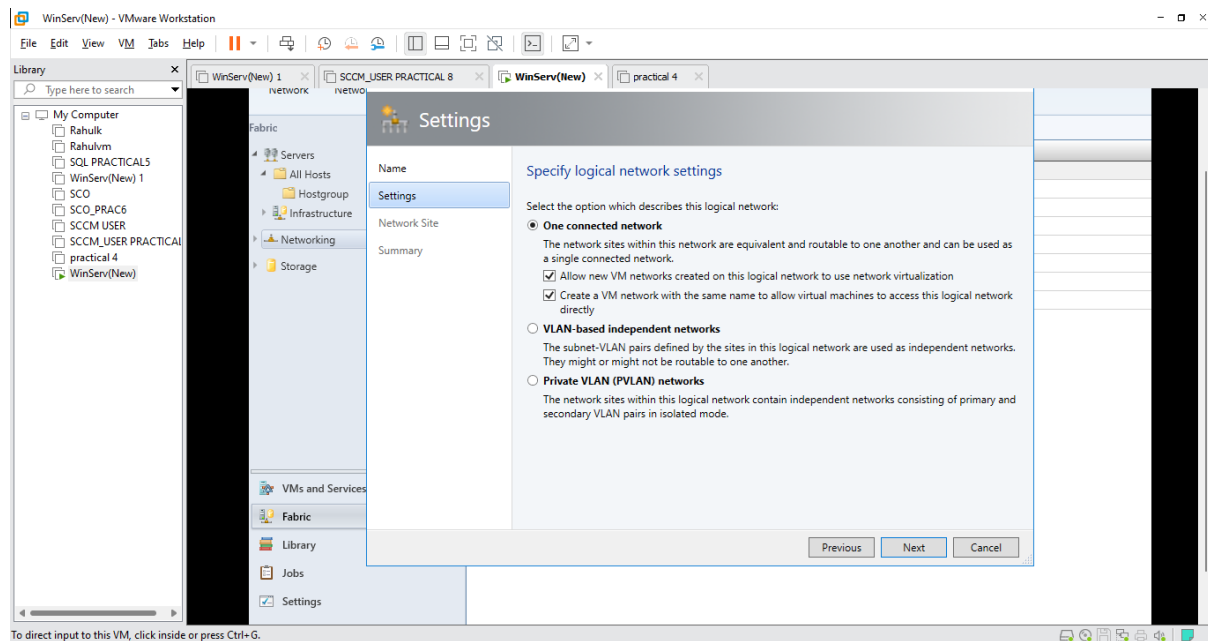
- Create a new logical network by Selecting the **Fabric Tab** in SCVMM and **Right-Click on Logical Networks** and Select **Create a Logical Network**



- Give Name for your Logical Network (Here it is: **SLB\_LOGICAL NETWORK**)  
And then **Click on Next**

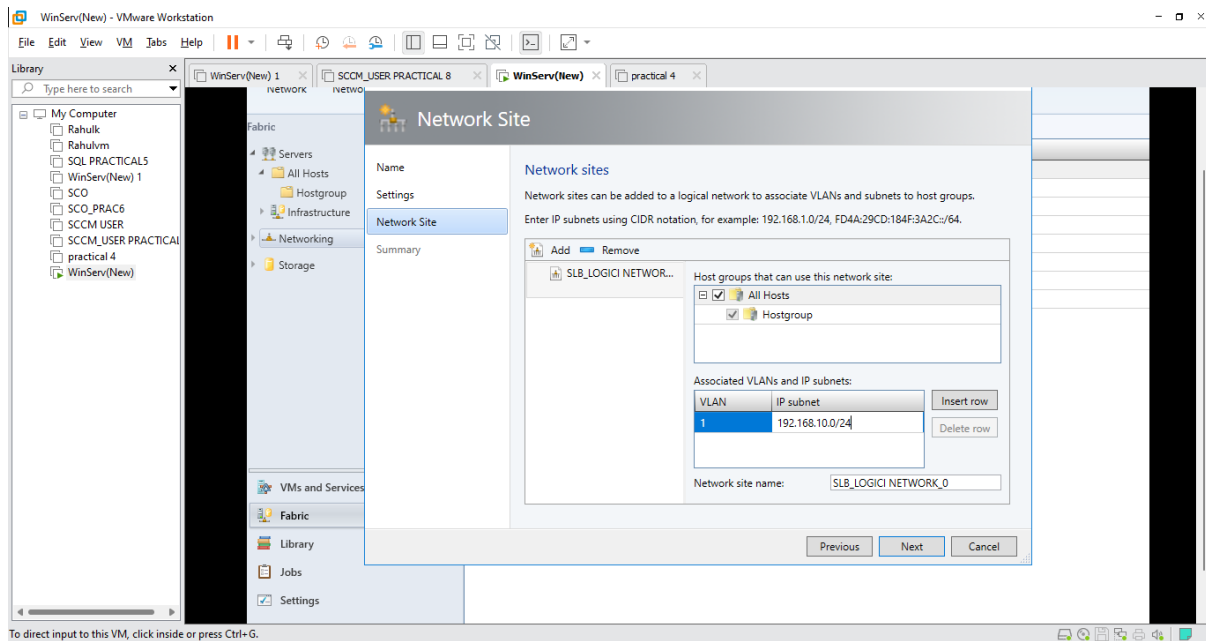


- Select **One-connected network** and Tick both **Allow new VM networks created on the logical network to use network virtualisation** and **Click on Next**

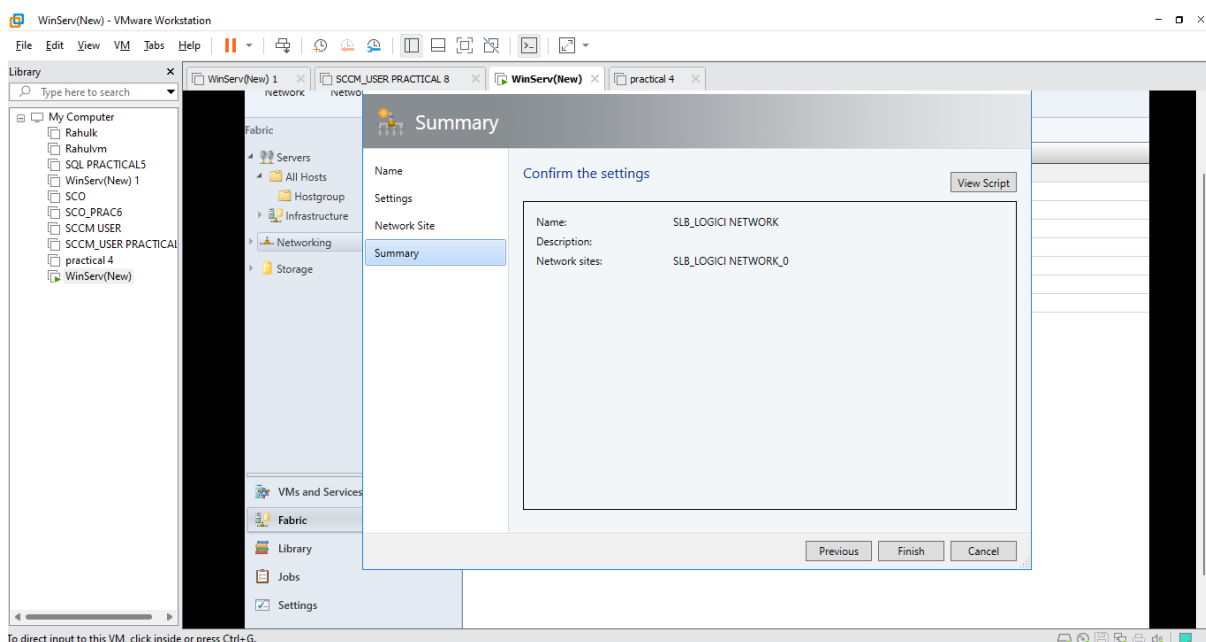




- Within Network Site Click on Add Select All Hosts and then Click on Insert row Enter VLAN and IP address Here it is: **192.168.10.0/24** and then Click on Next

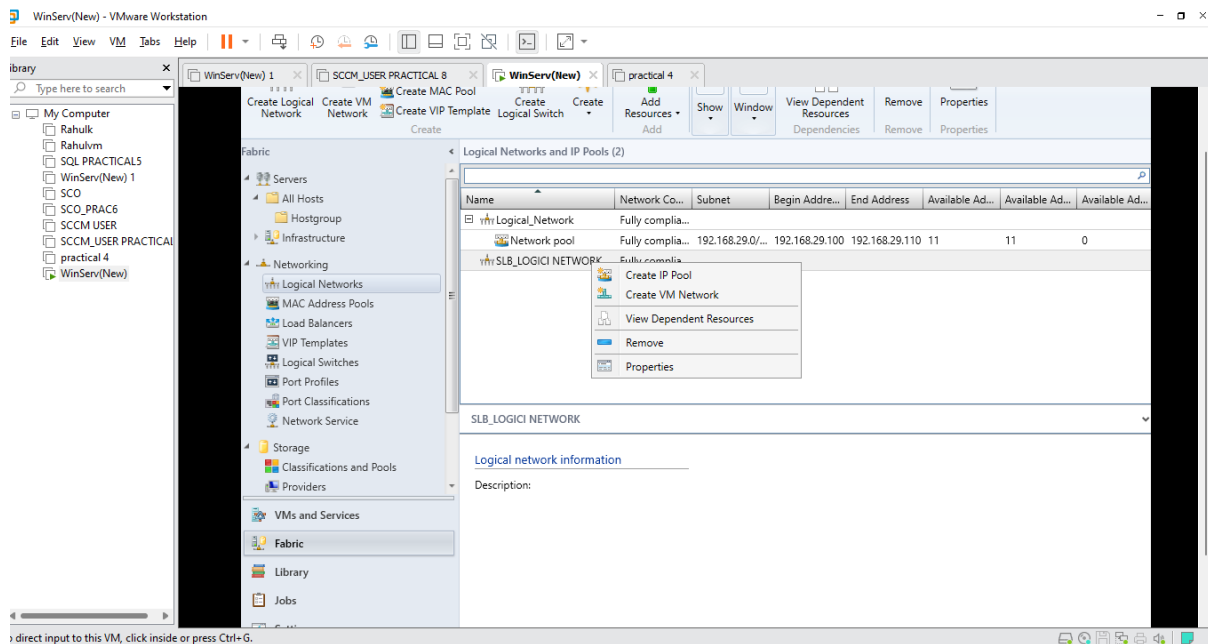


- Click on Finish

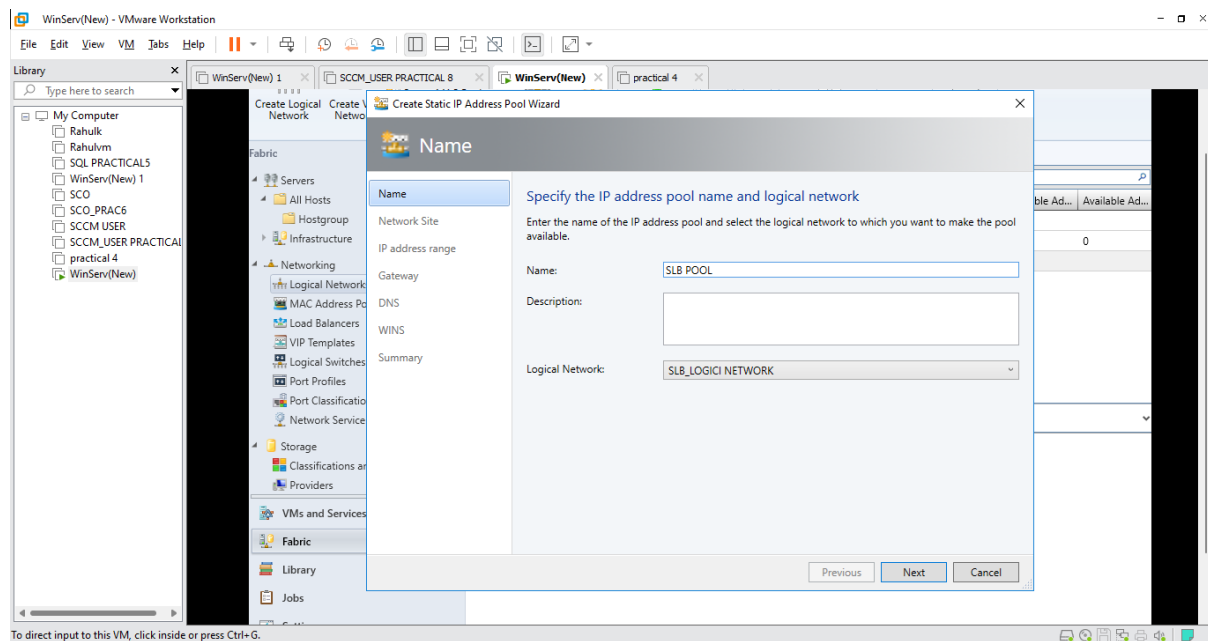


## Step 2: Create IP Pool

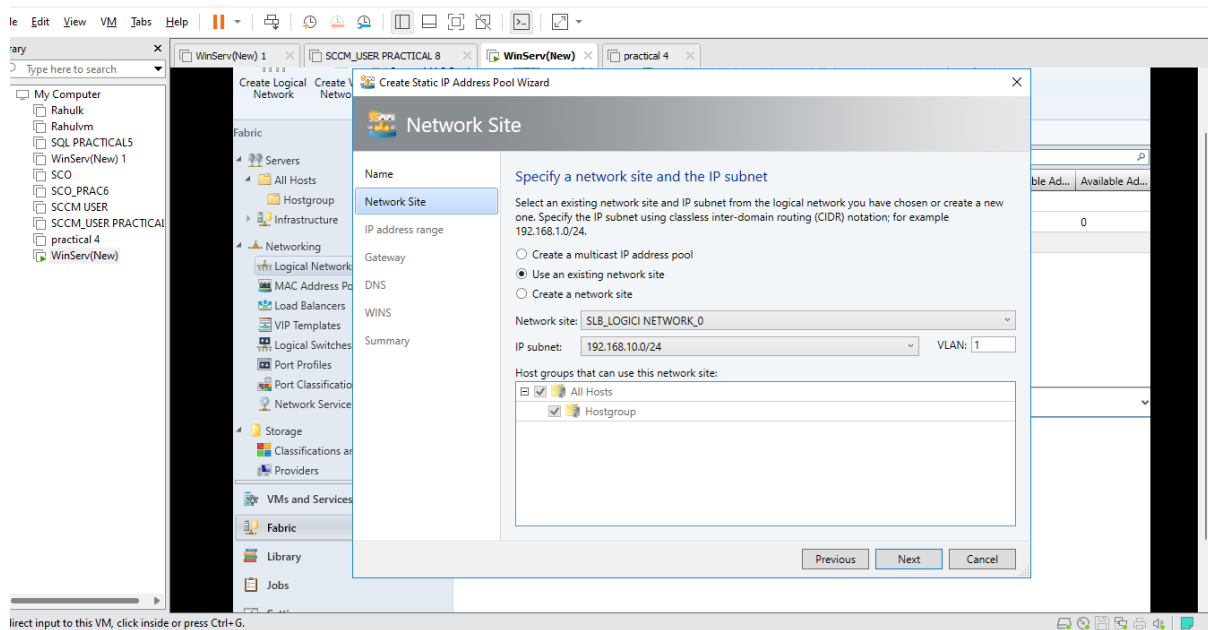
- Right-Click on **SLB\_LOGICAL NETWORK** and then Select **Create IP Pool**



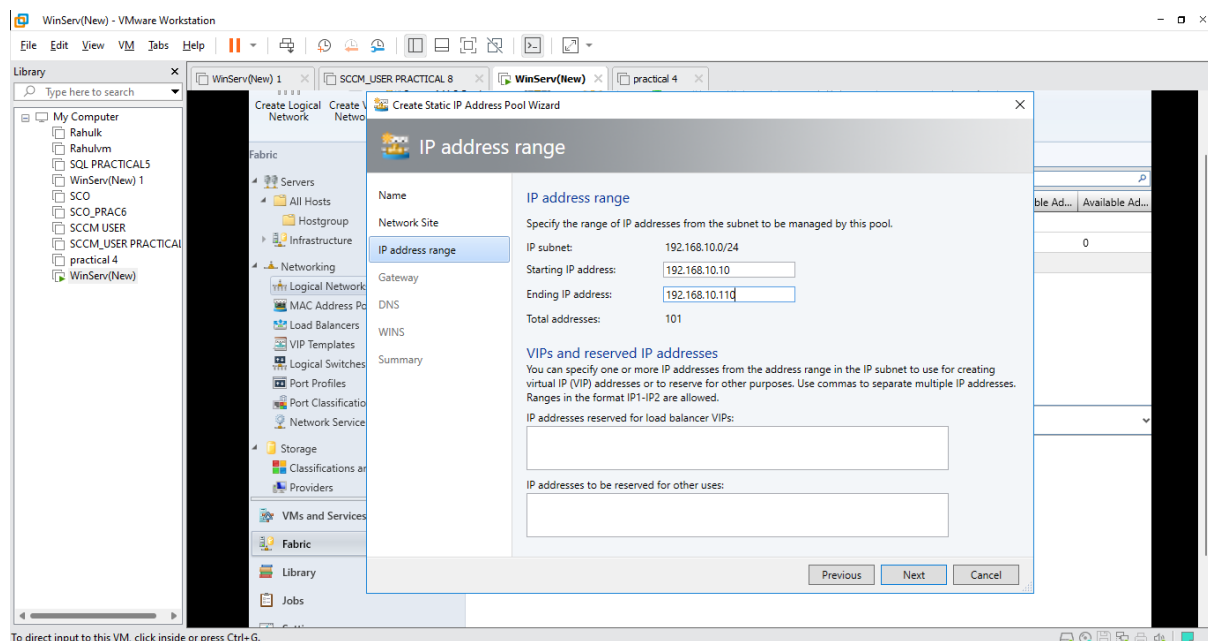
- Specify the IP address pool name and logical network (Here it is: **SLB POOL**) and then **Click on Next**



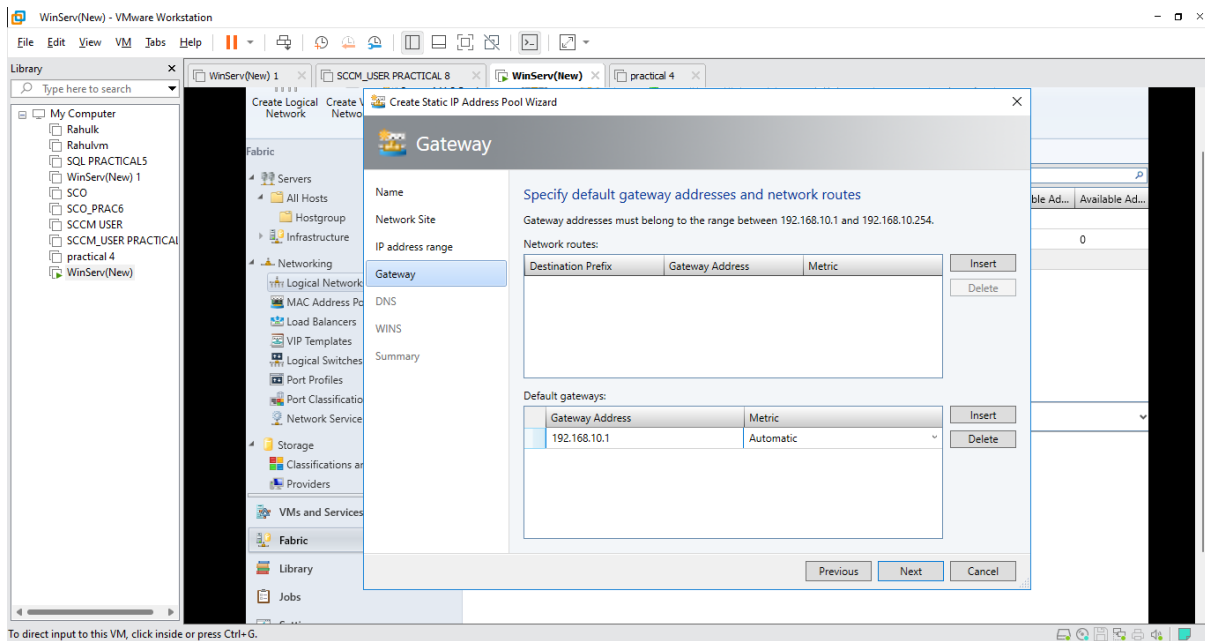
- Keep the default Value and Click on Next



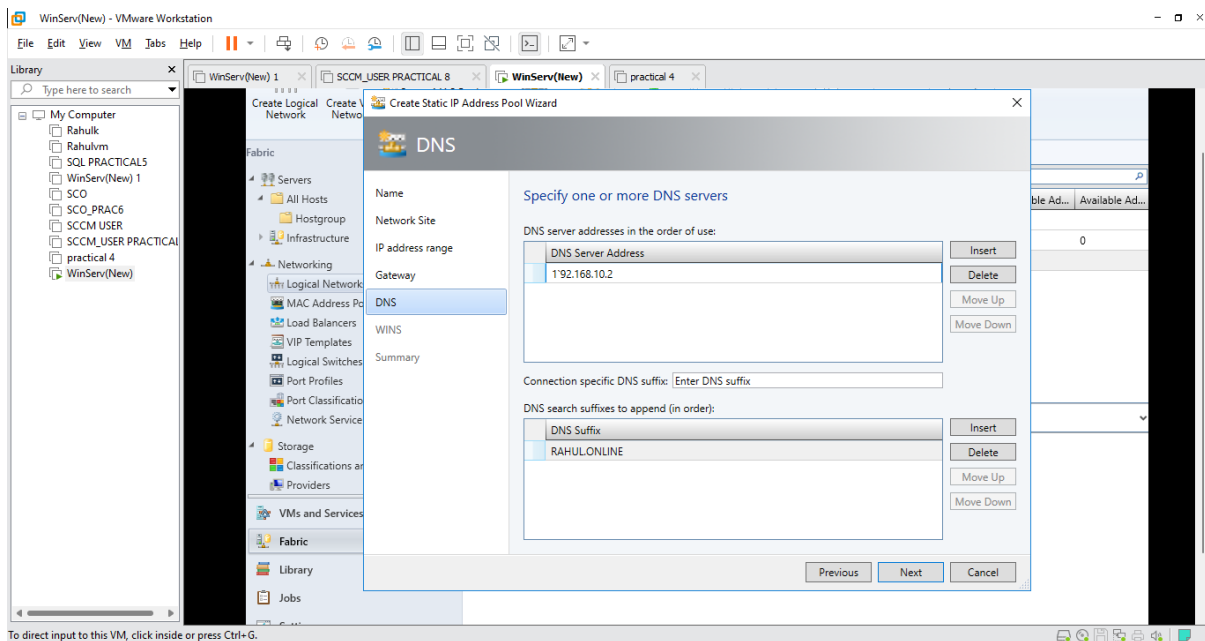
- Keep the Starting IP address as **192.168.10.10** and the Ending IP Address as **192.168.10.110** and Click Next



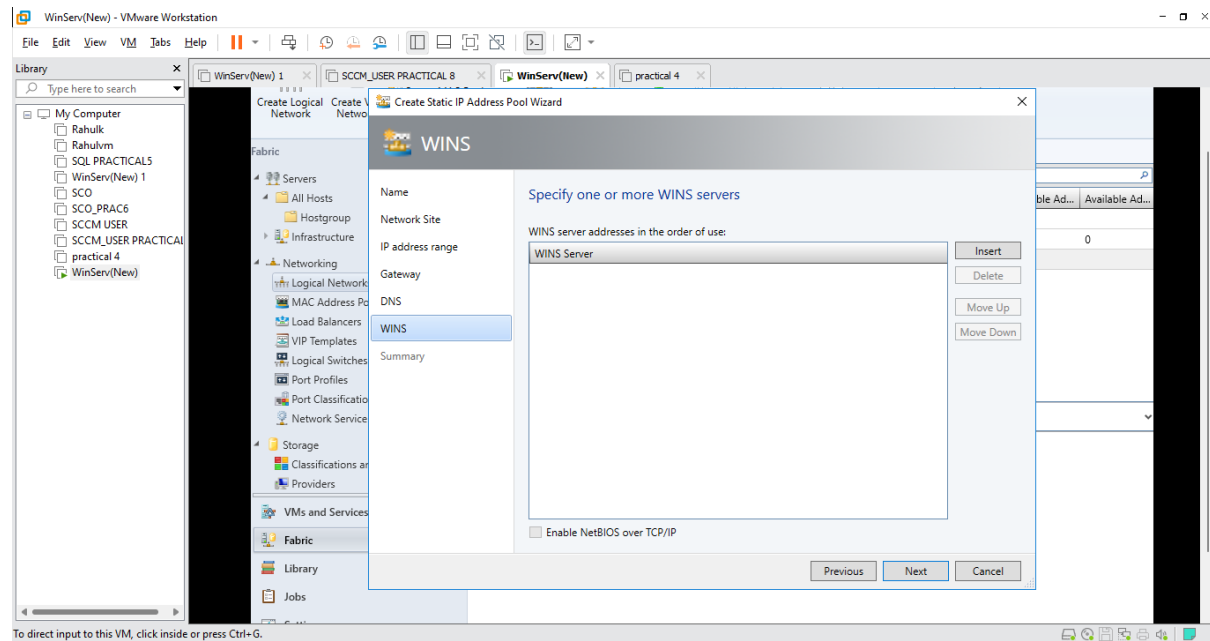
- Enter a Default gateway address **192.168.10.1** and Keep the Metric as **Automatic** and **Click on Next**



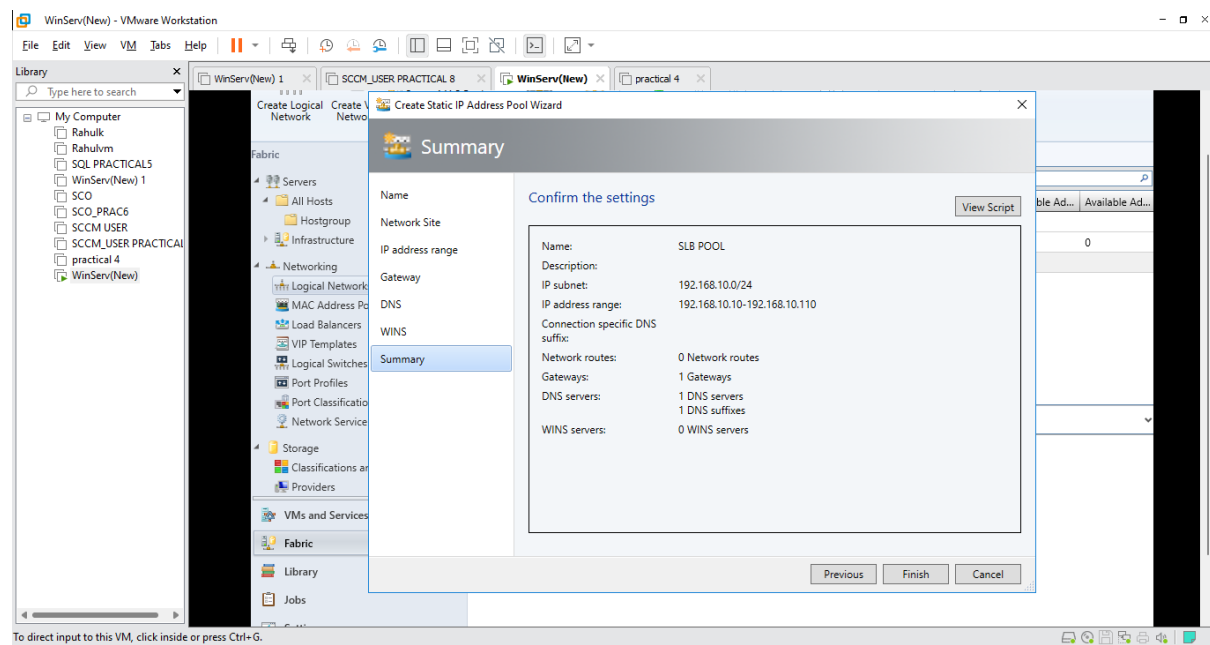
- Enter a DNS IP Address (Here it is **192.168.10.2**) and a DNS Suffix which is your DOMAIN\_NAME.ONLINE (Here it is **RAHUL.ONLINE**) and **Click on Next**



- Click on Next

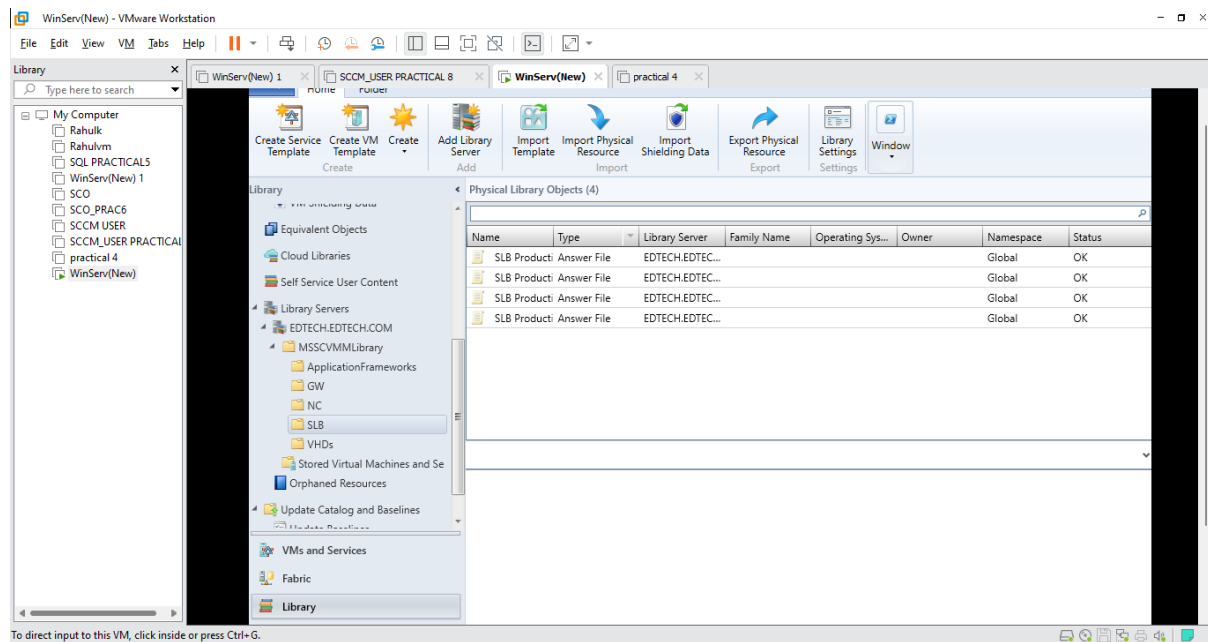


- Click on Finish

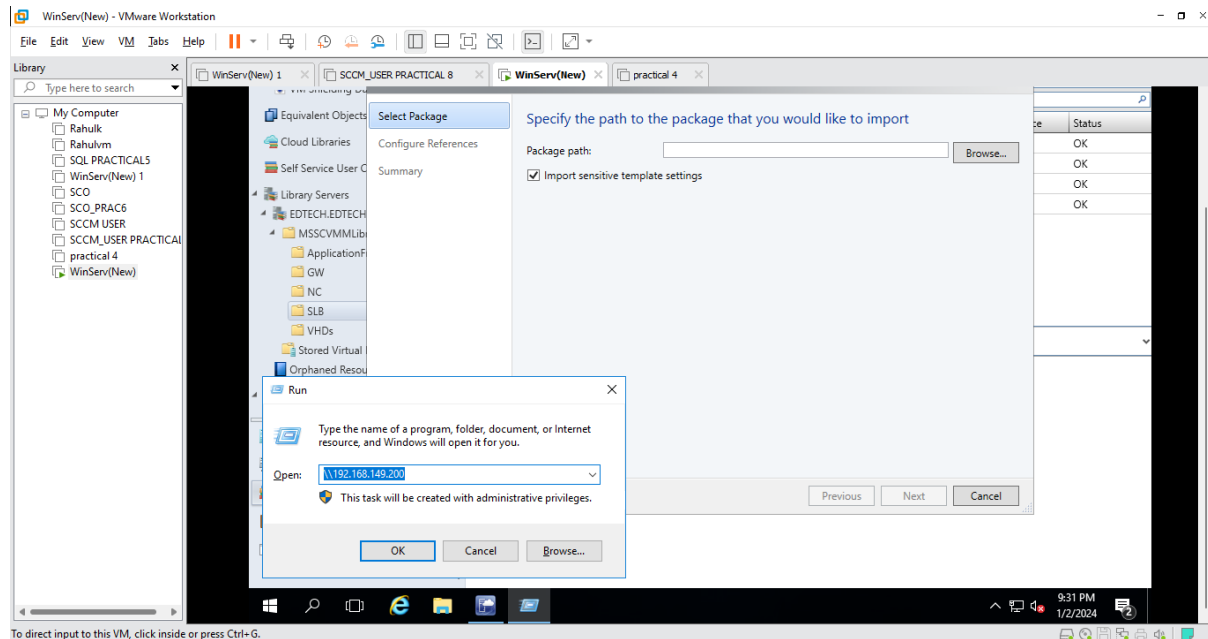


### Step 3: Creating an SLB Service Template

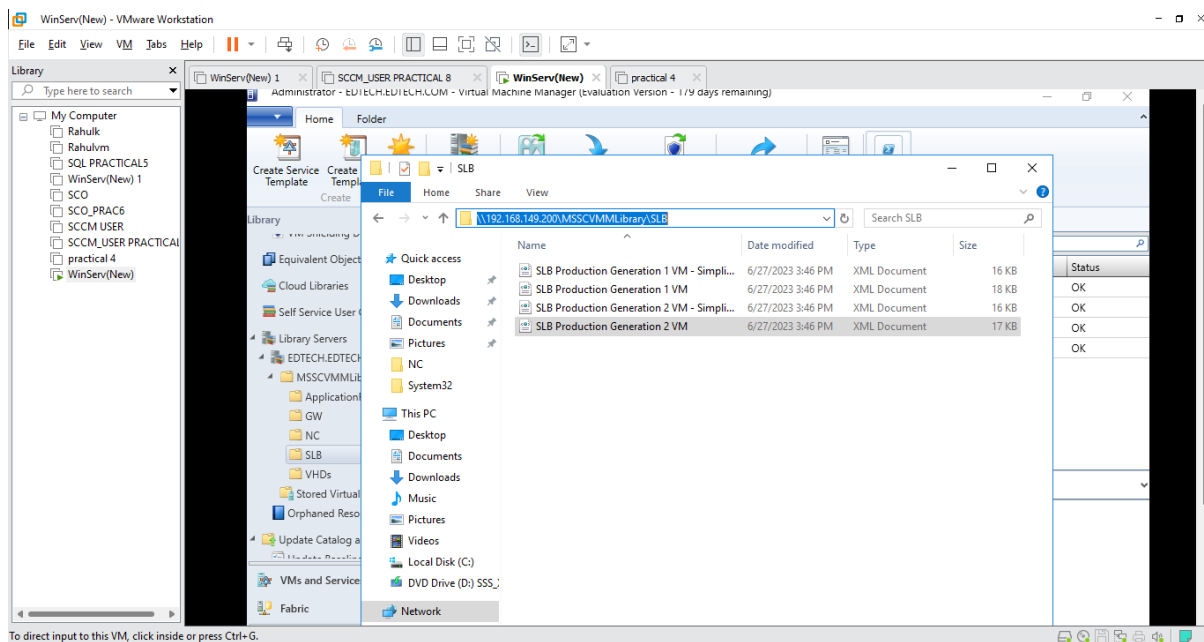
- Select the Library Tab and Click on Import Template



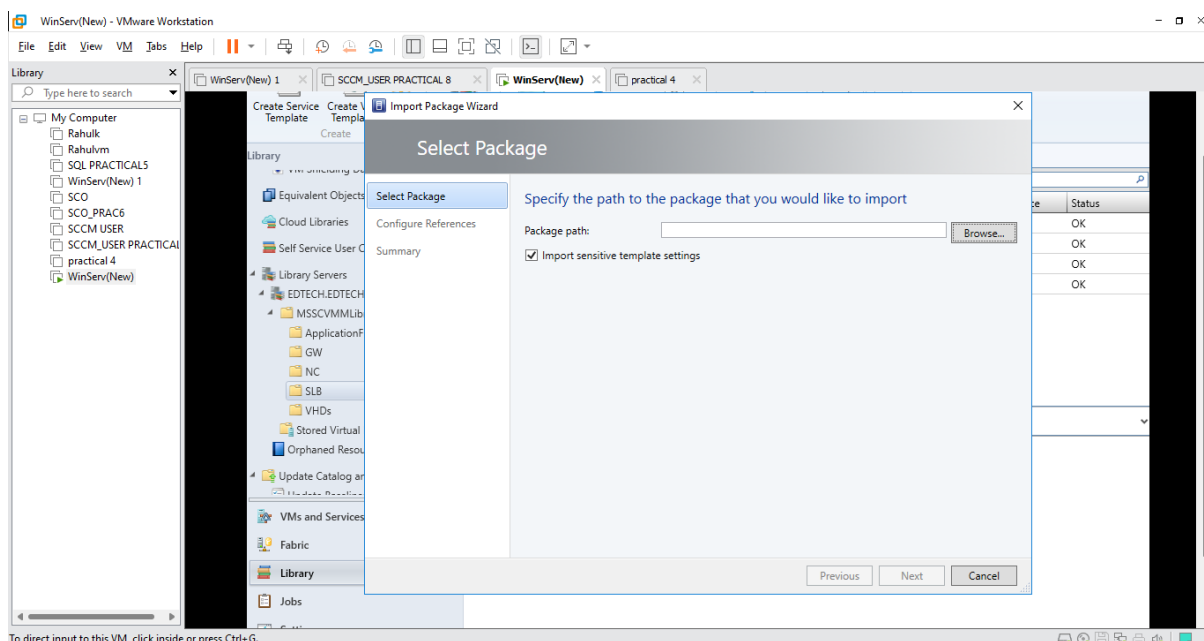
- In Select Package Open Run Window and then Type IP address of the WinServ(New) (Here it is \\192.168.149.200) Click on Ok



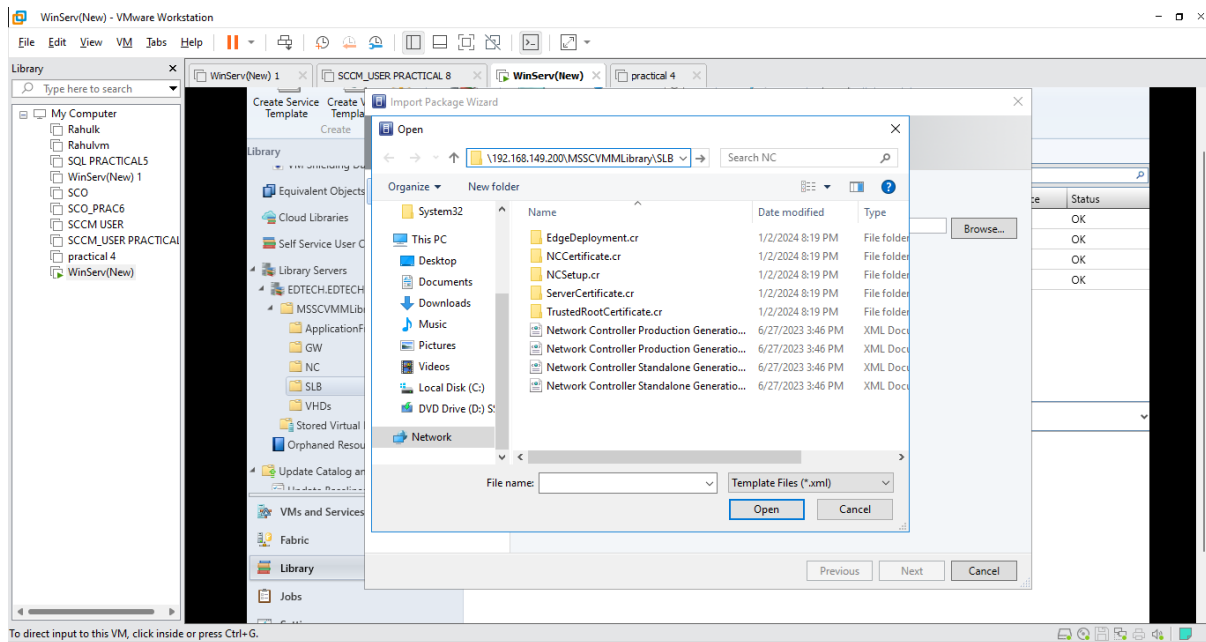
- Within **MSSCVMMLibrary** Select **SLB Folder** and then **Copy the Path**



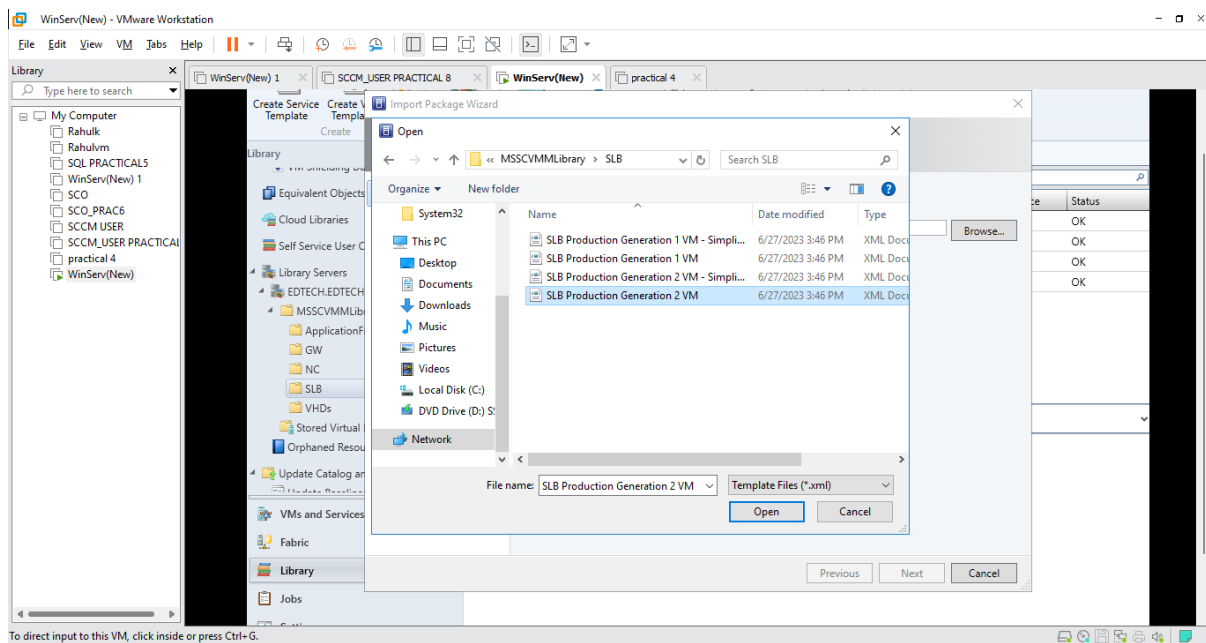
- Now again Go to the Select Package window **Click on Browse**



- Now paste the path for SLB and then **Click on Enter**

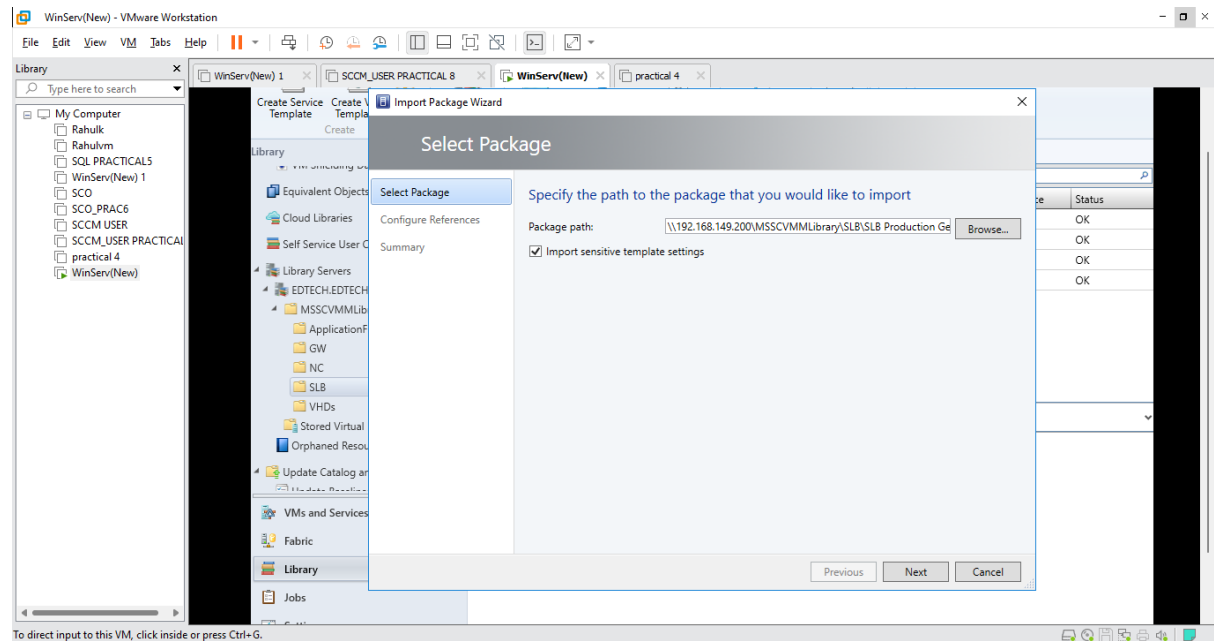


- Within SLB Folder Select **Network SLB Production Generation 2 VM** and then **Click on Open**

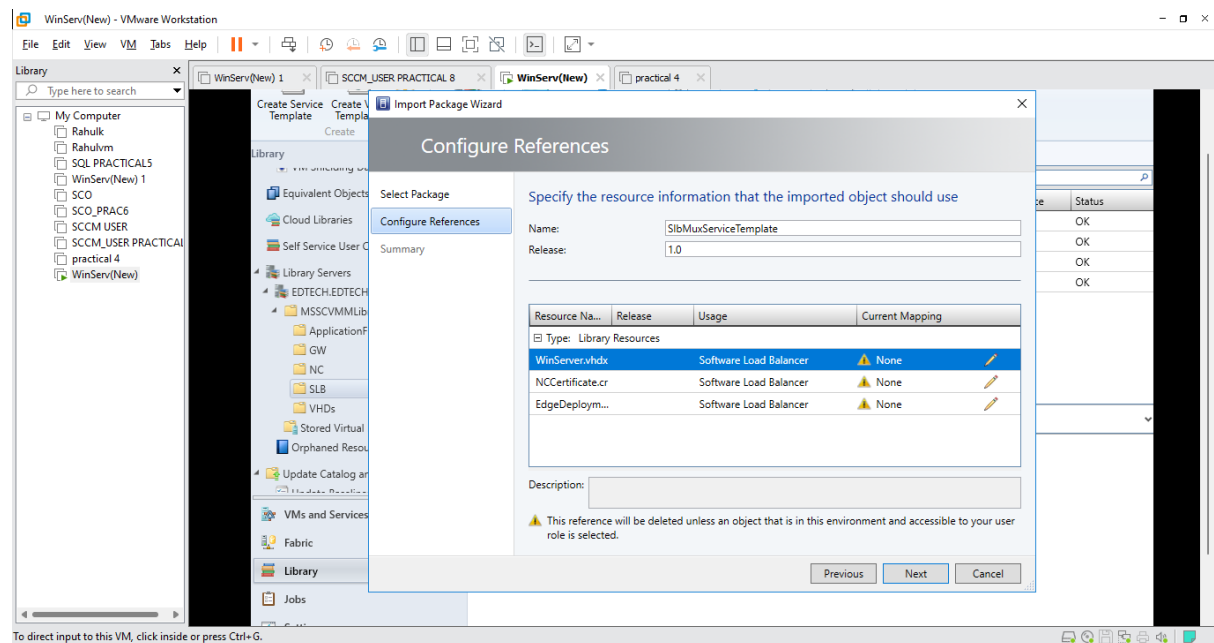




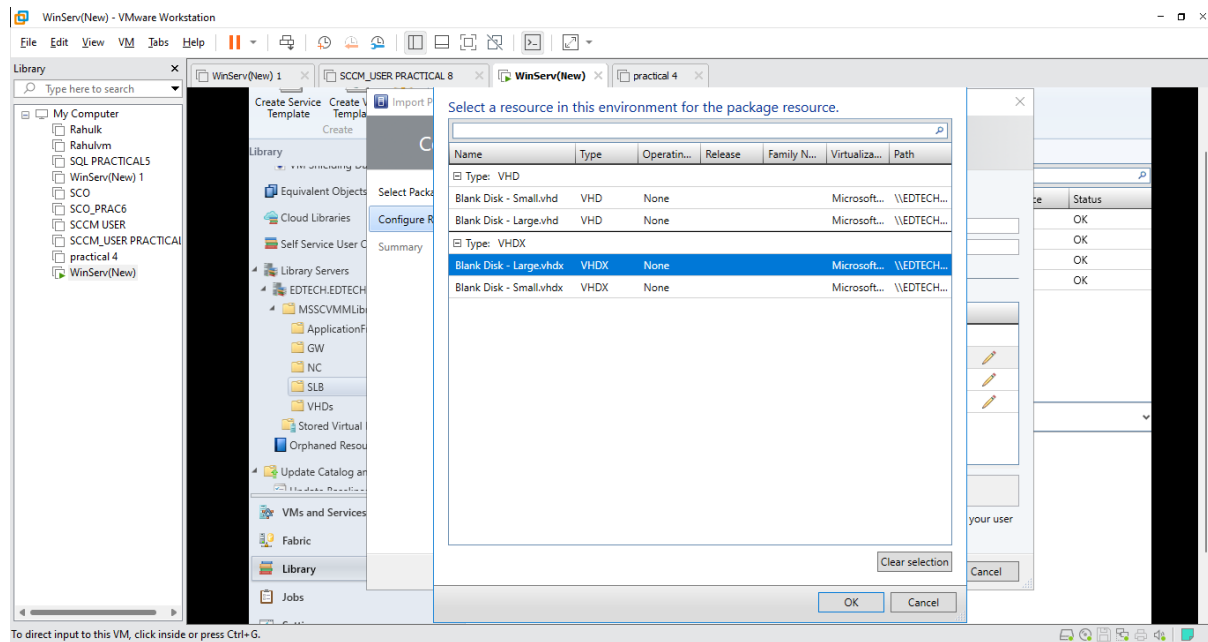
- **Click on Next**



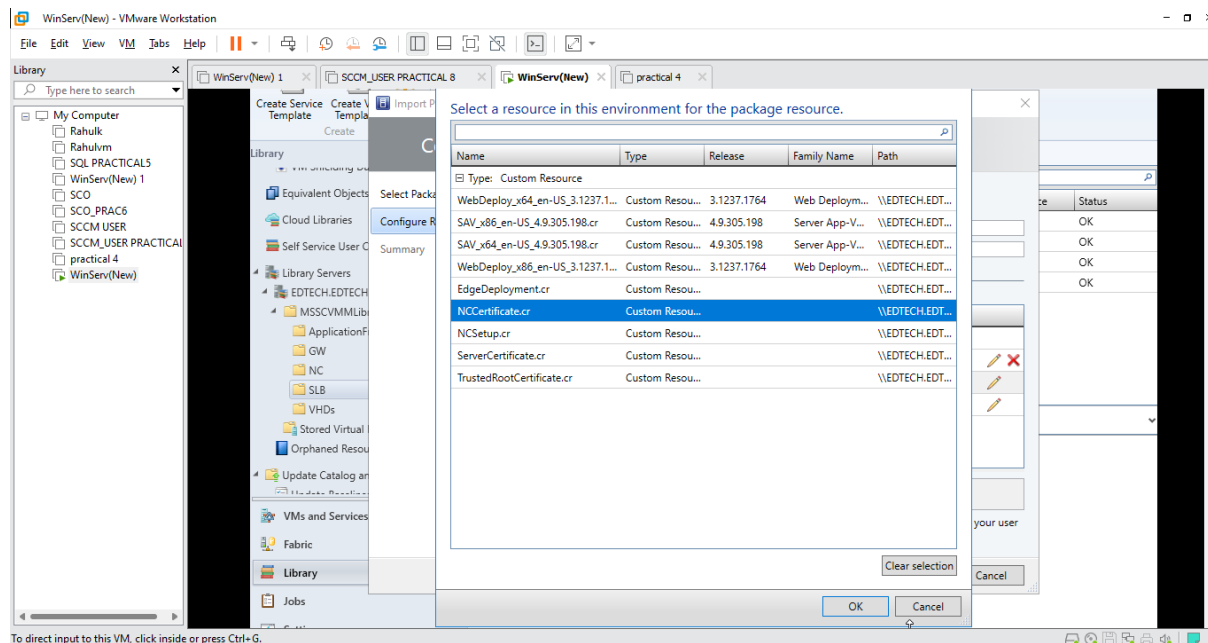
- **Within Configuration References Click on WinServer.vhdx and then Click on Edit**



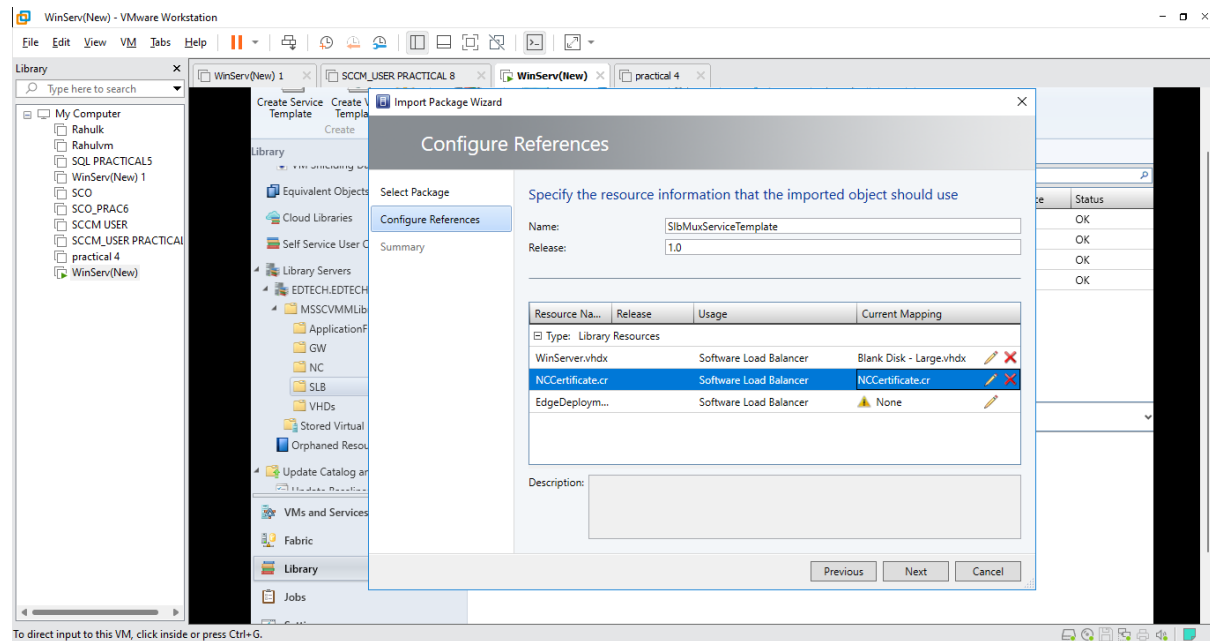
- **Within Type: VHDX Select Blank Disk – Large.vhdx and Click on OK**



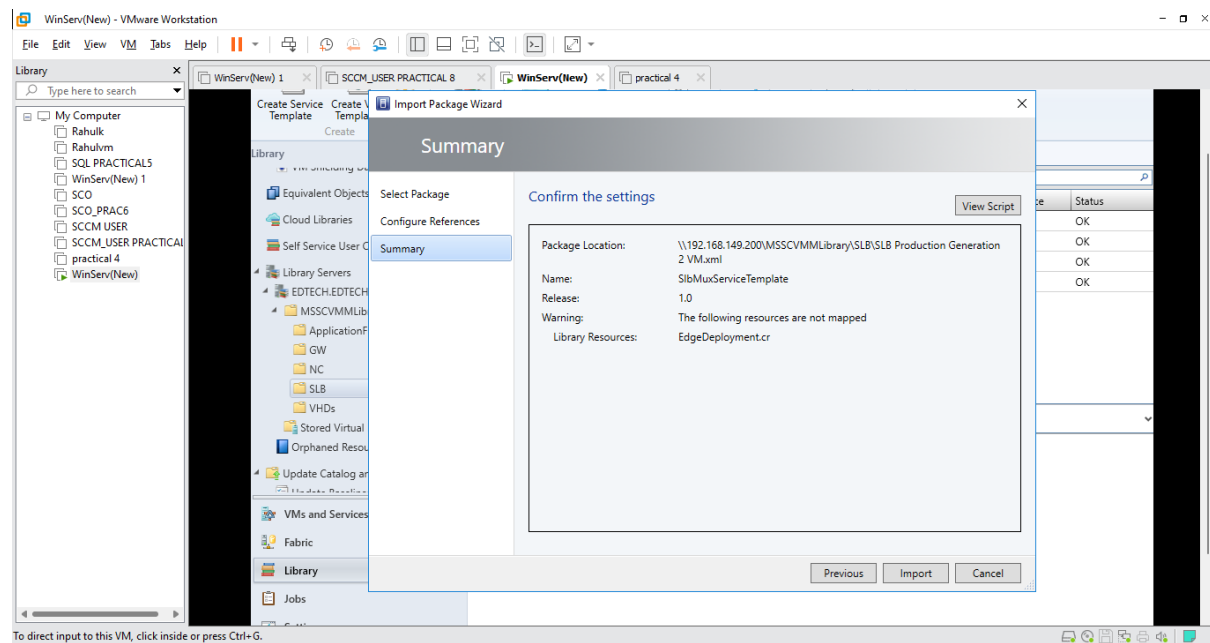
- **In NCCertificate, Click on the Custom Resource NCCertificate.cr and Click Ok**



- Click on Next

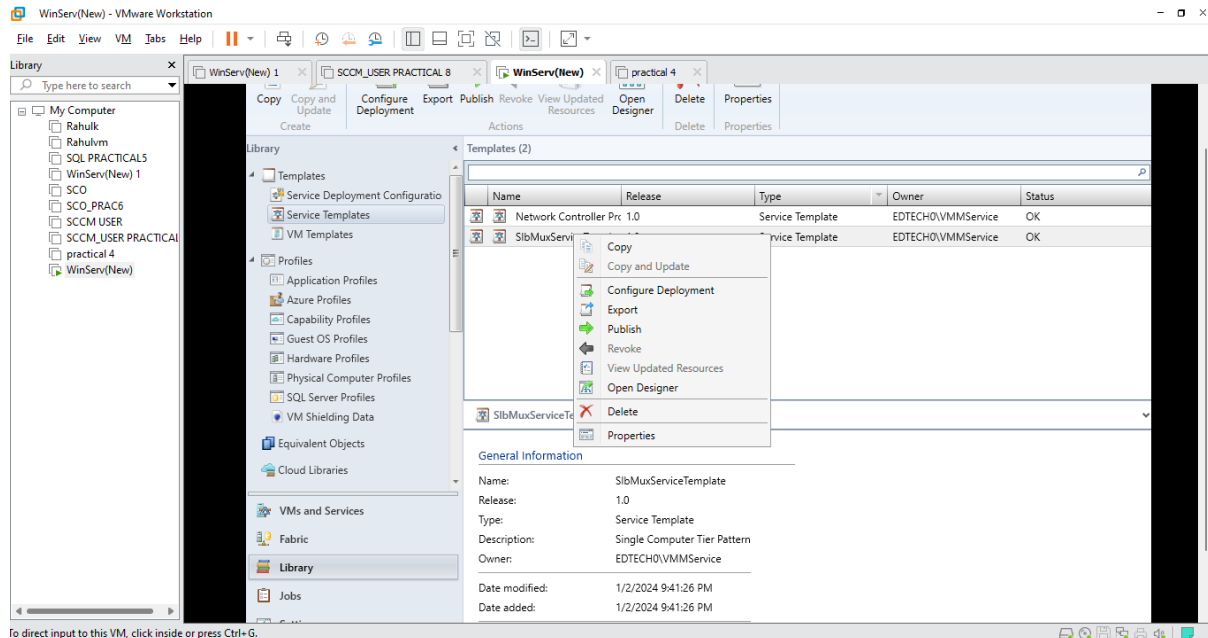


- Click on Import

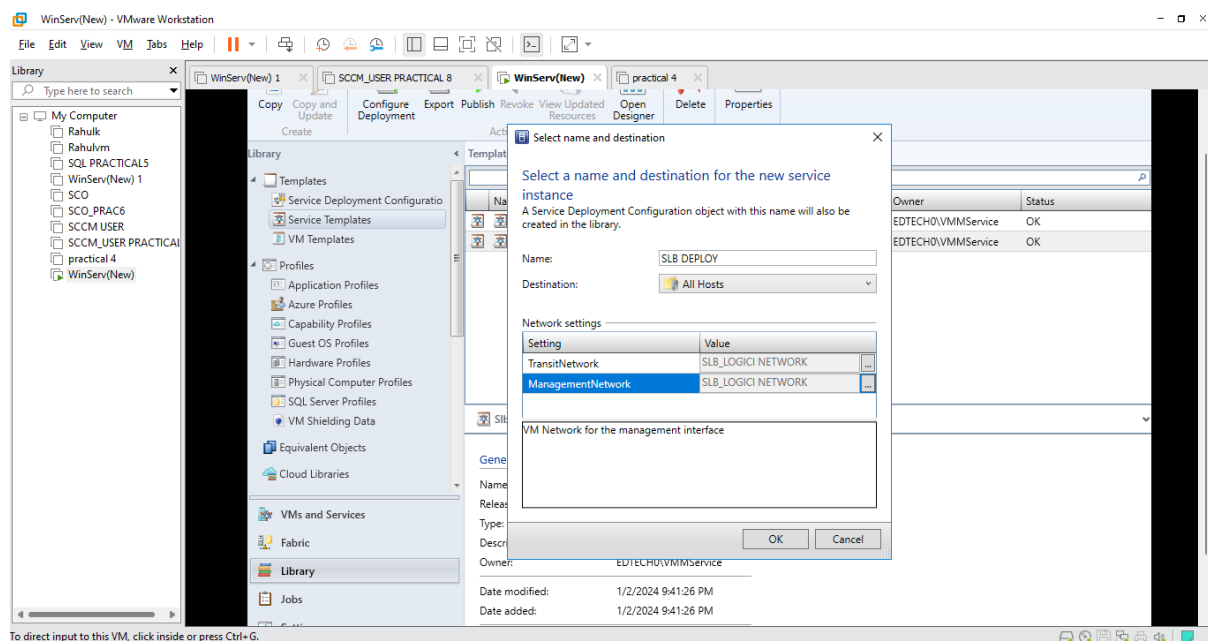


## Step 4: Deploying the SLB Service Template

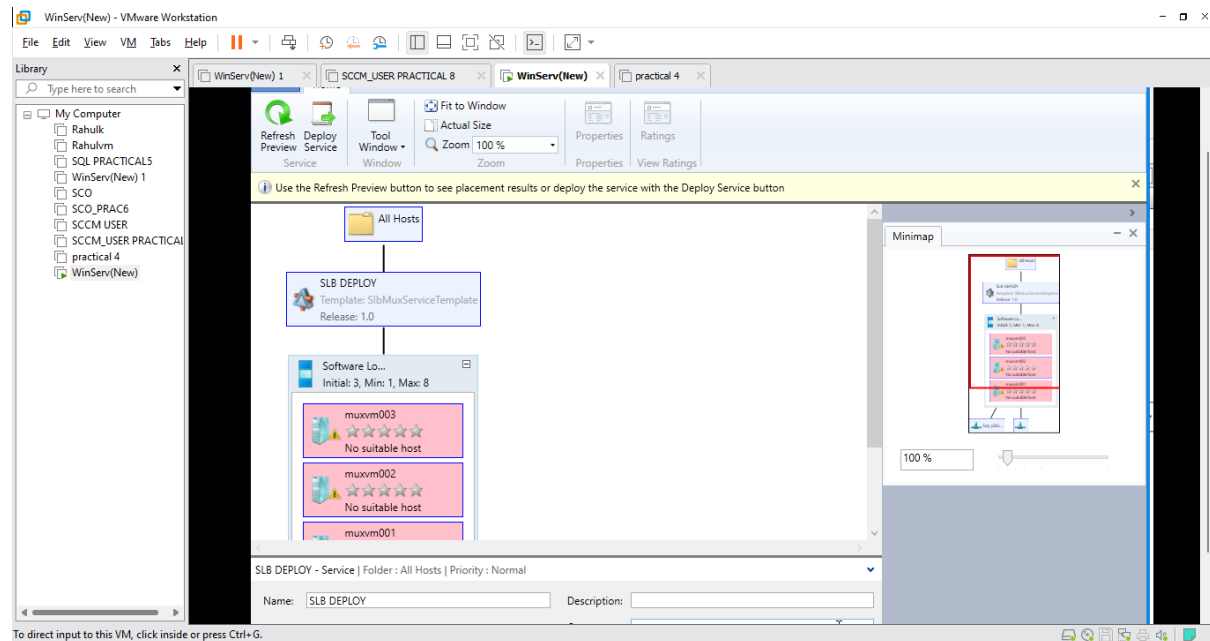
- Within Service Templates **Right-Click on SlbMuxServiceTemplate** and then **Click on Configure Deployment**



- Type in the name (Here it is **SLB\_DEPLOY**) and in the **Transit and Management Network** Select the newly created Logical Network (Here it is **SLB\_LOGICI NETWORK**) and **Click on Ok**



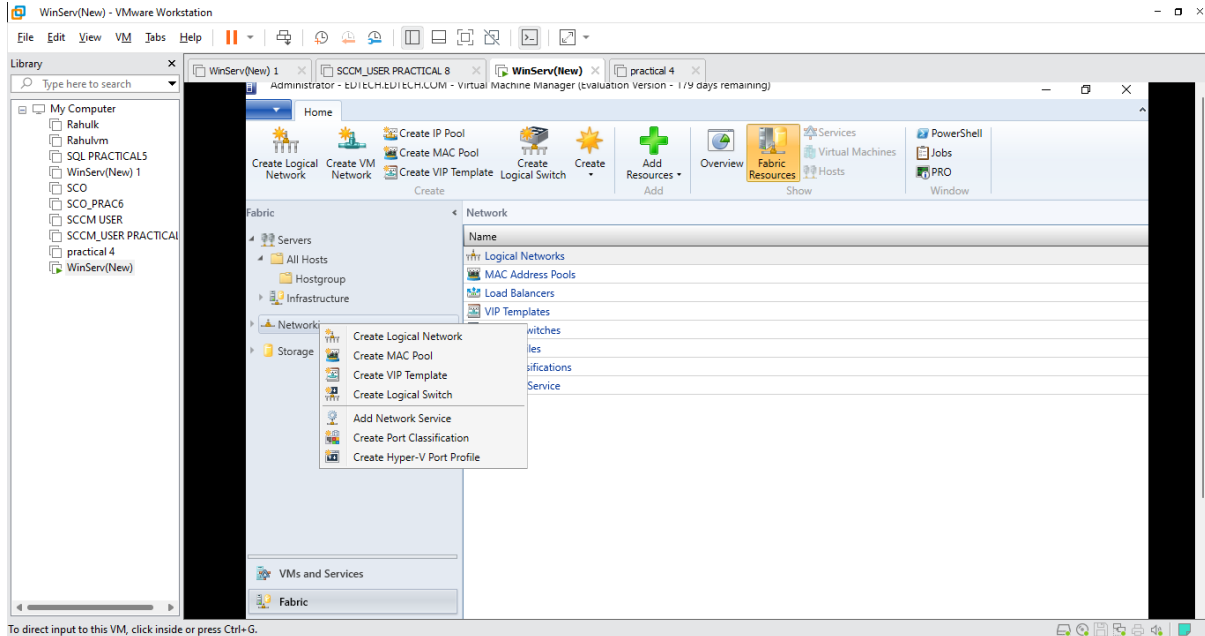
- **Now in Settings Select the Following:**
- **localAdmin: NT AUTHORITY\Local Service**
- **ManagementNetwork: SLB\_LOGI NETWORK**
- **MgmtDomainAccount: NT AUTHORITY\LocalService**
- **MgmtDomain FQDN: EDTECH.EDTECH.COM**
- **SelfSignedCertificate: True**
- **TransitNetwork: SLB\_LOGI NETWORK**



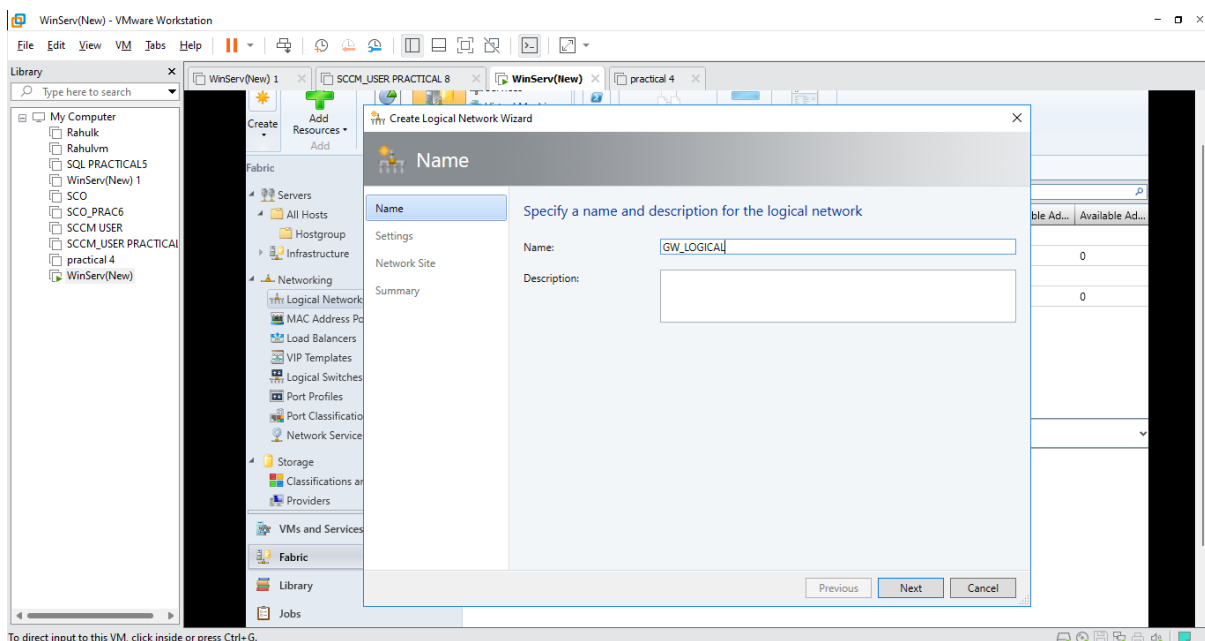
## Step 4: Deploying the GRE Service Template

### Step 1: Create a Logical Network

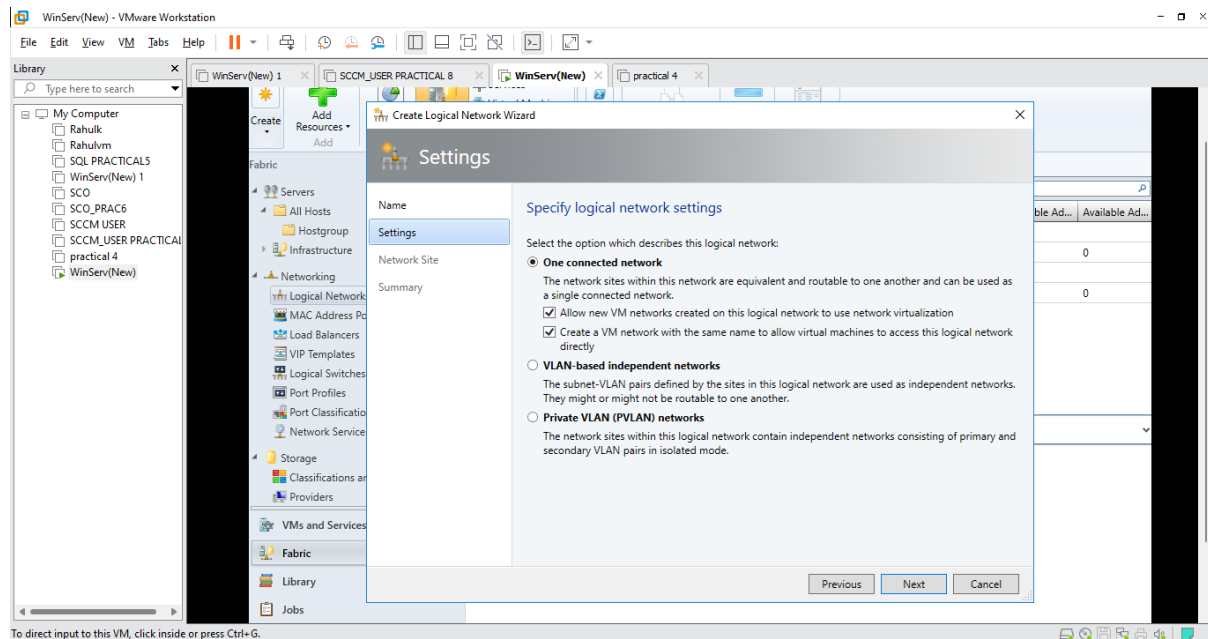
- **Create a new logical network by Selecting the Fabric Tab in SCVMM and Right-Click on Logical Networks and Select Create a Logical Network**



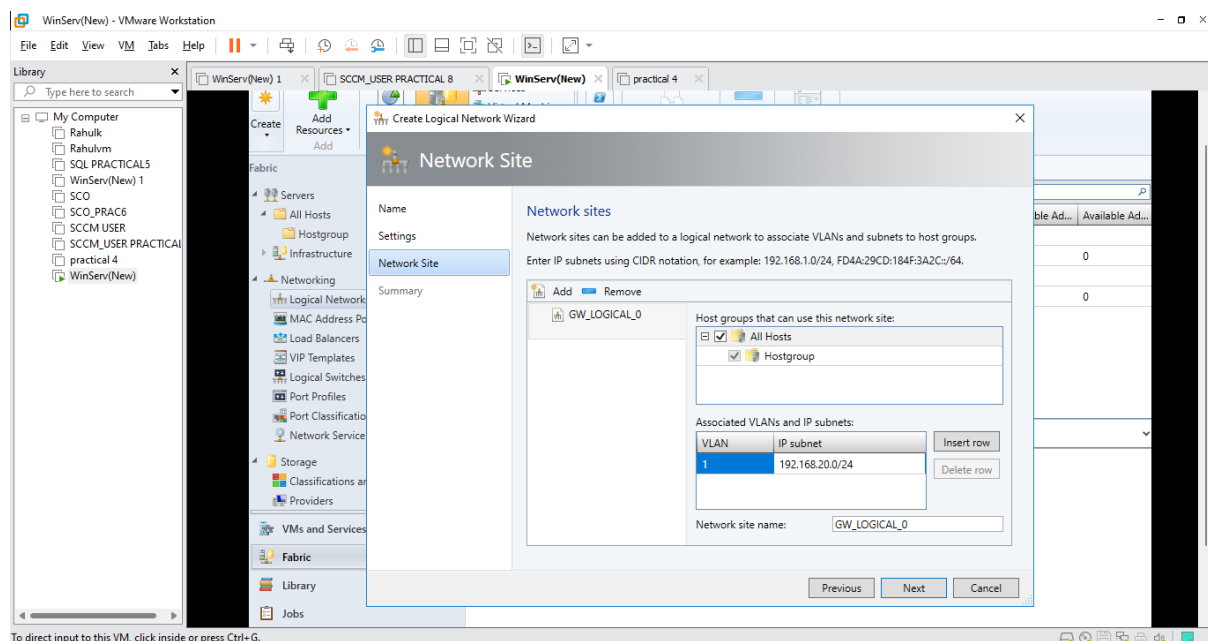
- Give Name for your Logical Network (Here it is: **GW\_LOGICAL**) And then **Click on Next**



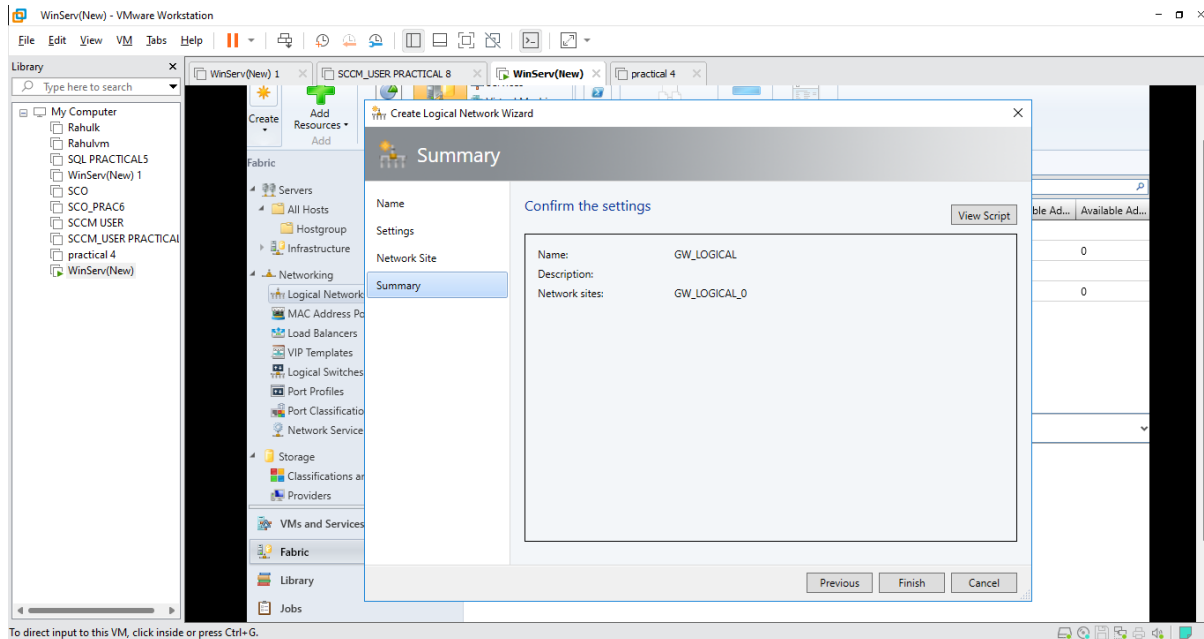
- Select **One-connected network** and Tick both **Allow new VM networks created on the logical network to use network virtualisation** and **Click on Next**



- Within Network Site Click on **Add Select All Hosts** and then **Click on Insert row** Enter VLAN and IP address Here it is: **192.168.20.0/24** and then **Click on Next**

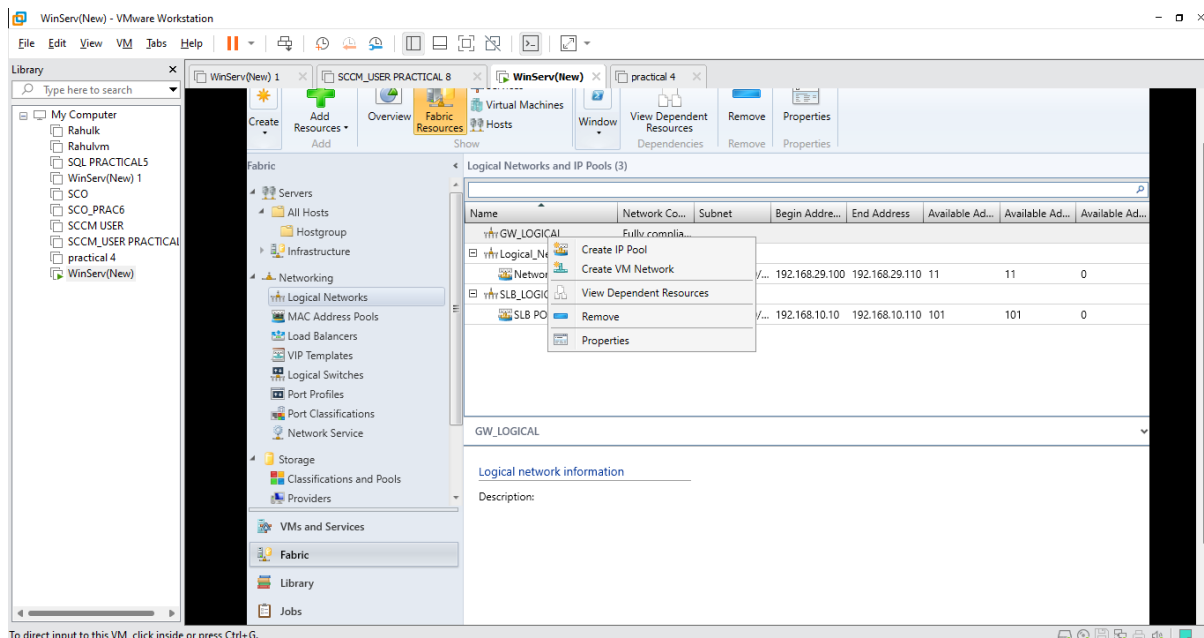


- **Click on Finish**



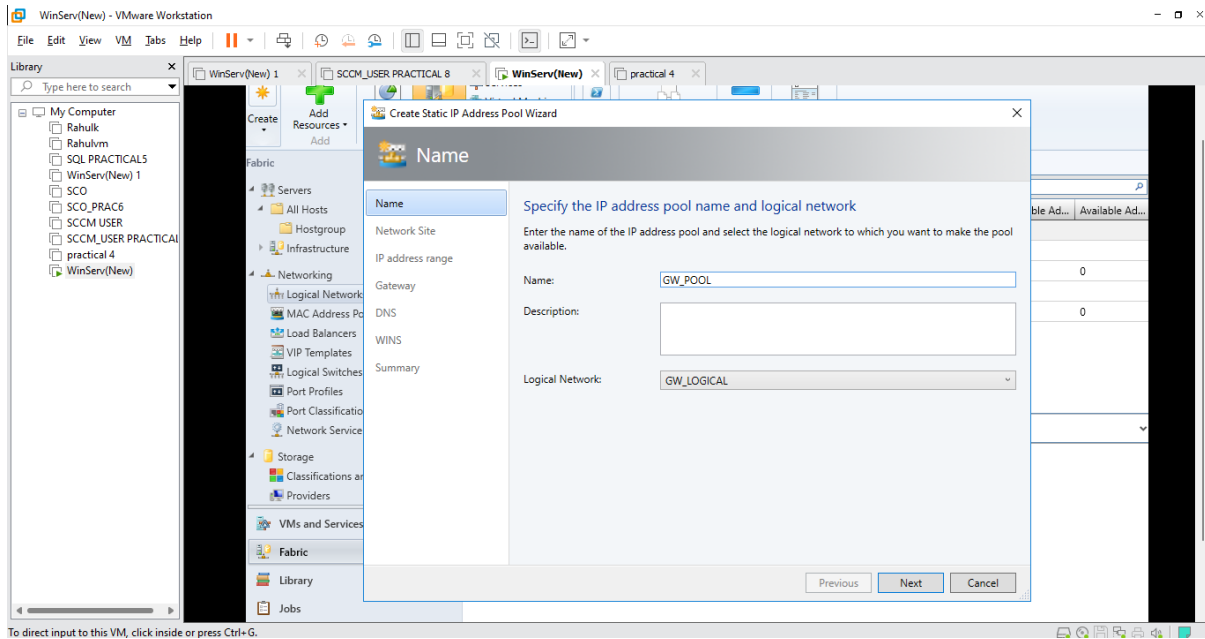
## Step 2: Create IP Pool

- **Right-Click on GW\_LOGICAL and then Select Create IP Pool**

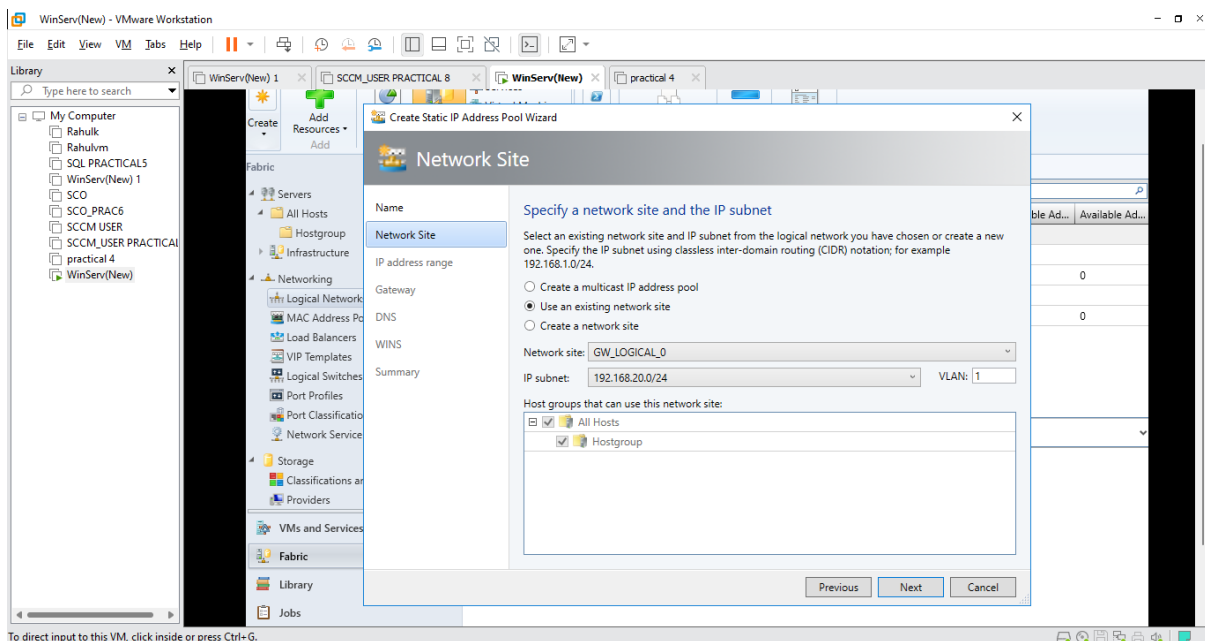




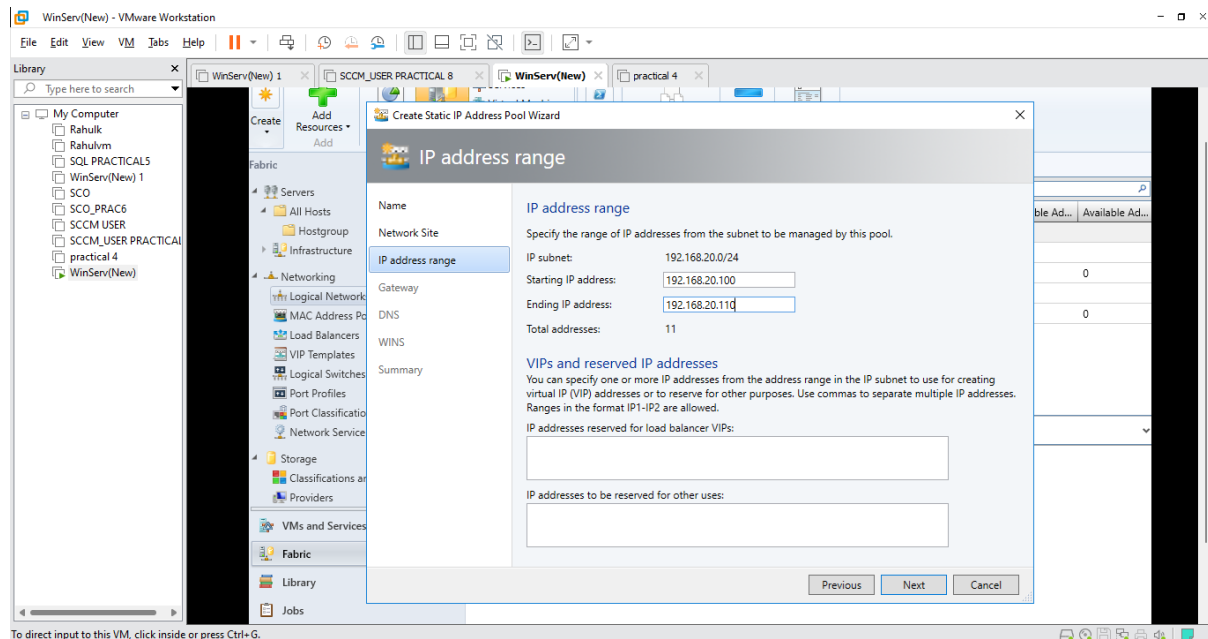
- Specify the IP address pool name and logical network (Here it is: **GW\_POOL**) and then **Click on Next**



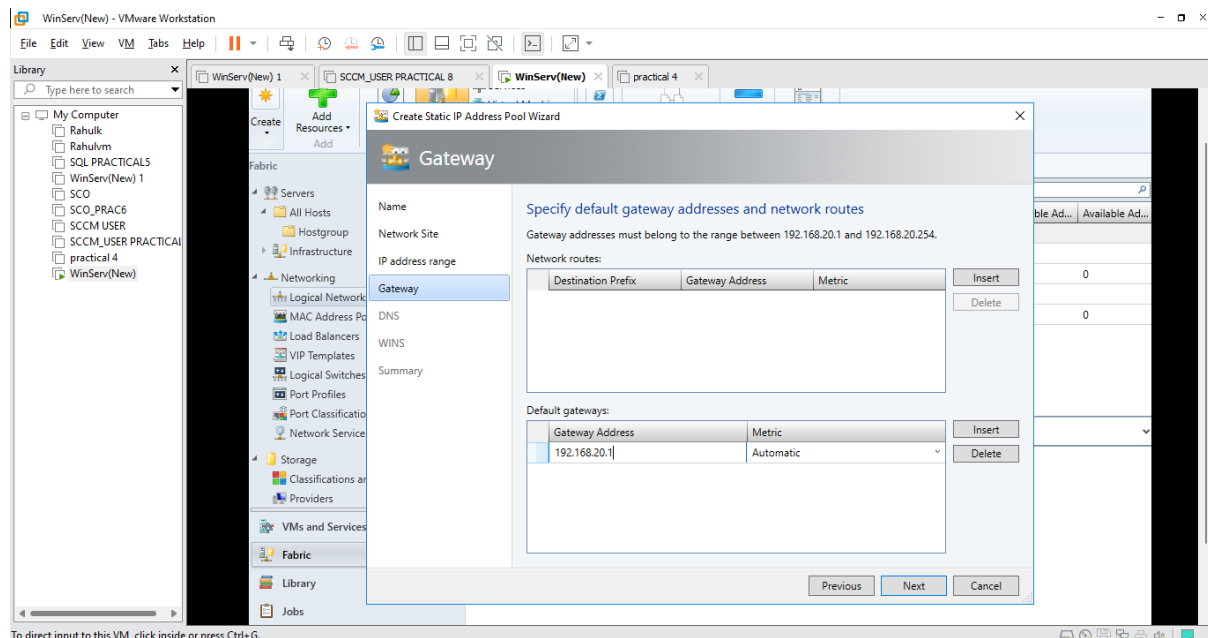
- Keep the default Value and **Click on Next**



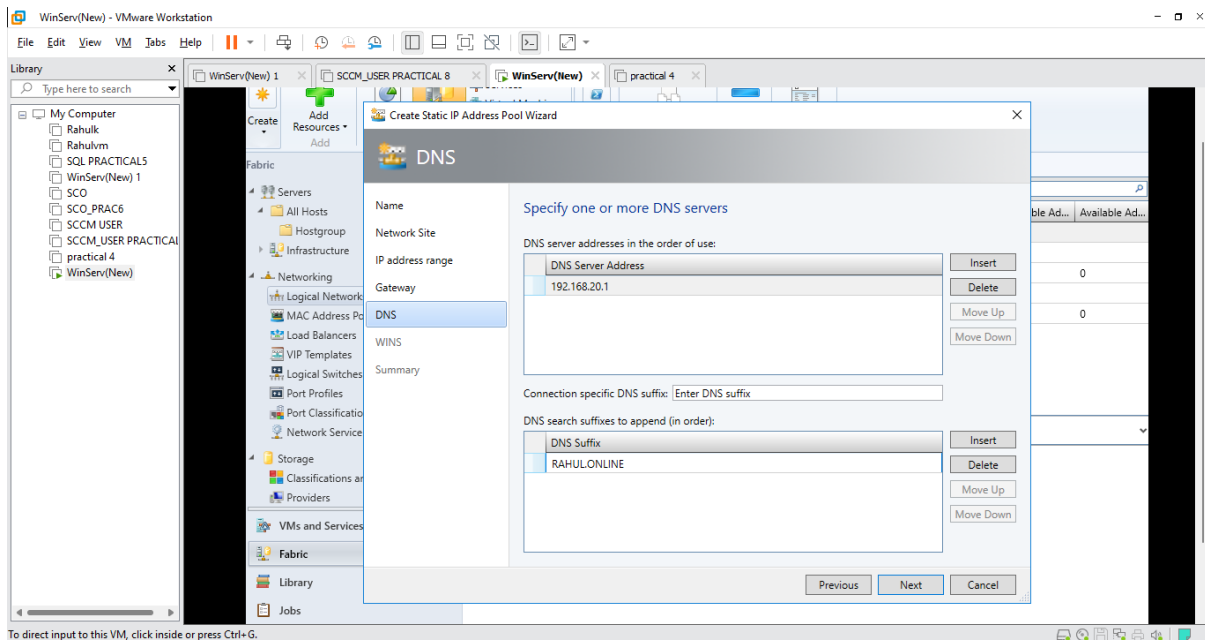
- Keep the Starting IP address as **192.168.20.100** and the Ending IP Address as **192.168.20.110** and Click Next



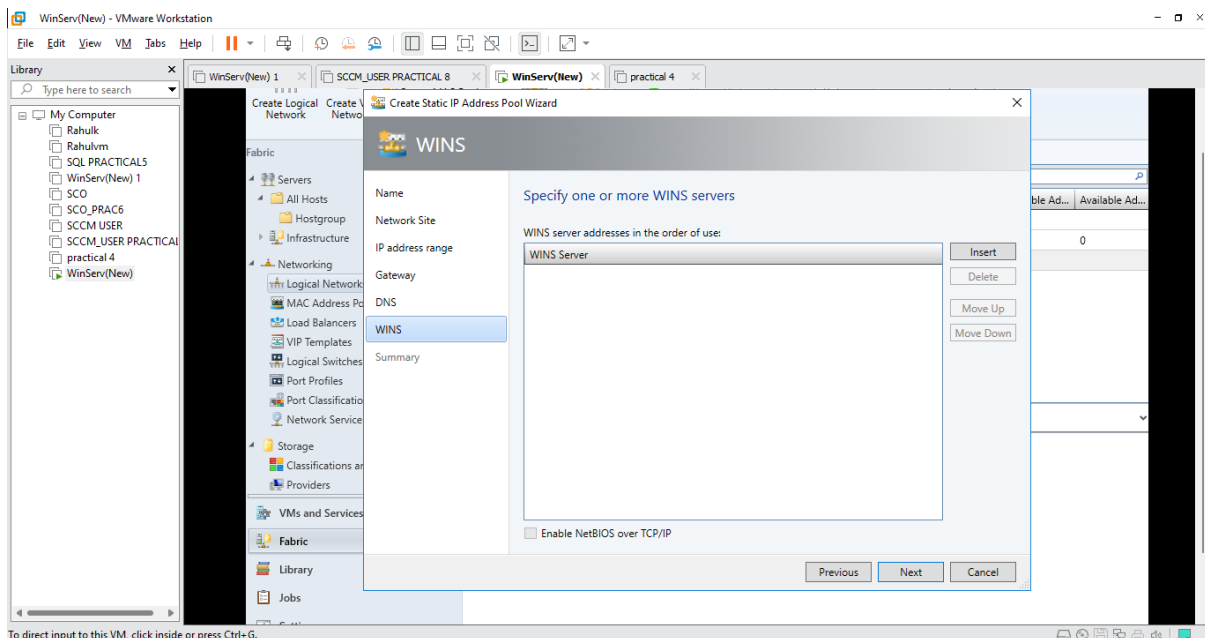
- Enter a Default gateway address **192.168.20.1** and Keep the Metric as **Automatic** and Click on Next



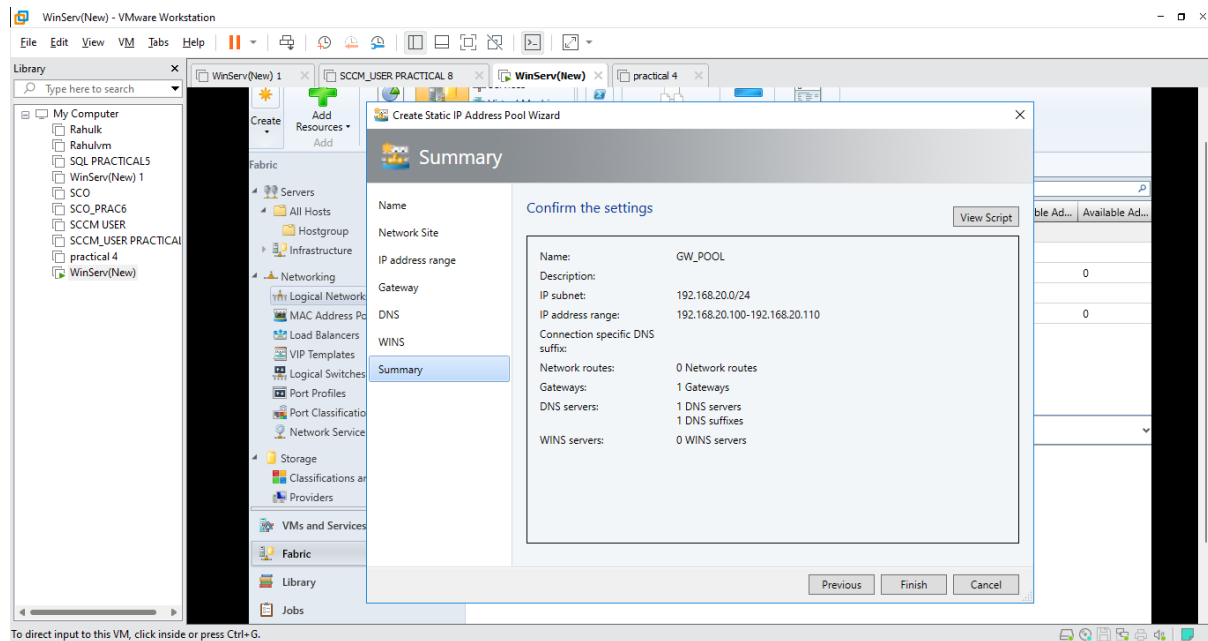
- Enter a DNS IP Address (**Here it is 192.168.20.1**) and a DNS Suffix which is your DOMAIN\_NAME.ONLINE (**Here it is RAHUL.ONLINE**) and Click on Next



- Click on Next

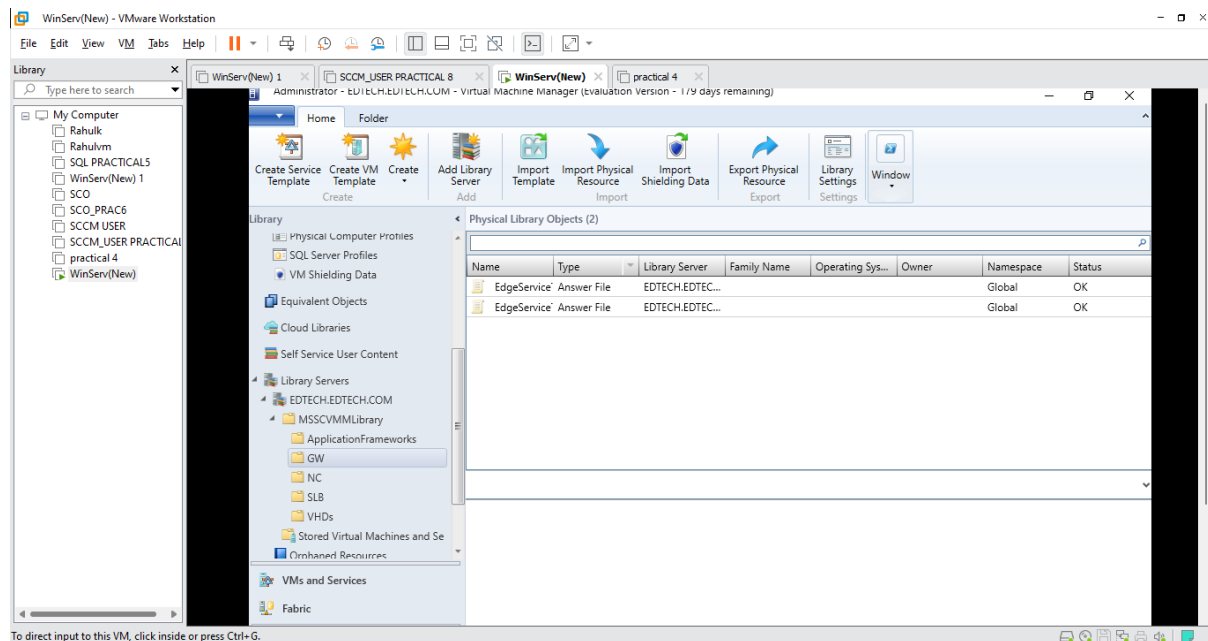


- **Click on Finish**

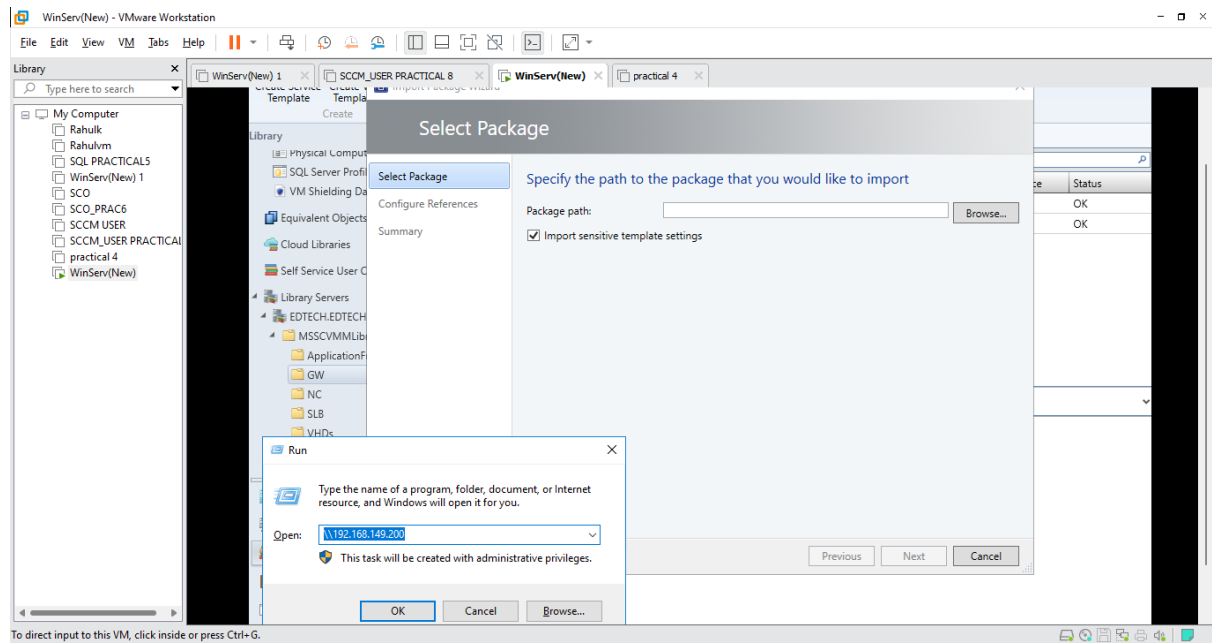


### Step 3: Creating a GW Service Template

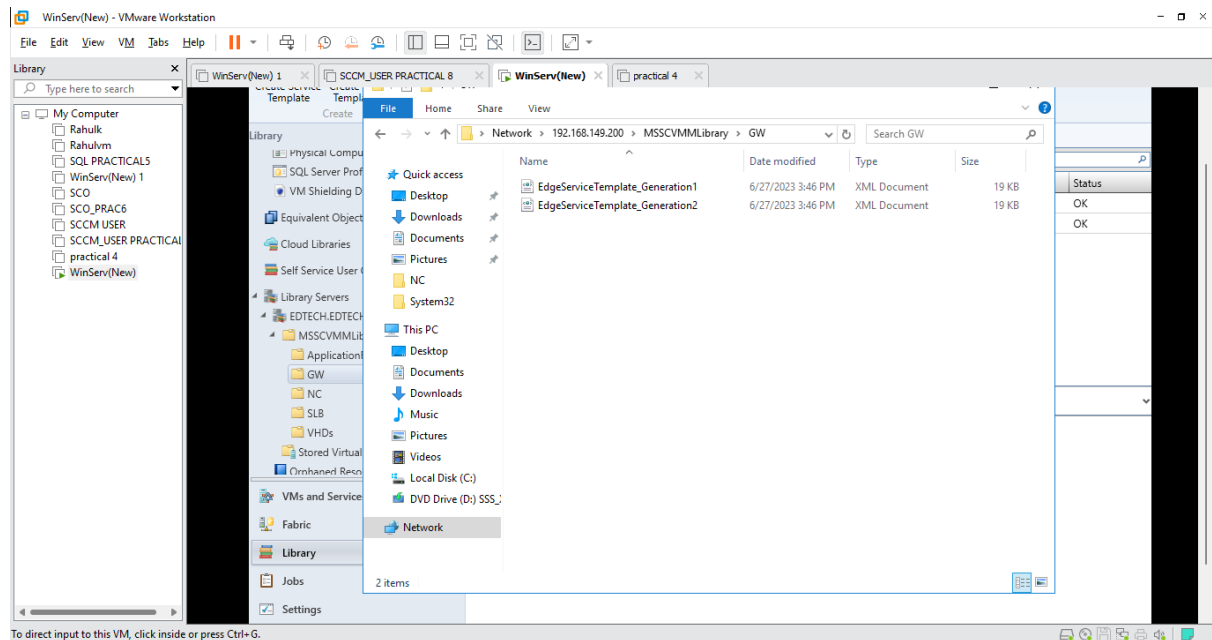
- **Select the Library Tab and Click on Import Template**



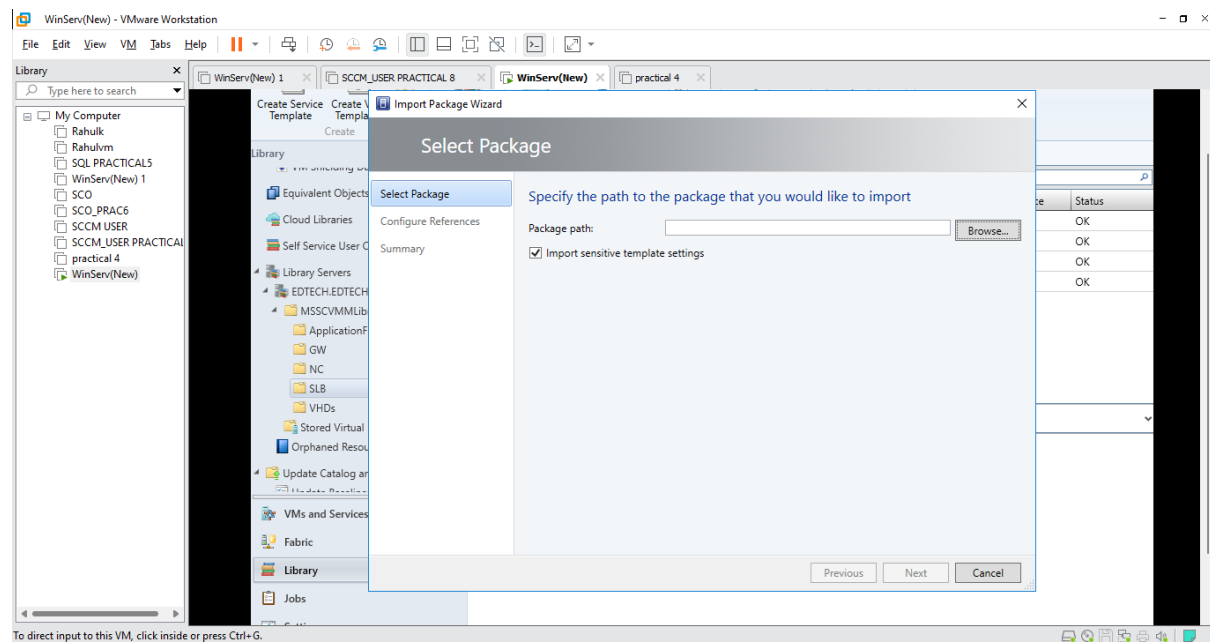
- In Select Package Open Run Window and then Type **IP address of the WinServ(New)** (Here it is **\\192.168.149.200**) Click on Ok



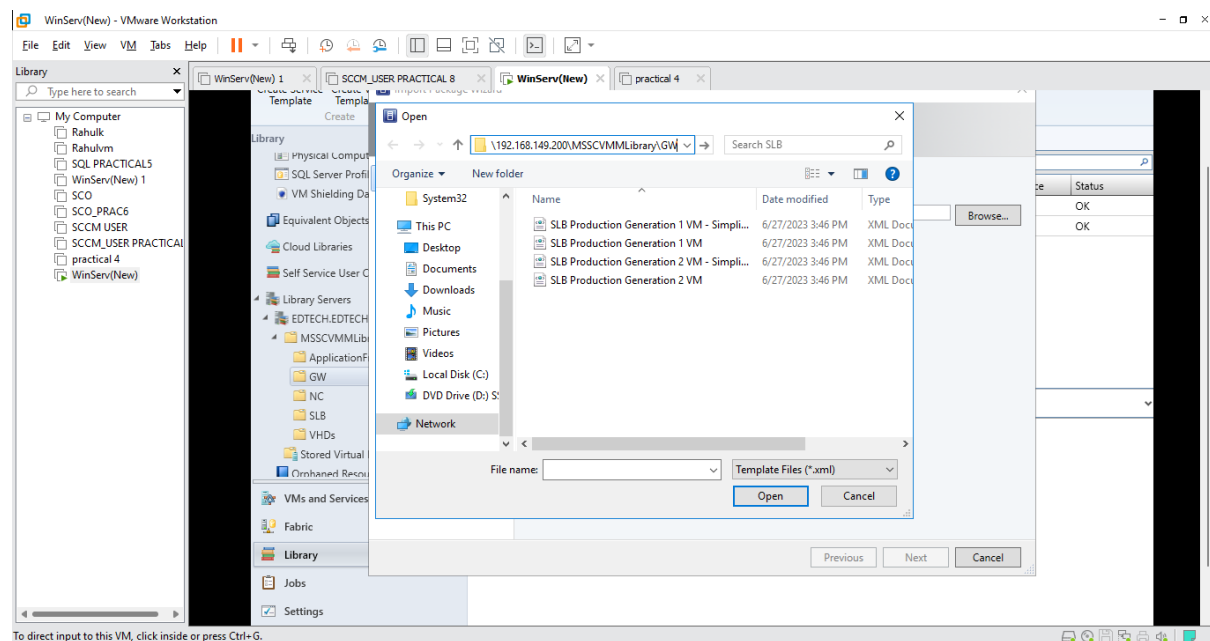
- Within MSSCVMMLibrary Select GW Folder and then Copy the Path



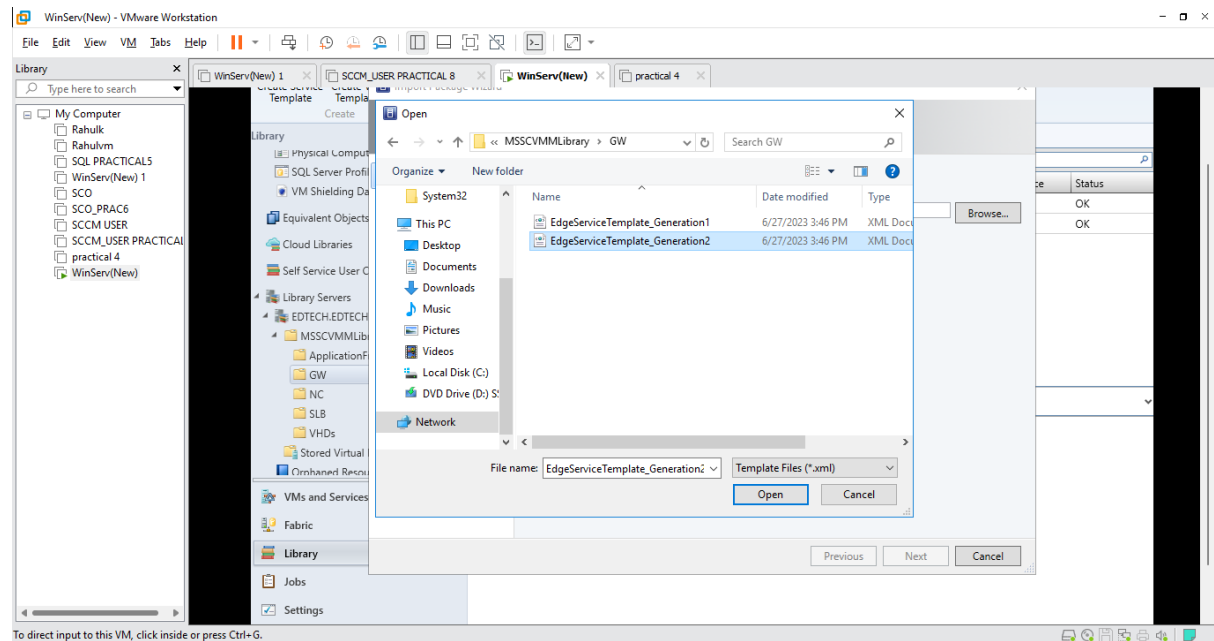
- Now again Go to the Select Package window **Click on Browse**



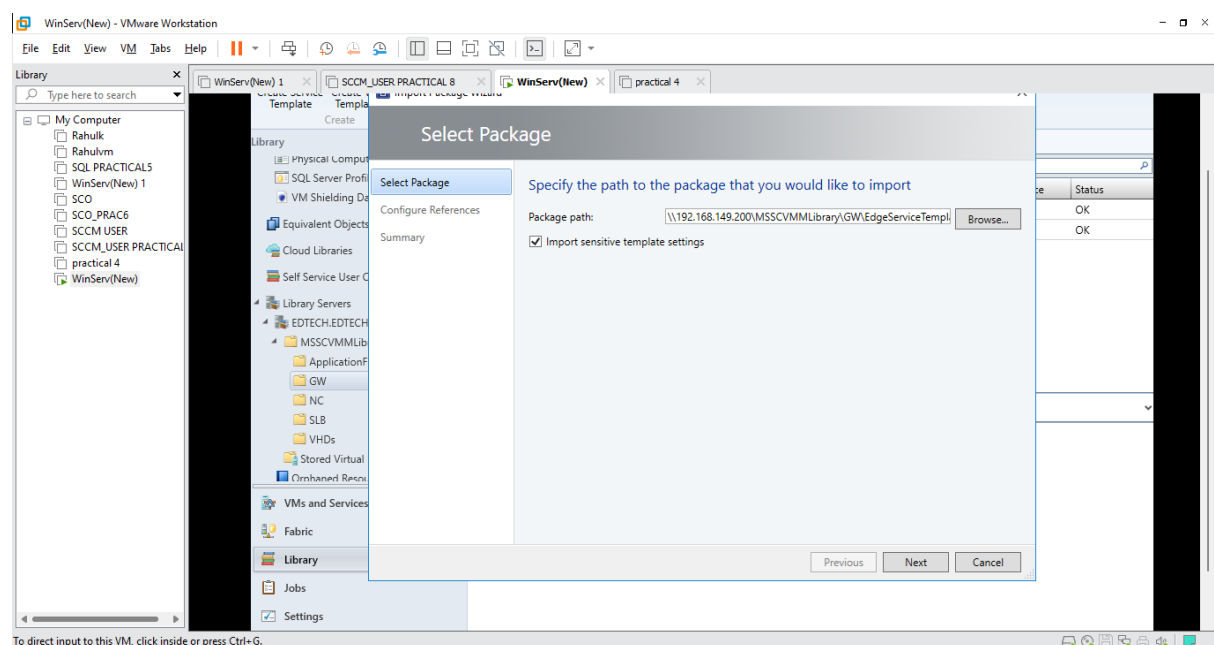
- Now paste the path for GW and then **Click on Enter**



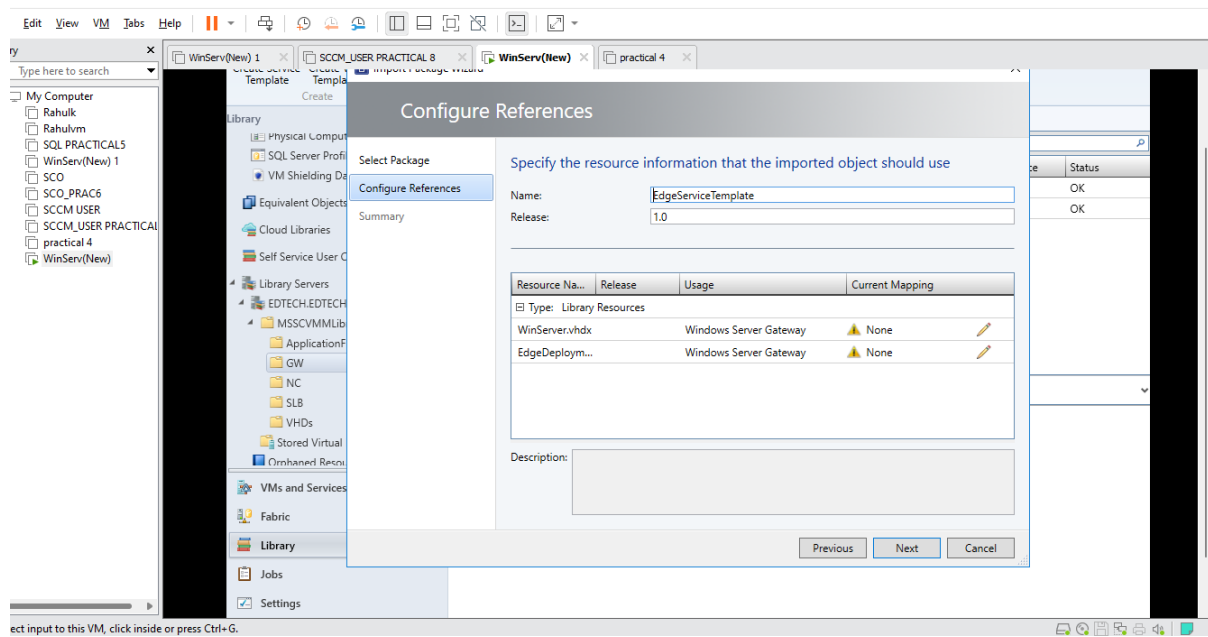
- Within GW Folder Select Network **EdgeServiceTemplate\_Generation 2 VM** and then **Click on Open**



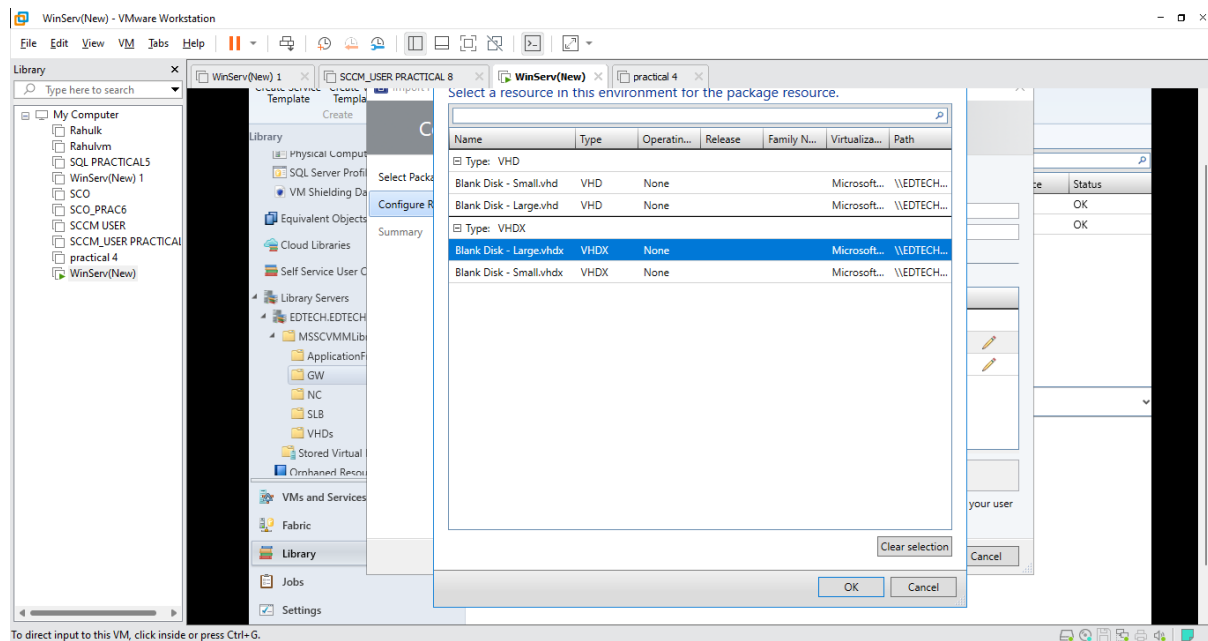
- **Click on Next**



- Within Configuration References Click on WinServer.vhdx and then Click on Edit

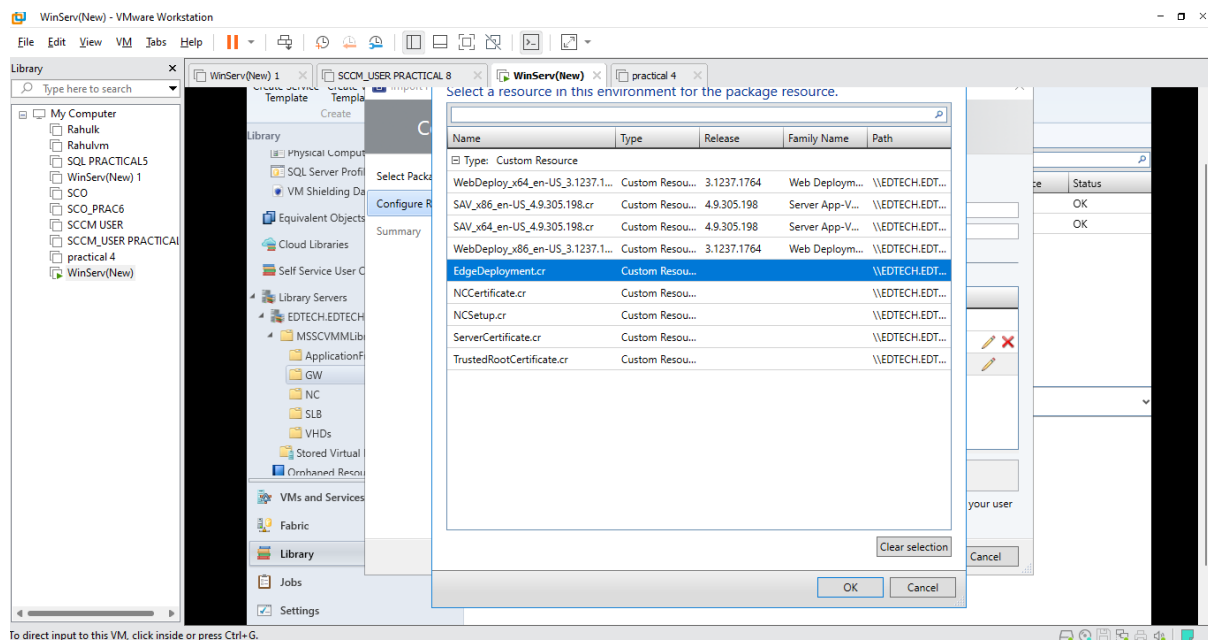


- Within Type: VHDX Select Blank Disk – Large.vhdx and Click on OK

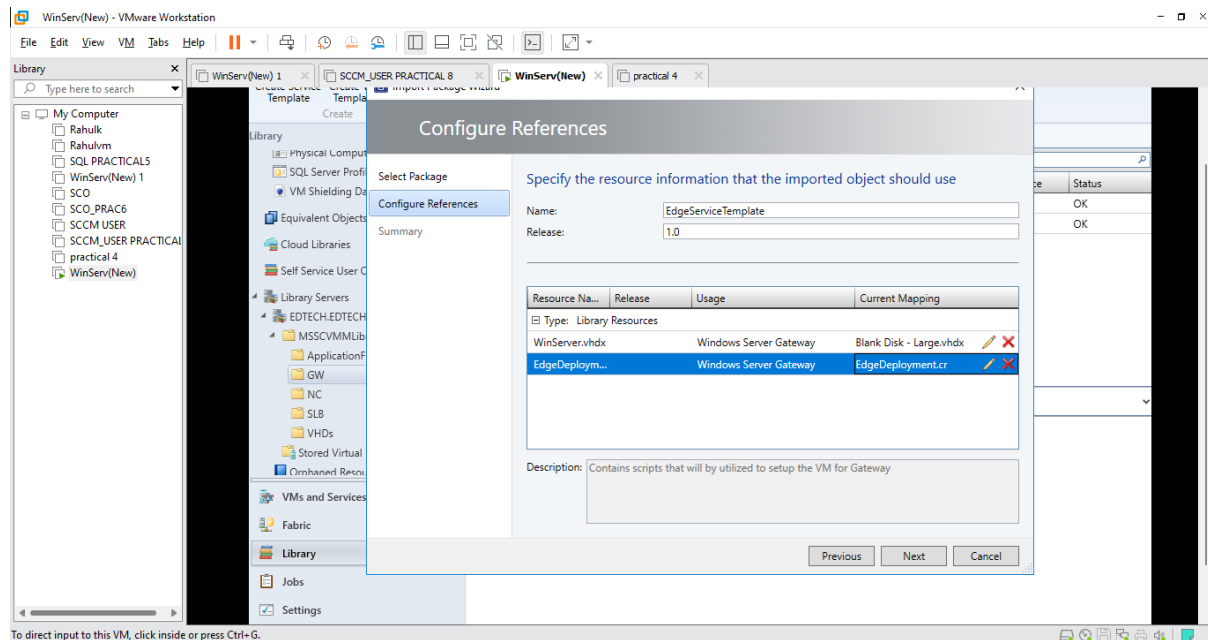




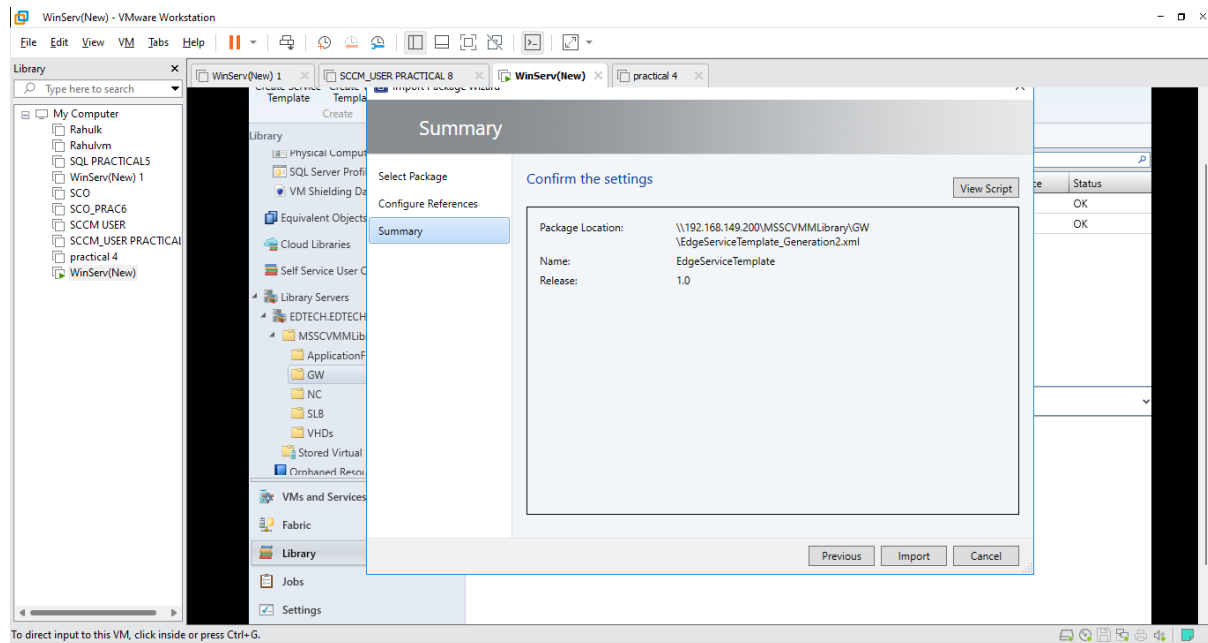
- Click on EdgeDeployment Windows Server Gateway and then Select EdgeDeployment.cr and Click on Ok



- Click on Next

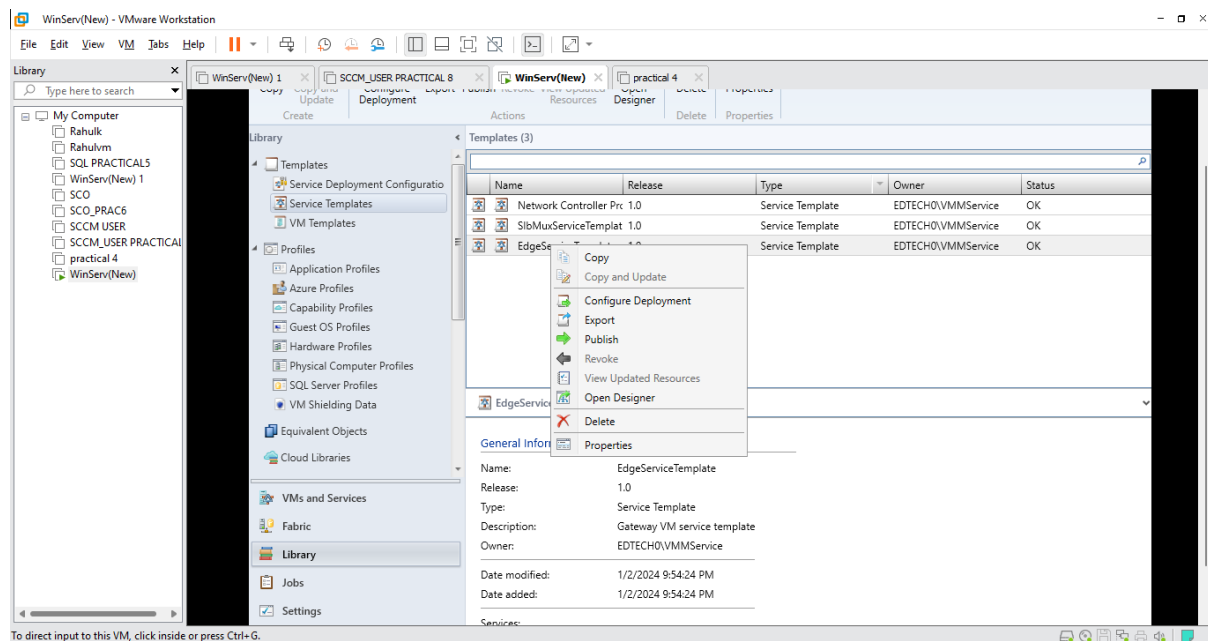


- **Click on Import**

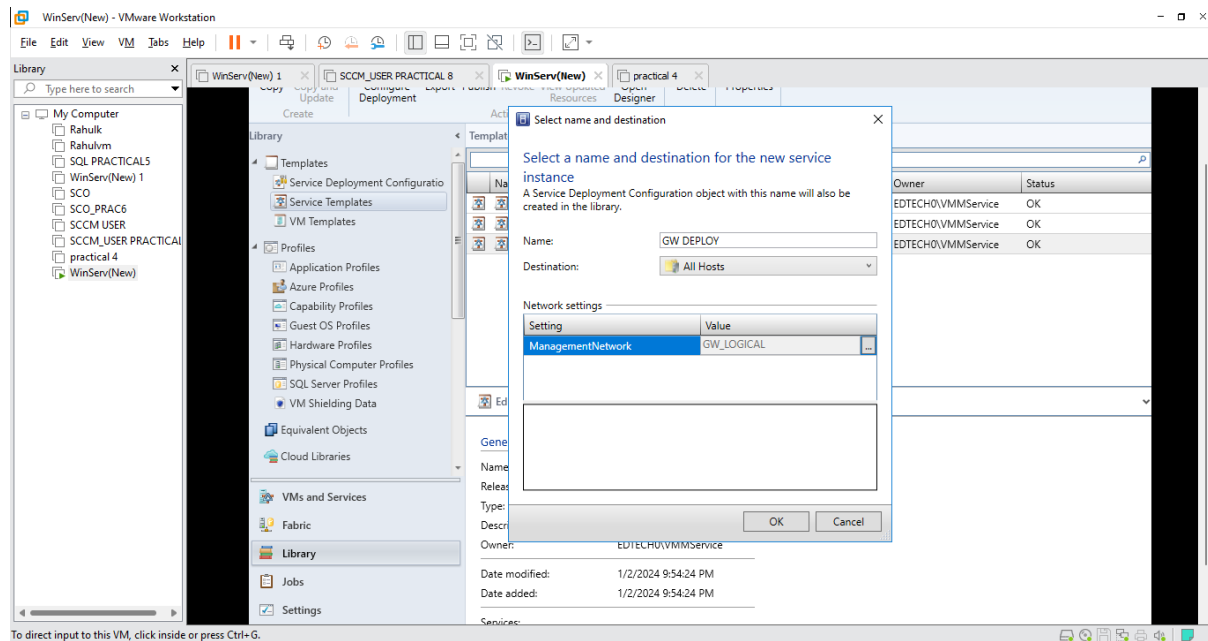


#### Step 4: Deploying the GW Service Template

- Within Service Templates **Right-Click on EdgeServiceTemplate** and then **Click on Configure Deployment**



- Type in the name (Here it is **GW DEPLOY**) and in the **Transit and Management Network** Select the newly created Logical Network (Here it is **GW\_LOGICAL**) and Click on Ok



- Here enter the following:
- AdminAccount: NT AUTHORITY\LocalService**
- MangementNetwork: GW\_LOGICAL**
- MgmtDomainAccount: NT AUTHORITY\Local Service**
- MgmtDomainFQDN: EDTECH.EDTECH.COM**

